

**SIGNAL DETECTION OF GRAVITATIONAL WAVES FROM
ECCENTRIC BINARIES**

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of

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by

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ABSTRACT

SIGNAL DETECTION OF GRAVITATIONAL WAVES FROM ECCENTRIC BINARIES

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Coalescing compact binaries have been pointed out as the most promising source of gravitational waves for kilometer-size interferometers such as the Laser Interferometers Gravitational-Wave Observatory. Gravitational wave signals are extracted from the noise in the detectors by matched filtering. This technique performs really well if an *a priori* theoretical knowledge of the signal is available. The information known about the possible sources is used to construct a model of the expected waveforms (templates). A common assumption made when constructing templates for coalescing compact binaries is that the companions move in a quasi-circular orbit. Some scenarios, however, predict the existence of eccentric binaries. In this thesis, we calculate the loss in signal-to-noise ratio induced by non-optimal filtering of eccentric signals.

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Chapter 1

Synopsis of gravitational waves and their detection

1.1 Opening remarks

This chapter is intended to explain the basics of gravitational waves and their detection. It is not destined to be a complete and rigorous treatment of the different theoretical aspects. The first topic discussed is gravitational waves: their generation and their interaction with matter (subsections 1.2.1 and 1.2.2 respectively). Once we are a little more familiar with this type of radiation, the kilometer-size interferometers presently under construction will be examined.

A detector is not complete without a technique of extraction of the signals from the noise. In the case of gravitational waves of known shape, the most promising extraction method is matched filtering. The basic ideas of matched filtering are discussed in subsection 1.3.2. Once these different aspects of wave generation and wave detection are established, various sources of gravitational waves will be described in subsection 1.3.3. One of the most promising sources are inspiraling compact binaries;

these are described in subsection 1.3.3.

The final portion of this chapter (section 1.4) contains justifications for the present work, as well as an outline of the thesis.

1.2 Gravitational waves: overview of the theory

1.2.1 Generation of gravitational waves

The starting point of any discussion on general relativity is the well known Einstein equation[1, 2]:

$$R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R = 8\pi T_{\mu\nu} . \quad (1.1)$$

This equation relates the matter stress-energy tensor ($T_{\mu\nu}$) to the spacetime curvature ($R_{\mu\nu}$ and R) in a dynamical way. Essentially it implies that matter and energy interact via gravitation and the manifestation of this interaction appears in the spacetime curvature. Gravitation is a geometric phenomenon.

In some situations Einstein's equation can be solved exactly, but the analytical solutions do not, typically, represent physical gravitational fields. To surmount this difficulty one often has recourse to an approximation, either in the form of an expansion in the characteristic velocity of the system or by perturbing a background spacetime[3, 4]. For the present work, the background spacetime is chosen to be flat. Thus the background geometry is described by the Minkowski metric $\eta_{\mu\nu} = \text{diag}(-1, 1, 1, 1)$. Gravitational waves enter the picture when this flat background spacetime is perturbed. The perturbation $h_{\mu\nu}$ is expressed as the difference between the perturbed and the unperturbed metric:

$$h_{\mu\nu} = g_{\mu\nu} - \eta_{\mu\nu} , \quad (1.2)$$

where $g_{\mu\nu}$ is the true metric of the spacetime. Linearizing Einstein's equation with respect to the auxiliary field $\bar{h}_{\mu\nu} = h_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}h^\alpha{}_\alpha$ results in the wave equation[5]

$$\square \bar{h}^{\mu\nu} = 16\pi T^{\mu\nu}, \quad (1.3)$$

where $\square \equiv \frac{\partial^2}{\partial t^2} - \nabla^2$ is the wave operator in flat spacetime. This wave equation is valid when $\bar{h}_{\mu\nu}$ satisfies the Lorentz gauge condition $(\bar{h}^{\mu\nu}{}_{,\nu} = 0)$ [6].

The gravitational field can be shown to have two radiative degrees of freedom. The formal solution of equation (1.3) reduced to the physical degrees of freedom is most conveniently expressed in the transverse-traceless gauge (TT). In this gauge, the radiation is traceless ($\bar{h}^\nu{}_\nu = 0$) and transverse to the direction of propagation of the wave, i.e. if $\mathbf{n}$ points in this direction, then $\bar{h}_{ij}n^j = 0$. The TT gauge is introduced formally in subsection 3.2.2. For now it suffice to state the solution of the wave equation[5]:

$$h_{ij}^{TT} = \frac{G}{c^4} \frac{2}{R} \ddot{\mathfrak{I}}_{ij}^{TT}, \quad (1.4)$$

where R is the distance between the source and the observer, G is Newton's constant, c is the speed of light, the superscript TT means that a traceless-transverse projection has been taken, the overbar on $\bar{h}_{\mu\nu}$ has been dropped since in the TT gauge $\bar{h}_{\mu\nu} = h_{\mu\nu}$, and $\mathfrak{I}_{ij}$ is the reduced quadrupole moment of the source defined in term of the usual quadrupole moment (I_{ij}) as

$$\mathfrak{I}_{ij} = I_{ij} - \frac{1}{3}\delta_{ij}I^k{}_k. \quad (1.5)$$

For a source with characteristic size $\bar{r}$ and mass M , the quadrupole moment is approximately $M\bar{r}^2$. If the system evolves on a time scale $\bar{t}$, the time derivatives in equation (1.4) can be replaced by $\bar{t}$. The quadrupole formula gives an order-of-magnitude

estimate of the gravitational-wave strength as

$$h \sim \frac{G}{c^2} \frac{M}{R} \left(\frac{\bar{r}}{c\bar{t}} \right)^2 \sim \frac{G}{c^2} \frac{M}{R} \left(\frac{\bar{v}}{c} \right)^2, \quad (1.6)$$

where $\bar{v} \equiv \frac{\bar{r}}{\bar{t}}$ is a characteristic velocity of the source. A source with characteristic parameters $M = 1 M_\odot$ and $\bar{v} = 0.15c$ emits gravitational waves with amplitude

$$h \sim \begin{cases} 10^{-18} & \text{for a source located at } R = 1\text{kpc} \\ 10^{-21} & \text{for a source located at } R = 1\text{Mpc}. \end{cases} \quad (1.7)$$

If the gravitational waves are traveling in the z direction, all the components h_{iz} vanish in the TT gauge. In this case the only non-vanishing radiative parts are

$$\begin{aligned} h_+ &= h_{xx} = -h_{yy} \\ h_\times &= h_{xy} = h_{yx} \end{aligned} \quad (1.8)$$

h_+ ($h_\times$) is the plus- (cross-) polarization mode of gravitational waves. The origin of the name will be made clear at the end of subsection (1.2.2).

1.2.2 Interaction of the waves with matter

In order to detect gravitational waves, we have to know in which way they affect matter. Consider for example two test masses A and B. Both masses follow a space-time geodesic. Assume that A is located at x_A^α , and B at $x_B^\alpha = x_A^\alpha + \xi^\alpha$. Hence ξ^α is the separation vector between the two masses. Choose a coordinates system centered at A and comoving with A, i.e. coordinates in which A is at rest. With this choice of coordinates, the vector ξ^α describes the motion of B with respect to A, i.e. $\xi^\alpha = x_B^\alpha$. An observer located at A will feel no gravitational force, but will see B move according to the geodesic deviation equation[6]

$$\frac{D^2 \xi^\mu}{D\tau^2} = -R^\mu{}_{\alpha\nu\beta} \xi^\nu \frac{dx_A^\alpha}{d\tau} \frac{dx_A^\beta}{d\tau}, \quad (1.9)$$

where τ is the proper time along the geodesic followed by A , $R^\mu{}_{\alpha\nu\beta}$ is the Riemann curvature tensor, and

$$\frac{DV^\mu}{D\tau} = \frac{dV^\mu}{d\tau} + \Gamma^\mu{}_{\alpha\beta} \frac{dx^\alpha}{d\tau} V^\beta$$

for any vector V^μ ; $\Gamma^\mu{}_{\alpha\beta}$ is the affine connection.

Specializing to the case of a perturbed flat spacetime (equation (1.2)), the geodesic deviation equation can be simplified by using the following approximate relations:

1. A 's four-velocity only has a time component, $\frac{dx^\alpha}{d\tau} = \delta_0^\alpha$;
2. to first order in the perturbation field,

$$\begin{aligned}\tau &= t, \\ \frac{D}{D\tau} &= \frac{d}{dt};\end{aligned}$$

3. spatial indices are raised and lowered freely, since $\eta_{ij} = \delta_{ij}$.

The geodesic deviation equation (equation (1.9)) then simplifies to[5]

$$\frac{d^2 x_B^j}{dt^2} = -R_{j0k0} x_B^k. \quad (1.10)$$

In the linearized theory of general relativity, the Riemann tensor can be shown to be proportional to the second time derivative of the perturbation field $h_{\mu\nu}$. The end result for the motion of B with respect to A is

$$\frac{d^2 x_B^j}{dt^2} = \frac{1}{2} \frac{\partial^2 h_{jk}^{TT}}{\partial t^2} x_B^k. \quad (1.11)$$

If the two masses are at rest with respect to each other before the arrival of the gravitational wave, equation (1.11) yields, to leading order,

$$x_B^j(t) = x_B^j(0) + \frac{1}{2} x_B^k(0) h_{jk}^{TT}(t), \quad (1.12)$$

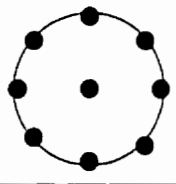
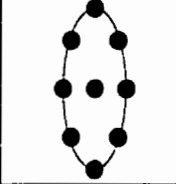
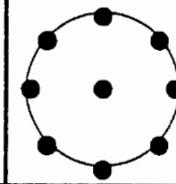
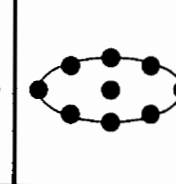
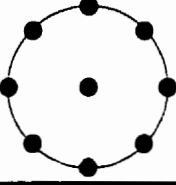
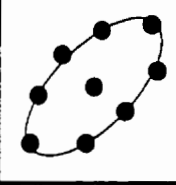
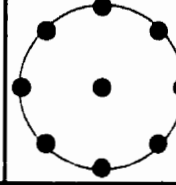
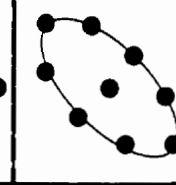
polarization h_+				
polarization h_x				
phase of the wave	0	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$

Figure 1.1: The effect of a gravitational wave passing through a circular array of test masses. The ring is located in the plane transverse to the direction of propagation of the wave. Both polarizations produce the same type of effect, but are at 45° from each other. The gravitational wave is assumed to be traveling in the z direction (out of the sheet). The phase describes the oscillating part of the wave.

where $x_B^j(0)$ is the position of B before the radiation reaches it. This describes the motion of B as seen by A . The wave affects only the transverse component of the separation between A and B .

Now that we know how to describe the interaction between gravitational waves and matter, we can understand the effect of the two polarizations mentioned previously. Take three masses and put them in the xoy plane transverse to the wave's direction of propagation, with mass A at the origin, mass B on the y axis and mass C on the x axis. If the only polarization present in the wave is the plus-polarization ($h_x = 0$), B and C move with respect to A according to

$$y_B(t) = y_B(0) \left[1 - \frac{1}{2} h_+(t) \right], \quad (1.13)$$

$$x_C(t) = x_C(0) \left[1 + \frac{1}{2} h_+(t) \right]. \quad (1.14)$$

Thus if B moves toward A, then C moves away and vice-versa. A schematic representation of the effect of the plus- and cross-polarization on a ring of test masses is shown in figure (1.1).

1.3 Gravitational wave detection

1.3.1 Laser interferometers as gravitational wave detectors

Quite generally, physical phenomena are detected through their effect on the surrounding environment. Electromagnetic waves, for example, are detected through their interaction with charged particles. The specifics of the interaction give information about the type of waves, the possible sources, the physical processes involved, etc ... The detection of gravitational waves is not different: the goal is to measure the strain produced by the passing gravitational wave as it interacts with matter (equation (1.11)) and to extract knowledge about the source from this strain. Because gravitational waves interact with matter weakly, the effects are tiny, and the experimental challenge is enormous.

Presently, many groups are working on gravitational-wave detectors. The projects underway, such as LIGO, VIRGO, TAMA, and GEO, use laser interferometry to measure the strain produced by gravitational waves. The kilometer-size interferometers seem to be the most efficient detectors at measuring this strain. They are adapted to the quadrupolar nature of the wave (equation (1.4)). They essentially are interferometers similar to the Michelson-Morley apparatus with the single corner mass replaced by two masses (see figure (1.2)). The motion of the two end masses is measured in the reference frame of the fixed corner masses. A laser is used to monitor the motion of the end masses. The laser beams travel in both arms, they bounce off the end masses

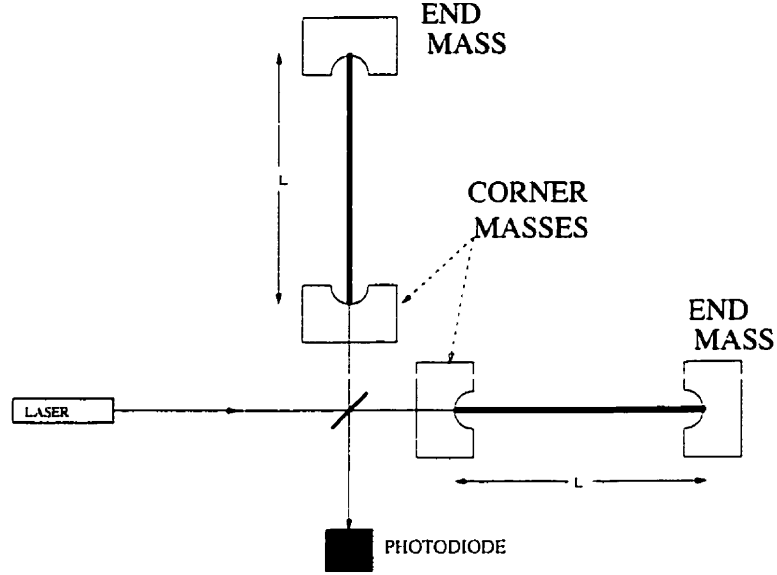


Figure 1.2: The LIGO interferometers under construction are Fabry-Perot type beam detectors. Represented is a top view of this type of detectors. Both arms have length L . The passing of gravitational wave causes the end masses to oscillate with the wave. The interference pattern produced is then converted to an electrical current by the photodiode.

and the resulting interference is converted to an electrical signal by the photodiode. If no gravitational wave is passing through the detector, both arms have the same length L and there is no interference between the beams.

Now suppose there is a plus-polarized gravitational wave passing through the detector. Furthermore suppose that the detector is oriented in such a way that both arms lay in the plane transverse to the direction of propagation of the wave; one along the x axis and one along the y axis. The effect of the gravitational wave is then given by equation (1.14). The interference between the two beams depends on the difference in path length (δl). This difference is given by subtracting the length of both arms when the wave is present. This is given by

$$\delta l(t) = \delta x(t) - \delta y(t), \quad (1.15)$$

where δx (δy) is the change in position of the mass on the x (y) axis. These are given by equation (1.14) with the initial position of the two masses set to L . Thus the detector will measure a path difference

$$\delta l(t) = h_+(t)L \quad (1.16)$$

or a strain

$$\frac{\delta l(t)}{L} = h_+(t). \quad (1.17)$$

In general, the signal will be a linear superposition of both polarization modes

$$h(t) = F_+ h_+(t) + F_\times h_\times(t), \quad (1.18)$$

where F_+ and $F_\times$ are the beam factors which depend on the relative orientation of the detector and the direction of propagation of the gravitational wave. For realistic gravitational waves equation (1.17) must be replaced by[7]

$$\frac{\delta l(t)}{L} = h(t). \quad (1.19)$$

This is the strain that must be monitored by the laser interferometers. Note that the sensitivity of the detector increases when the length of each arm is increased. However, there is a limit on how long the arms of the detector can be: the arms must be smaller than the characteristic wavelength λ_c of the radiation. For most sources the radiation will have $\lambda_c \geq 50$ km, but the wavelength could be as small as 5 km. The arms of the interferometers under construction are smaller than this: the LIGO interferometers have an arm length $L = 4$ km.

Gravitational waves are expected to produce a strain $\frac{\delta l}{L}$ in the interferometers of the order of 10^{-18} for sources in our own Galaxy and 10^{-21} for sources a few Mpc away (including nearby clusters)[8, 9]. These correspond to the estimates displayed

in equation (1.7). The noise in the interferometers is roughly of the same order of magnitude; a typical signal-to-noise ratio of 10 is expected. The initial LIGO interferometers are limited at low frequencies ($f \leq 40$ Hz) by seismic noise and at high frequencies ($f \geq 1000$ Hz) by photon shot noise. The initial LIGO will therefore be sensitive to signals having a frequency between 40 Hz and 1000 Hz. The advanced LIGO will have a good sensitivity between 10 Hz and 1000 Hz due to a better suspension of the masses that will eliminate seismic noise more thoroughly. The laser interferometers are *broad-band* detectors.

1.3.2 Matched filtering as a detection method

The comparable strengths of the expected signal and the noise motivates the general belief that matched filtering will be needed to extract the signals from the output of the detectors[7]. When the signals are of known shape, this technique produces the highest signal-to-noise ratio[10]. Suppose a gravitational wave $h(t)$ reaches the detector. The output of the detector $o(t)$ is then a superposition of the useful signal $h(t)$ and the noise $n(t)$. For matched filtering, the signal is extracted by using a theoretical template (theoretical model) that mimics the signal as completely as possible; we shall call this template $m(t, \mathbf{\Omega})$. The vector $\mathbf{\Omega}$ represents the parameters that characterize the template. If the template were a perfect copy of the signal these would be the real parameters of the source, such as its mass and distance from earth. If the templates are not a perfect copy of the real signal, the parameters $\mathbf{\Omega}$ are just phenomenological parameters.

The natural way to do signal detection is to work in frequency space. The Fourier

transform of a real function $F(t)$ is

$$\tilde{F}(f) = \int_{-\infty}^{\infty} dt F(t) e^{2\pi i f t}. \quad (1.20)$$

Working in frequency space allows the introduction of the inner product for matched filtering. For two functions $\tilde{a}(f)$ and $\tilde{b}(f)$, the inner product is defined as

$$(a|b) = 2 \int_0^{\infty} df \frac{\tilde{a}^*(f) \tilde{b}(f) + \tilde{a}(f) \tilde{b}^*(f)}{S_n(f)}, \quad (1.21)$$

where a “*” denotes complex conjugation and $S_n(f)$ is the one-sided spectral density of the detector’s noise. In terms of this inner product, the average signal-to-noise (SNR) ratio can be expressed as[11]

$$\langle \rho \rangle = \frac{(m(\boldsymbol{\Omega})|h)}{\sqrt{(m(\boldsymbol{\Omega})|m(\boldsymbol{\Omega}))}}. \quad (1.22)$$

The average is taken over all possible realizations of the noise $n(t)$, which is assumed to have stationary and Gaussian statistical properties. The signal-to-noise ratio can be shown to be a Gaussian random variable, whose mean value is given by equation (1.22), and whose variance is equal to one. If $\langle \rho \rangle \gg 1$, then the actual signal-to-noise ratio is never very different from the mean. Throughout this thesis we will work exclusively in terms of $\langle \rho \rangle$.

In practice, the parameters $\boldsymbol{\Omega}$ are varied until the signal-to-noise ratio is maximized. Of course this is the maximum signal-to-noise ratio achievable given the choice of template $\tilde{m}(f, \boldsymbol{\Omega})$. The signal-to-noise ratio of equation (1.22) does not give any information about the quality of the templates, or equivalently, about how well the template models the signal. The Schwartz inequality provides an answer to this question[11]. The absolute maximum value the signal-to noise ratio can take is achieved when the template is a *perfect* match of the signal and the parameters $\boldsymbol{\Omega}$ correspond to the parameters of the source ($\tilde{m}(t, \boldsymbol{\Omega}) \equiv \tilde{h}(f)$). The optimal SNR is

$$\langle \rho \rangle_{max} = \sqrt{(h|h)}. \quad (1.23)$$

By dividing the signal-to-noise ratio (equation (1.22)) with the value achieved by optimal filtering (equation (1.23)), we construct the ambiguity function $\mathcal{A}(\Omega)$:

$$\begin{aligned}\mathcal{A}(\Omega) &\equiv \frac{\langle \rho \rangle}{\langle \rho \rangle_{max}} \\ &= \frac{(m(\Omega)|h)}{\sqrt{(m(\Omega)|m(\Omega))(h|h)}}.\end{aligned}\tag{1.24}$$

This function takes values between 0 and 1. It is equal to 1 when the optimal template is used.

The value of the parameters Ω are varied until $\mathcal{A}(\Omega)$ is maximized. The maximum value of the ambiguity function is the fitting factor:

$$FF = \max_{\Omega} \mathcal{A}(\Omega).\tag{1.25}$$

The fitting factor is a direct measure of the template's quality since it can be related to the loss of event rate, i.e. the number of events missed by using an inappropriate set of templates. This loss is calculated according to $1 - FF^3$ [12]. For example, for a given set of templates, if the fitting factor is 0.8, then 48.8% of the events would be mistaken for noise.

1.3.3 Astrophysical sources of gravitational waves

In order to construct accurate templates, as required by matched filtering, we must have some idea of the important sources of gravitational waves. A great deal of work went into evaluating the possible sources and the construction of adequate templates.

An interesting source of gravitational waves are rotating neutron stars. Such stars emit gravitational waves if their shape is not axially symmetric. Deviations from axial symmetry could arise in various ways. The process that led to the formation of the star might have left some large deformation on the solid crust; or if the neutron

stars have a strong internal magnetic field, the magnetic pressure could support such deformations. The faster the star rotates and the larger the deformations are, the stronger the emitted radiation will be. Of course, the emission of gravitational waves will cause the star eventually to become axisymmetric, and it will slow down the rotation. Unfortunately, most known pulsars have an extremely low slowdown rate, implying that they emit very weak gravitational waves[7].

The most promising source of gravitational waves for the laser interferometers is without any doubt coalescing compact binaries[8, 9, 7]. These binaries are made of neutron star and/or black holes. Those are expected to be fairly abundant since they constitute an end result of stellar evolution. As the companions travel around each other, the emission of gravitational waves drives them closer together. The binary's orbital frequency sweeps upward until the two companions merge. For a binary system made of two neutron stars, the gravitational wave's frequency just prior to merger is just below 1000 Hz, placing them in the right frequency range for the LIGO interferometers.

Compact binaries are probably one of the simplest astrophysical systems to model. In a first approximation the waveforms can be calculated by considering the two stars to be point masses. Spin-orbit, spin-spin, and tidal couplings are higher order phenomena[13, 14]. The coalescence consists of three stages. The first stage is an adiabatic inspiral well modeled by a post-Newtonian expansion of the general-relativistic equations. The second stage is the free-fall of the two companions toward each other after the orbit has passed beyond the innermost stable circular orbit. The last stage is the merger. The gravitational waveforms emitted during the last two stages need to be obtained by numerically evolving the full general-relativistic equations; this is especially true of the merger.

The complete knowledge of the waveforms associated with coalescing binaries will allow accurate measurements of their masses, their distance and their spin. These precise measurements will constitute the strongest tests of general relativity[15] and make possible the measurements of astrophysical quantities such as the Hubble constant[8, 7].

1.4 Scope of the thesis

1.4.1 LIGO and eccentric compact binaries

Coalescing compact binaries have been pointed out as one of the most promising source of gravitational wave as long as accurate templates can be constructed for matched filtering. One strong assumption made when constructing these templates is that the binaries are old enough to have radiated most of their eccentricity away, i.e. the companions move in a quasi-circular orbit. As a result of the radiation reaction, the eccentricity e decreases with the semi-major axis a according to the approximate relation $(e/e_o) \approx (a/a_o)^{19/12}$, where e_o (a_o) is the eccentricity (semi-major axis) at some prior time[16]. Thus, a reduction of the semi-major axis by a factor of 2 is accompanied by a reduction of the eccentricity by a factor of 3. The radiation reaction therefore quickly circularizes the orbit. Using this assumption, the circular waveforms have been calculated up to order $(\frac{v}{c})^5$ (2.5PN order) beyond the leading-order result obtained from the quadrupole formula[14, 13, 3, 4]. For most compact binaries the assumption of circularity is valid, but exceptions can occur.

It came as a surprise to many when AGN's (Active Galactic Nuclei) were first observed. AGN's are optically active and are the site of star formation. They have strong X - and γ -ray spectra, which are thought to originate from high power plasma

jets. One of the possible explanations for the jets is the presence of a supermassive black hole ($10^6 - 10^9 M_\odot$) at the center of the cluster, combined with an accreting disc[17]. The question remains as to what processes led to the formation of those supermassive black holes.

One possible scenario was pointed out by Shapiro and Teukolsky[18]. They considered a dense Newtonian cluster whose evolution is dominated by star-star collision and secular collapse. The central density increases until it creates a dynamical instability that causes the cluster to collapse catastrophically to a supermassive black hole. Quinlan and Shapiro looked at the same scenario, but followed the formation of binaries in greater detail[19]. In such a cluster, binaries form according to two different schemes: two- and three-body encounters. In two-body encounters, the dissipative force comes from the emission of gravitational waves, while in three-body encounters, some of the kinetic energy is removed from the system by the third star. In both cases the binaries harden, once formed, by emitting gravitational waves. This ensures that they will not be disrupted by other encounters and that they will coalesce. Hence, the collapse of a cluster to a supermassive black hole might lead to a copious number of eccentric binary systems.

Whether or not radiation coming from these eccentric binaries can be detected is the question that we wish to examine in this work. One way to address this issue would be to incorporate eccentric templates in the bank of templates available to the LIGO interferometers. This solution comes with problems. It would give rise to an extra computational burden that might forbid real time detection. The computer burden is already fairly heavy because of the number of templates involved; the bank of circular templates is huge even though techniques to maximize the template spacing and minimize the number of templates are starting to be developed[20, 21].

Another drawback is the need to increase the threshold on the signal-to-noise ratio when the number of templates is increased, because the probability of false detection increases when the number of templates is increased. In order to counterbalance this effect, the minimum value the signal-to-noise ratio must take before a signal can be concluded to be present has to be increased. This could lead to the dismissal of circular signals that were really present in the detector.

In this thesis we choose to opt for another strategy: search for eccentric signals using circular templates. Our main goal is to assess the loss in signal-to-noise ratio induced by non-optimal filtering of the eccentric signals.

1.4.2 Outline

This thesis is divided as follow. Chapter 2 presents the basic elements of detection theory that was discussed briefly in subsection 1.3.2. This includes a derivation of the signal-to-noise ratio and the fitting factor. A derivation of the quadrupole moment approximation for gravitational waves and the radiation reaction is provided in chapter 3. We then move on to present and explain our results. Chapter 4 contains a derivation of the waveforms for eccentric binaries. These are needed to construct the ambiguity function, the corner stone of the present work. In chapter 5 we calculate the Fourier transform of the various waveforms. In this chapter, we describe a method for constructing and maximizing the ambiguity function, and results are presented for the optimal signal-to-noise ratio and the fitting factor. Finally, chapter 6 contains a brief summary of our results and conclusions. It also contains suggestions for future improvements of the work presented here.

Throughout this thesis, we assume summation over repeated indices and that $G = c = 1$. Latin indices run from 1 to 3, Greek indices from 0 to 3 with $\{x_0, x_1, x_2, x_3\} \Rightarrow \{t, x, y, z\}$.

Chapter 2

Signal Detection: Elements of the Theory

2.1 Introduction

Signal processing involves two very different steps. The first one is signal detection. At this stage, the goal is to assess whether a signal is present or not in the output of the detector. Once it is known that a signal is present, signal extraction begins. Both processes involve separating the useful signal from the noise as cleanly as possible. To accomplish this, a filter is passed through the output of the detector to eliminate the noise as much as possible.

The LIGO interferometers are expected to be collecting data at all times, giving access to the entire signal during the detection process. This property can be used to eliminate the noise as thoroughly as possible. Consequently, the filter should achieve two things: mimic the useful signal as accurately as possible and discard the regions of the spectrum where the noise is comparable in size to, or larger than, the useful signal. The template is the crude model of the gravitational waves used to construct the filter. It is thus an approximation of the waveforms expected to be emitted by a given astrophysical source. When such a

template is used, the extraction process is called matched filtering. This technique has been identified as the most promising one for the extraction of radiation emitted by known and well modeled astrophysical sources. Coalescing compact binaries are among such sources[7].

The chapter is divided as follows. Subsection 2.2.1 introduces linear filters and provides a definition of their action on a signal. In order to make a full treatment of the detection process, some properties of the noise must be known. The noise is a statistical process and as such it needs a special treatment to determine how it is affected by the filter. This is the purpose of subsection 2.2.2. Because the noise is a random variable, the various quantities to be measured are random variables as well. The presence (absence) of a signal is decided through the use of the signal-to-noise ratio introduced in subsection 2.2.3. Even though this quantity is sufficient to decide if a signal is present or not, it is insufficient to assess the quality of a given template. Subsection 2.2.3 introduces the fitting factor. It is a direct measure of the quality of the templates, i.e. it measures how well the templates (theoretical models of the radiation) model the signal.

2.2 Detection of signals of known shape

2.2.1 Signal processing with linear filters

In this section linear filters are defined and some of their basic properties are established. Suppose that a violent astrophysical event takes place, emitting gravitational waves in the process. The gravitational waves emitted will depend on the source's parameters. Thus, we denote the useful signal by $h(t, \boldsymbol{\vartheta})$, where $\boldsymbol{\vartheta}$ denotes the set of unknown parameters characterizing the source. If the source is described by N_s parameters, then $\boldsymbol{\vartheta} = \{\vartheta_1, \dots, \vartheta_{N_s}\}$. These are the parameters one is ultimately interested in measuring.

Upon reception, the signal $h(t, \boldsymbol{\vartheta})$ will get mixed up with the noise $n(t)$ in the detector.

This means that before filtering, the input signal $i(t, \boldsymbol{\vartheta})$ is of the form

$$i(t, \boldsymbol{\vartheta}) = h(t, \boldsymbol{\vartheta}) + n(t). \quad (2.1)$$

Beforehand it is unknown if a signal is really present. In order to decide if a signal is present, a filter $T(t, \boldsymbol{\Omega})$ is adopted, where $\boldsymbol{\Omega}$ is the set of parameters characterizing the filter. The filtered signal $O(t, \boldsymbol{\Omega}, \boldsymbol{\vartheta})$ is obtained by acting on $i(t, \boldsymbol{\vartheta})$ with the filter. For linear filters the action is defined as[22]

$$O(t, \boldsymbol{\Omega}, \boldsymbol{\vartheta}) = \int_{-\infty}^{\infty} T(t' + t, \boldsymbol{\Omega}) i(t', \boldsymbol{\vartheta}) dt'. \quad (2.2)$$

This equation justifies the name “linear filter”. It is obvious that $O(t, \boldsymbol{\Omega}, \boldsymbol{\vartheta})$ is a linear combination of the signal $i(t, \boldsymbol{\vartheta})$ at all times with each contribution weighted by the filter $T(t, \boldsymbol{\Omega})$ [11]. It is also linear in the usual sense: filtering the sum of two signals results in the sum of two filtered signals. We denote the action of a linear filter on a signal $s(t)$ by $\mathcal{D}[s(t)]$; then $\mathcal{D}[s_1(t) + s_2(t)] = \mathcal{D}[s_1(t)] + \mathcal{D}[s_2(t)]$.

Thus, it is meaningful to ask what the effect of the filter is on the useful signal $h(t, \boldsymbol{\vartheta})$. This is trivially seen by using the property mentioned above. Filtering the output of the detector results in

$$H(t, \boldsymbol{\Omega}, \boldsymbol{\vartheta}) = \int_{-\infty}^{\infty} T(t' + t, \boldsymbol{\Omega}) h(t', \boldsymbol{\vartheta}) dt'. \quad (2.3)$$

The signal and the set of templates can be decomposed in their spectral components by means of a Fourier transform. For a function $F(t)$, the inverse Fourier transform is

$$F(t) = \int_{-\infty}^{\infty} \tilde{F}(f) e^{-2\pi i f t} df, \quad (2.4)$$

where a tilde “ $\sim$ ” denotes the Fourier transform. Remembering that

$$\int_{-\infty}^{\infty} e^{-2\pi i (f+f')t} dt = \delta(f + f'), \quad (2.5)$$

and that for a real function $F(t)$, $\tilde{F}^*(f) = \tilde{F}(-f)$, the effect of filtering on the useful signal can be expressed as

$$H(t, \boldsymbol{\Omega}, \boldsymbol{\vartheta}) = \int_{-\infty}^{\infty} \tilde{T}^*(f, \boldsymbol{\Omega}) \tilde{h}(f, \boldsymbol{\vartheta}) e^{-2\pi i f t} df, \quad (2.6)$$

where a “ $\cdot$ ” denotes complex conjugation.

2.2.2 The noise in the initial LIGO

Because the noise is a random process, its Fourier transform is undefined. It does not seem to matter since, in the time domain, the effect of the filter on the noise is clear. Unfortunately, the calculations are most efficiently carried out in the frequency domain. It is the goal of this section to define the effect of the filter on the noise in the frequency domain and to establish some of the noise’s properties.

Because the noise is a random process, some assumptions must be made about its behavior; we assume that the noise in the LIGO interferometers is stationary, i.e. that its statistical properties are time-independent, and we assume that it is a random process with Gaussian statistics. The various realizations of the noise in the interferometers follow a Gaussian distribution.

A specific realization $n(t)$ of the noise is drawn from an ensemble. All the statistical quantities should thus be calculated by ensemble averaging. Fortunately, the ergodic hypothesis allows us to replace the ensemble averages by time averages. In this spirit, the mean value of the noise is defined as

$$\overline{n(t)} = \frac{1}{T} \int_{-T/2}^{T/2} n(\tau + t) d\tau \quad (2.7)$$

By monitoring the noise over a long period of time, we can get rid of its mean by first calculating it (using equation (2.7)) and then subtracting it off. This is possible because the noise is stationary and has a constant mean value $\overline{n(t)}$. This means that $\overline{n(t)}$ is not an important statistical quantity. The quantity in terms of which the noise can be described is the autocorrelation function[11],

$$\begin{aligned} R_n(t) &= \overline{n(\tau - t)n(\tau)} \\ &= \overline{n(-t)n(0)} = \overline{n(0)n(t)}, \end{aligned} \quad (2.8)$$

where the last line is valid for any stationary random process. For $t = 0$, the autocorrelation function is just the mean squared value of the noise before filtering, $\overline{n^2}$. Note that since the mean is zero, this quantity corresponds to the variance of the Gaussian distribution.

Contrary to the noise itself, the autocorrelation is not a random function and it admits a spectral decomposition. Its Fourier representation is defined by

$$\tilde{R}_n(f) = \int_{-\infty}^{\infty} e^{2\pi i f t} R_n(t) dt. \quad (2.9)$$

Note that equation (2.9) combined with equation (2.8) implies that $R_n(f) = R_n(-f)$. Because of this identity, it is conventional to define the power spectral density of the noise for positive frequencies only. In terms of $\tilde{R}_n(f)$, the power spectral density is defined as

$$S_n(f) = 2\tilde{R}_n(f) = 2 \int_{-\infty}^{\infty} e^{2\pi i f t} R_n(t) dt, \quad (2.10)$$

where the factor of 2 in the definition of $S_n(f)$ is conventional. The inverse of equation (2.10) is

$$R_n(t) = \int_0^{\infty} S_n(f) e^{-2\pi i f t} df, \quad (2.11)$$

where we used the fact that $\tilde{R}_n(f)$ is even to change the limits of integration to $[0, \infty]$. It is clear from equation (2.11) that the variance of the noise is given as

$$\begin{aligned} \overline{n^2} &\equiv R_n(0) \\ &= \int_0^{\infty} S_n(f) df. \end{aligned} \quad (2.12)$$

The power spectral density of the noise in the initial LIGO interferometers is not totally known, since they are still under construction, but it is estimated that at low frequencies ($f < 40$ Hz) the dominant contribution is due to seismic noise and at high frequencies, it is due to photon shot noise (fluctuations in the number of photons). Between these two frequencies, the noise will be caused by various sources (see L. S. Finn and D.F. Chernoff[23] for a description of the different sources). For the present purpose, we adopt an analytic fit

to the one-sided power spectral density $S_n(f)$ [24]:

$$S_n(f) = \begin{cases} \infty & 0 \leq f < 40 \text{ Hz} \\ S_o \left(2 + 2 \left(\frac{f}{f_m} \right)^2 + \left(\frac{f}{f_m} \right)^{-4} \right) & f \geq 40 \text{ Hz} \end{cases}, \quad (2.13)$$

where f_m is the frequency at which the noise reaches its minimum value ($f_m = 200$ Hz for the initial LIGO) and S_o is a constant amplitude left unspecified (the actual value is irrelevant for our calculations). This is not the only possible analytic fit to the power spectrum[10], but it is sufficient for the present work.

We now are in a position to evaluate how the noise is affected by the filtering process in the frequency domain. After filtering, the noise becomes (equation 2.2)

$$N(t, \mathbf{\Omega}) = \int_{-\infty}^{\infty} T(\tau, \mathbf{\Omega}) n(\tau - t) d\tau. \quad (2.14)$$

where we changed variable of integration from t' to $\tau = t' + t$. Because $\overline{n(\tau - t)} = 0$, we have $\overline{N(t, \mathbf{\Omega})} = 0$. On the other hand the mean squared average of $N(t, \mathbf{\Omega})$ is

$$\overline{N(t, \mathbf{\Omega})^2} = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} d\tau dt' T(\tau, \mathbf{\Omega}) T(t', \mathbf{\Omega}) \overline{n(\tau - t) n(t' - t)}. \quad (2.15)$$

The term $\overline{n(\tau - t) n(t' - t)}$ is just the autocorrelation of the noise $R_n(\tau - t')$ which in turn is related to the power spectral density (equation (2.11)) of the noise. Inserting equation (2.11) yields

$$\begin{aligned} \overline{N(t, \mathbf{\Omega})^2} &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} d\tau dt' T(\tau, \mathbf{\Omega}) T(t', \mathbf{\Omega}) \int_0^{\infty} df S_n(f) e^{-2\pi i f(\tau - t')} \\ &= \int_0^{\infty} df S_n(f) \int_{-\infty}^{\infty} d\tau T(\tau, \mathbf{\Omega}) e^{-2\pi i f \tau} \int_{-\infty}^{\infty} T(t', \mathbf{\Omega}) e^{2\pi i f t'} \\ &= \int_0^{\infty} df S_n(f) \tilde{T}(-f, \mathbf{\Omega}) \tilde{T}(f, \mathbf{\Omega}) \\ &= \int_0^{\infty} df S_n(f) \tilde{T}^*(f, \mathbf{\Omega}) \tilde{T}(f, \mathbf{\Omega}). \end{aligned} \quad (2.16)$$

2.2.3 The signal-to-noise ratio and the fitting factor

Quite generally, the probability of a signal being present or absent in the noisy output of the detectors can be linked to the signal-to-noise ratio (SNR)[23]. The SNR is defined as

the ratio of the strength of the useful signal to the strength of the noise after filtering. The standard theory of detection relates the probability that a signal has been detected to the signal-to-noise ratio. The higher the signal-to-noise ratio is, the more confident we can be that a signal is present in the detector. The threshold value the signal-to-noise ratio has to achieve before a signal can be concluded to be present is fixed by the observer. The threshold is chosen so that a signal will be concluded to be present with a certain confidence level, judged satisfactory by the observer. Formally, we define the squared average SNR $(\langle \rho \rangle^2)$ as

$$\langle \rho \rangle^2 = \frac{\overline{H(t, \mathbf{\Omega}, \mathbf{\vartheta})}^2}{\overline{N(t, \mathbf{\Omega})}^2}. \quad (2.17)$$

In terms of $H(t, \mathbf{\Omega}, \mathbf{\vartheta})$ and $\overline{N(t)^2}$ defined in equations (2.6) and (2.16) respectively, the squared SNR is written as

$$\langle \rho \rangle^2 = \frac{\left| \int_{-\infty}^{\infty} df \tilde{T}^*(f, \mathbf{\Omega}) \tilde{h}(f, \mathbf{\vartheta}) \right|^2}{\int_0^{\infty} df S_n(f) \tilde{T}(f, \mathbf{\Omega}) \tilde{T}^*(f, \mathbf{\Omega})}. \quad (2.18)$$

The question now arises as to which filter $\tilde{T}^*(f, \mathbf{\Omega})$ maximizes $\langle \rho \rangle$. To clarify this issue, it is worthwhile to introduce the notion of Wiener's optimal filter.

Consider the expression

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} df df' |\tilde{A}(f) \tilde{B}^*(f') - \tilde{A}^*(f') \tilde{B}(f)|^2 \geq 0,$$

where $\tilde{A}(f)$ and $\tilde{B}(f)$ are two arbitrary functions of f and $|\tilde{A}(f)|^2 \equiv \tilde{A}(f) \tilde{A}^*(f)$. The inequality is satisfied because the term on the left-hand side is positive definite and the equality is achieved only when $\tilde{A}(f) = \tilde{B}(f)$. Expanding the left hand side results in a sum of two different terms

$$\int_{-\infty}^{\infty} df |A(f)|^2 \int_{-\infty}^{\infty} df |B(f)|^2 - \left| \int_{-\infty}^{\infty} df \tilde{A}(f) \tilde{B}(f) \right|^2 \geq 0 \quad (2.19)$$

or

$$\left| \int_{-\infty}^{\infty} df \tilde{A}(f) \tilde{B}(f) \right|^2 \leq \int_{-\infty}^{\infty} df |A(f)|^2 \int_{-\infty}^{\infty} df |B(f)|^2. \quad (2.20)$$

This is the Schwartz inequality[11]. If we set

$$\begin{aligned}\tilde{A}(f) &= \tilde{T}^*(f, \boldsymbol{\Omega}) \sqrt{S_n(f)}, \\ \tilde{B}(f) &= \frac{\tilde{h}(f, \boldsymbol{\vartheta})}{\sqrt{S_n(f)}},\end{aligned}\tag{2.21}$$

and divide equation (2.20) by $\int_0^\infty df S_n(f) \tilde{T}(f, \boldsymbol{\Omega}) \tilde{T}^*(f, \boldsymbol{\Omega})$ we get

$$\langle \rho \rangle^2 \leq \langle \rho \rangle_{max}^2 = 4 \int_0^\infty df \frac{\tilde{h}(f, \boldsymbol{\vartheta})^* \tilde{h}(f, \boldsymbol{\vartheta})}{S_n(f)},\tag{2.22}$$

where we used $\tilde{h}(f, \boldsymbol{\vartheta}) = \tilde{h}^*(-f, \boldsymbol{\vartheta})$ to change the limits of integration to $[0, \infty]$. The maximum value of $\langle \rho \rangle^2$ is achieved for the filter

$$\tilde{T}(f, \boldsymbol{\Omega} = \boldsymbol{\vartheta}) = \frac{\tilde{h}(f, \boldsymbol{\vartheta})^*}{S_n(f)}.\tag{2.23}$$

This defines Wiener's optimal filter. Two choices need to be made to produce this filter. The first one is to choose a template that perfectly models the signal, and the second one is to weigh the template by $S_n(f)^{-1}$ in order to eliminate the regions of the spectrum where the noise is very loud. For the initial LIGO, the regions below 40 Hz and above 1000 Hz will be eliminated (equation (2.13)). Because we have some information about the noise, the template can always be weighted by the power spectral density of the noise, so that the second choice is always made. On the other hand, it is reasonable to assume that the template will not be an exact representation of the true signal. At best, it will be obtained by some approximation scheme (perturbation theory, post-Newtonian expansion, etc...). Accordingly, the parameter space of the template does not have to be of the same dimension as the signal's parameter space, i.e if $\boldsymbol{\Omega}$ has N_p components, then in general $N_p \neq N_s$. For example, a signal built using a post-Newtonian expansion will not have the same number of parameters as the waveforms built using the quadrupole formula. The post-Newtonian signal can be detected by the Newtonian template, but the extraction of the parameters is impossible, i.e. the parameters extracted do not match the parameters used to prepare the signal[25]. Consequently, we take the filter to be

$$\tilde{T}(f, \boldsymbol{\Omega}) = \frac{\tilde{m}(f, \boldsymbol{\Omega})}{S_n(f)},\tag{2.24}$$

where $\tilde{m}(f, \mathbf{\Omega})$ is the Fourier transform of the template used to model the gravitational waveforms. Wiener's theorem then states that $\tilde{T}(f, \mathbf{\Omega})$ maximizes the signal-to-noise ratio if $\tilde{m}(f, \mathbf{\Omega})$ is an accurate representation of the true signal $\tilde{h}(f, \mathbf{\vartheta})$, and if the template's parameters are matched to the source's parameters, $\Omega_i = \vartheta_i$, and $N_p = N_s$.

If the template $\tilde{m}(f)$ is not a perfectly accurate version of the true signal the product $\tilde{m}(f)^* \tilde{h}(f)$ might not be real. To get around this, we introduce a scalar product in the template space as

$$(a(\mathbf{\Omega})|b(\mathbf{\Omega}')) = 2 \int_0^\infty df \frac{\tilde{a}^*(f, \mathbf{\Omega}) \tilde{b}(f, \mathbf{\Omega}')}{S_n(f)} + c.c., \quad (2.25)$$

where $\tilde{a}(f, \mathbf{\Omega})$ and $\tilde{b}(f, \mathbf{\Omega}')$ are two arbitrary functions and *c.c.* denotes the complex conjugate. In terms of this scalar product the squared mean average SNR is expressed as

$$\langle \rho \rangle^2 = \frac{(m(\mathbf{\Omega})|h(\mathbf{\vartheta}))^2}{(m(\mathbf{\Omega})|m(\mathbf{\Omega}))} \quad (2.26)$$

or if we divide $\langle \rho \rangle^2$ by its maximum value (equation (2.22)) and take the square root

$$\frac{\langle \rho \rangle}{\langle \rho \rangle_{max}} = \frac{(m(\mathbf{\Omega})|h(\mathbf{\vartheta}))}{\sqrt{(m(\mathbf{\Omega})|m(\mathbf{\Omega}))(h(\mathbf{\vartheta})|h(\mathbf{\vartheta}))}} \quad (2.27)$$

The left hand side of equation (2.27) is the ambiguity function

$$\mathcal{A}(\mathbf{\Omega}, \mathbf{\vartheta}) = \frac{(m(\mathbf{\Omega})|h(\mathbf{\vartheta}))}{\sqrt{(m(\mathbf{\Omega})|m(\mathbf{\Omega}))(h(\mathbf{\vartheta})|h(\mathbf{\vartheta}))}} \quad (2.28)$$

and it takes values between 0 and 1. It will achieve a value of 1 if and only if:

1. the template $\tilde{m}(f, \mathbf{\Omega})$ is an accurate representation of the true signal $\tilde{h}(f, \mathbf{\vartheta})$,
2. the parameters $\mathbf{\Omega}$ are the same as the parameters of the source, $\Omega_i = \vartheta_i$.

Even if the template is not an exact copy of the true signal, it is possible to find a maximum of the ambiguity function. This is achieved by varying the parameters $\mathbf{\Omega}$ until a global maximum is found. Once the ambiguity function is maximized over $\mathbf{\Omega}$, it is called the fitting factor:

$$FF = \max_{\mathbf{\Omega}} \mathcal{A}(\mathbf{\Omega}, \mathbf{\vartheta}). \quad (2.29)$$

The fitting factor is a direct measure of the quality of the template, i.e. the way in which it mimics the true signal. A fitting factor close to unity implies a high quality template, since in this case the filter is close to Wiener's optimal filter. The fitting factor can also be related to the number of events missed by the detection process when an inadequate template is used. The number of signal missed by non-optimal filtering is given by $1 - FF^3$ [12]. For example, if we fix the minimum value the fitting factor can take to 0.9, then 27% of the signals at the output of the detectors will be missed. The value $FF = 0.9$ is the threshold we adopt in this work: below $FF = 0.9$ the circular templates are inadequate for the detection of eccentric signals.

Chapter 3

Gravitationally Bound Systems

3.1 Introduction

The Einstein equations (1.1) are difficult to deal with in their full non-linear form. It is better to adopt some approximation scheme appropriate to the problem at hand. This is reinforced by the fact that most exact solutions are physically unlikely. It is the purpose of this chapter to develop a formalism adequate to the problem of bound gravitational systems in the weak field limit. The results obtained will be used for the case of interest in this thesis: eccentric coalescing compact binaries.

The weak field approximation is introduced in section 3.2. This approximation allows the Einstein equations to be linearized with respect to the flat spacetime metric. Like in electromagnetism, the gravitational field is determined up to a gauge choice. The Lorentz gauge is imposed in subsection 3.2.2. This gauge choice results in a flat spacetime wave equation for the perturbation field. Subsection 3.3.2 contains the solution to the wave equation: the leading-order solution is the well known quadrupole formula. The energy and angular momentum carried away by the radiation are also calculated to leading-order. The derivation is sketched in subsection 3.3.2. The results of the first two sections (3.2 and 3.3) are valid for any system satisfying the weak field approximation. These results will be used

in chapter 4 to calculate the waveforms emitted by binary systems.

3.2 Linearized Theory of General Relativity

3.2.1 The weak field approximation

In the weak field approximation, a perturbation is applied to a background spacetime. The background spacetime is described by the metric $g_{\mu\nu}^B$, while the perturbed spacetime is described by $g_{\mu\nu}^P$. If the perturbation is small enough, the new metric will differ from the background by a small amount. The perturbation is then expressed as the difference between the reference and the perturbed spacetime[14]:

$$h_{\mu\nu} = g_{\mu\nu}^P - g_{\mu\nu}^B. \quad (3.1)$$

In the case at hand the background geometry is assumed to be flat. Thus, it is sufficient to define the perturbation field as

$$h_{\mu\nu} = g_{\mu\nu}^P - \eta_{\mu\nu}. \quad (3.2)$$

where $\eta_{\mu\nu} = \text{diag}(-1, 1, 1, 1)$ is the Minkowski metric. This closely follows the definition of C.W. Misner, K.S. Thorne and J.A. Wheeler for the weak field approximation[5].

Using the definition for the potential $h_{\mu\nu}$, the Einstein equations can be re-casted in a convenient form. In order to do this, we first note that since $\eta_{\mu\nu}$ has constant diagonal elements, all its derivatives are zero. Thus, taking derivatives of the metric $g_{\mu\nu}$ to evaluate the Christoffel symbols and the Riemann tensor (see B.F. Schutz [2] for a definition of these quantities) is equivalent to taking derivatives of the perturbation field and if no derivatives of the perturbed metric elements are taken, they are replaced by their flat spacetime equivalent:

$$\begin{aligned} g_{\mu\nu,\alpha} &\rightarrow h_{\mu\nu,\alpha}, \\ g_{\mu\nu} &\rightarrow \eta_{\mu\nu}. \end{aligned} \quad (3.3)$$

This way of linearizing the equations implies that all indices on linear quantities in the perturbation field must be raised and lowered using the flat space time metric $\eta_{\mu\nu}$. Inserting the relations (3.3) in Einstein's equation (1.1) leaves one with[6]

$$\square h_{\mu\nu} + h^\alpha{}_{\alpha,\mu\nu} - h_{\mu\alpha,\nu}{}^\alpha - h_{\nu\alpha,\mu}{}^\alpha + \eta_{\mu\nu} (h^{\alpha\beta}{}_{,\alpha\beta} - h^\alpha{}_{\alpha,\beta}{}^\beta) = -16\pi T_{\mu\nu} + 2 \left(G_{\mu\nu} - G_{\mu\nu}^{(1)} \right), \quad (3.4)$$

where $G_{\mu\nu}$ is the exact Einstein tensor, $G_{\mu\nu}^{(1)}$ is the Einstein tensor formed from the first order terms in the perturbation field and " $\square$ " is the wave operator in flat spacetime. In this form equation (3.4) is still exact and is not an approximation[5, 14]. This cumbersome expression can be simplified by introducing the auxiliary field $\bar{h}_{\mu\nu} = h_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}h^\alpha{}_\alpha$ to express equation (3.4) as[5]

$$\square \bar{h}_{\mu\nu} + \eta_{\mu\nu} \bar{h}^\alpha{}_{\alpha,\beta}{}^\beta - \bar{h}_{\mu\alpha,\nu}{}^\alpha - \bar{h}_{\nu\alpha,\mu}{}^\alpha = -16\pi T_{\mu\nu} + 2G_{\mu\nu}^{(2)}, \quad (3.5)$$

where only the second order terms were kept on the right hand side ($G_{\mu\nu}^{(2)} \approx G_{\mu\nu} - G_{\mu\nu}^{(1)}$).

3.2.2 Gauge invariance

In electromagnetism, the wave equation is typically solved by making a specific gauge choice (for example the Coulomb or the Lorentz gauge)[26]. The gauge choice, even though restrictive, does not affect the physics of the problem and simplifies the solution. In a similar way, the gravitational wave field possesses extra degrees of freedom associated with arbitrary coordinate transformations; adopting specific coordinates is analogous to a choice of gauge. The solution to equation (3.5) can be simplified by adopting the Lorentz gauge. In this gauge, the field is divergence free:

$$\bar{h}^{\mu\nu}{}_{,\nu} = 0. \quad (3.6)$$

These are four conditions on $\bar{h}_{\mu\nu}$ and they correspond to four choices of coordinates x^α . This choice of gauge is equivalent to the Lorentz gauge of electrodynamics where the vector potential is chosen to be divergence free[26].

To see that it is always possible to choose a divergence-free field, perform the infinitesimal coordinate transformation[6]

$$x^\mu \rightarrow x'^\mu = x^\mu + \varepsilon^\mu(x^\alpha). \quad (3.7)$$

The ε^μ are functions of the old coordinates and are at most of the same order of magnitude as the field $\bar{h}^{\mu\nu}$; they are otherwise arbitrary. Under a general coordinates transformation, the metric in the new coordinates ($g'^{\mu\nu}$) is given by

$$g'^{\mu\nu} = \frac{\partial x'^\mu}{\partial x^\rho} \frac{\partial x'^\nu}{\partial x^\sigma} g^{\rho\sigma}. \quad (3.8)$$

Inserting the inverse of equation (3.2), $g^{\mu\nu} \approx \eta^{\mu\nu} - \bar{h}^{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}\bar{h}^\alpha{}_\alpha$, we obtain

$$\bar{h}_{\mu\nu}^{(new)} = \bar{h}_{\mu\nu}^{(old)} - \varepsilon_{\mu,\nu} - \varepsilon_{\nu,\mu} + \eta_{\mu\nu}\varepsilon^\alpha{}_{,\alpha}. \quad (3.9)$$

The transformation is infinitesimal and leaves equation (3.5) invariant, i.e. if $\bar{h}^{old}$ is a solution of equation (3.5), so is $\bar{h}^{new}$. This can be used to fix the gauge.

Performing the previous coordinate transformation and imposing the generators of the transformation (equation 3.9) to satisfy

$$\square \varepsilon^\mu = \bar{h}^{\mu\nu}{}_{,\nu}^{(old)}, \quad (3.10)$$

we obtain a new field that is divergence-free. This was obtained by taking the divergence of equation (3.9) and by imposing $\bar{h}^{(new)\mu\nu}{}_{,\nu} = 0$. If the field is already in the Lorentz gauge, this type of infinitesimal transformation will leave it in the Lorentz gauge if the generators satisfy the harmonic equation $\square \varepsilon^\mu = 0$.

In general 8 constraints need to be imposed to fix the gauge completely. This reduces the degrees of freedom of the gravitational field to two, the right number since it can be shown that $\bar{h}_{\mu\nu}$ has only two degrees of freedom[6]. For the time being, the only gauge choice we make is the Lorentz gauge.

This gauge does not exhaust the gauge freedom associated with the gravitational field (4 constraints out of 8), but it is sufficient for the present purpose. Of course, some restrictions

must be imposed on the final result to keep only the physical components of $\bar{h}_{\mu\nu}$, but this poses no problem at all since the remaining gauge fixing freedom can then be used. The Lorentz gauge is convenient to calculate the solution to equation (3.5), since in this gauge it becomes a simple wave equation[5]:

$$\square \bar{h}^{\mu\nu} = -16\pi S^{\mu\nu} . \quad (3.11)$$

where the tensor $S_{\mu\nu} = T_{\mu\nu} - \frac{1}{8\pi} G_{\mu\nu}^{(2)}$ was introduced. The tensor $S_{\mu\nu}$ represents the total energy and momentum present in the spacetime. The contribution $T_{\mu\nu}$ is the energy-momentum of the matter, while $-\frac{1}{8\pi} G_{\mu\nu}^{(2)}$ is the energy momentum of the gravitational waves, which is quadratic in $\bar{h}_{\mu\nu}$. As it stands equation (3.11) is non-linear in $\bar{h}_{\mu\nu}$; if $\bar{h}_{\mu\nu}$ is sufficiently small, then $G_{\mu\nu}^{(2)}$ can be ignored and equation (3.11) reduces to

$$\square \bar{h}^{\mu\nu} = -16\pi T^{\mu\nu} . \quad (3.12)$$

3.3 Quadrupole Approximation

3.3.1 Solving the wave equation

The wave equation (3.12) obtained in the previous section is very similar to the wave equation of electrodynamics. The main difference is that in the gravitational case, the radiation acts as its own source. This a consequence of the non-linear terms in Einstein's equation. As is the case for electrodynamics, the solution is found through the use of a Green's function. The complete solution involves the advanced and the retarded Green's functions. Because we are concerned with systems *decaying* through the emission of gravitational waves, and because the advanced Green's function corresponds to radiation coming in from infinity, only the retarded Green's function is included in the solution:

$$\begin{aligned} \bar{h}^{\mu\nu}(t, \mathbf{x}) &= 4 \int_{space} \frac{T^{\mu\nu}(t - |\mathbf{x} - \mathbf{x}'|, \mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|} d^3x' \\ &\approx \frac{4}{R} \int_{space} T^{\mu\nu}(t - R, \mathbf{x}') d^3x' , \end{aligned} \quad (3.13)$$

where $u = t - |\mathbf{x} - \mathbf{x}'|$ is the retarded time and $R \equiv |\mathbf{x}| \gg |\mathbf{x}'|$ because the source is assumed to be spatially bounded and the field is evaluated at a point $\mathbf{x}$ far away from it.

Before pursuing, it is worth recalling that not all the components of $\bar{h}_{\mu\nu}$ represent radiative degrees of freedom. The components $\bar{h}^{0\mu}$ are time-independent and are not associated with radiation. The time-time component is associated with the total mass of the spacetime (the Coulomb field) and the time-space components are associated with the total momentum in the spacetime. They are expressed as

$$\begin{aligned}\bar{h}^{00} &= \frac{1}{4} \frac{M}{R}, \\ \bar{h}^{0i} &= \frac{1}{4} \frac{P^i}{R},\end{aligned}\tag{3.14}$$

where the total mass (M) and the total momentum (P^i) of the spacetime are defined as

$$M = \int_{space} d^3x' T^{00},$$

and

$$P^i = \int_{space} d^3x' T^{0i}.$$

These quantities are conserved: $\frac{dM}{dt} = \frac{dP^i}{dt} = 0$. This means that the only radiative components of equation (3.13) are the $\bar{h}^{ij}$. Gravitational waves are thus included in the space-space components of the perturbation field:

$$\bar{h}^{ij}(t, \mathbf{x}) = \frac{1}{R} \int_{space} T^{ij}(t - R, \mathbf{x}') d^3x'.\tag{3.15}$$

This equation describes the emission of gravitational waves by a matter source.

The solution is not in a very useful form yet. To simplify equation (3.15), we note that in the Lorentz gauge the left hand side of equation (3.12) obeys $(\square \bar{h}^{\mu\nu})_{,\nu} = 0$, because ordinary derivatives commute. This in turn implies that the right hand side of equation (3.12) obeys, to leading order, the conservation equation $T^{\mu\nu}_{,\nu} = 0$ [5]. Mathematically, this can be written as

$$T^{00}_{,0} = -T^{0l}_{,l}\tag{3.16}$$

and

$$T^{m0}{}_{,0} = -T^{ml}{}_{,l}. \quad (3.17)$$

These identities can be used to simplify the expression obtained for the gravitational wave field. Consider the following expression:

$$\begin{aligned} (T^{00}x^ix^j)_{,00} &= (T^{00}{}_{,0})_{,0}x^ix^j \\ &= (-T^{0l}{}_{,l})_{,0}x^ix^j && \text{(using eq.(3.16))} \\ &= (T^{lm}{}_{,lm})x^ix^j && \text{(using eq.(3.17))} \\ &= (T^{lm}x^ix^j)_{,lm} + 2T^{ij} - 4(T^{mi}x^j + T^{mj}x^i)_{,m}. \end{aligned} \quad (3.18)$$

Solving for T^{ij} in the previous expression yields:

$$T^{ij} = \frac{1}{2}(T^{00}x^ix^j)_{,00} - \frac{1}{2}(T^{ml}x^ix^j)_{,ml} + 2(T^{mi}x^j + T^{mj}x^i)_{,m}. \quad (3.19)$$

Upon insertion in equation (3.15), all the terms containing pure spatial derivatives can be eliminated by converting the volume integrals to surface integrals. Because the source is bounded, these terms vanish upon integration on a surface at infinity. For example

$$\int_{space} d^3x' (T^{ml}x^ix^j)_{,ml} = \oint_{surface} (T^{ml}x^ix^j)_{,m} dS_l = 0, \quad (3.20)$$

where dS_l is the oriented surface element. The final expression for the gravitational field in the quadrupole approximation is

$$\begin{aligned} \bar{h}^{ij} &= \frac{2}{R} \int_{space} \frac{d^2}{dt^2} (T^{00}x^ix^j) d^3x' \\ &= \frac{2}{R} \frac{d^2}{dt^2} \int_{space} T^{00}x^ix^j d^3x' \\ \bar{h}^{ij} &= \frac{2}{R} \frac{d^2 I^{ij}}{dt^2}, \end{aligned} \quad (3.21)$$

where I_{ij} is the quadrupole moment of the matter source.

The solution can be reduced to its two relevant radiative degrees of freedom by switching to the *transverse-traceless* gauge. In this gauge, the gravitational waves are traceless and

transverse to their direction of propagation:

$$\bar{h}^i_i = 0, \quad (3.22)$$

$$\bar{h}_{ij}n^j = 0, \quad (3.23)$$

where $\hat{n}$ is the vector pointing in the direction of propagation. Its components are $n^i = \frac{x^i}{R}$ in Cartesian coordinates. In this gauge $\bar{h}_{\mu\nu} = h_{\mu\nu}$ since the trace-free condition implies $\bar{h}^i_i = -h^i_i$. Hence, the overbar can be dropped from the equations involving $\bar{h}_{ij}$. The conditions (3.22) and (3.23) can be imposed by considering the infinitesimal coordinate transformation defined in equation (3.9). Since we wish to stay in the Lorentz gauge, the generators of the transformation must satisfy $\square \varepsilon^\mu = 0$. At a distance R far away from the source, the generators are of the form

$$\varepsilon^\mu = \frac{1}{R} A^\mu (t - R). \quad (3.24)$$

Inserting this in equation (3.9) and imposing the two gauge conditions (3.22) and (3.23) allows us to impose constraints on A^μ that reduce the field to its TT form. The result is

$$h_{ij}^{TT} = P_{il} h_{lm} P_{mj} - \frac{1}{2} P_{ij} P_{lm} h_{lm}, \quad (3.25)$$

where $P_{ij} = \mathbb{I}_{ij} - n_i n_j$. This leads to the final expression for the field in the TT gauge[5]:

$$h_{ij}^{TT} = \frac{2}{R} \frac{d^2 \mathfrak{I}_{ij}^{TT}}{dt^2}, \quad (3.26)$$

where the reduced quadrupole moment ($\mathfrak{I}_{ij} = \mathbb{I}_{ij} - \frac{1}{3} \delta_{ij} \mathbb{I}^k_k$) was introduced.

3.3.2 The loss of energy and angular momentum

The quadrupole formula for the gravitational wave field was obtained by keeping the leading order terms in the solution to the wave equation (equation (3.12)), i.e. to leading order the radiation is generated by a change in the source's quadrupole moment. The source's energy and angular momentum are expected to decrease due to the emission of gravitational waves

and consequently, we expect the radiation reaction to be described in terms of the field $(\bar{h}^{\mu\nu})$ introduced in the previous sections.

In electromagnetism, the energy and angular momentum carried away by the fields $\mathbf{E}$ and $\mathbf{H}$ can be calculated in many different ways. All these methods agree when the energy emitted, on average, is considered. The radiation of energy by electromagnetic waves is most conveniently calculated through the Poynting vector[26]. This corresponds to calculating the net flow of energy crossing a closed surface. In the weak field approximation, it is possible to obtain the energy and angular momentum loss by a similar technique. As for the gauge choice, it is found that the energy and angular momentum carried away by the radiation are most conveniently calculated in the Lorentz gauge. The TT gauge is dropped for the moment. This subsection closely follows P.C. Peters[16] in the derivation of the radiation reaction.

The starting point in deriving the energy and angular momentum loss is to note that for an arbitrary stress-energy tensor $S^{\mu\nu}$ that is divergence free (see equation (3.16)) and symmetrical, the existence of global conservation laws is guaranteed[16, 27]. Two of the conserved quantities can be identified as the total energy (E) and the total angular momentum ($\mathbf{L}$). They are defined as

$$E = \int_{space} S^{00} d^3x'. \quad (3.27)$$

$$L_i = \varepsilon_{ijk} \int_{space} S^{0j} x^k d^3x'. \quad (3.28)$$

where ε_{ijk} is the Levi-Civita tensor equal to one if (x, y, z) appear in this order or in an even number of permutations, -1 if the number of permutations is odd, and zero if two indices are the same.

We have already seen that $-\frac{1}{8\pi}G_{\mu\nu}^{(2)}$ represents the stress-energy of the radiation. The energy and angular momentum of gravitational waves can be expressed as

$$E^{GW} = -\frac{1}{8\pi} \int_{space} d^3x G^{(2)00},$$

$$L_i^{GW} = -\frac{1}{8\pi}\varepsilon_{ijk}\int_{space}d^3x G^{(2)0j}x^k. \quad (3.29)$$

By taking a time derivative of both equation and using the conservation law

$G^{(2)\mu\nu}{}_{,\nu} = 0$, the expressions in equation (3.29) can be rewritten as

$$\left(\frac{dE}{dt}\right)^{GW} = \frac{1}{8\pi}\int_{space}d^3x' G^{(2)0j}{}_{,j}, \quad (3.30)$$

$$\left(\frac{dL_i}{dt}\right)^{GW} = \frac{1}{8\pi}\varepsilon_{ijk}\int_{space}d^3x' G^{(2)jm}{}_{,m}x^k. \quad (3.31)$$

The energy and angular momentum carried away by the radiation must be coming from the matter source. Thus, on physical grounds, we can conclude that

$$\begin{aligned} \left(\frac{dE}{dt}\right)^{source} &= -\left(\frac{dE}{dt}\right)^{GW} \\ &= -\frac{1}{8\pi}\oint_{boundary}dS_j \left\langle G^{(2)0j} \right\rangle_t, \end{aligned} \quad (3.32)$$

and

$$\begin{aligned} \left(\frac{dL_i}{dt}\right)^{source} &= -\left(\frac{dL_i}{dt}\right)^{GW} \\ &= -\frac{1}{8\pi}\varepsilon_{ijk}\int_{boundary}dS_m \left\langle G^{(2)jm} \right\rangle_t, \end{aligned} \quad (3.33)$$

where the volume integrals were converted to surface integral, dS_j is the oriented surface element, and we took a time average $\langle \dots \rangle_t$ to ensure that the various methods to calculate the energy and angular momentum loss agree.

To calculate the energy and angular momentum lost to gravitational waves, an explicit expression for $G_{\mu\nu}^{(2)}$ is needed. It can be found by extracting the second-order terms in $\bar{h}^{\mu\nu}$ from the Ricci tensor. Using the method described in subsection 3.2.1 and discarding the terms proportional to a divergence of the field ($\bar{h}^{\mu\nu}{}_{,\nu} = 0$), one gets

$$G_{\mu\nu}^{(2)} = R_{\mu\nu}^{(2)} - \frac{1}{2}\eta_{\mu\nu}\eta^{\alpha\beta}R_{\alpha\beta}^{(2)}, \quad (3.34)$$

where the second order Ricci tensor is given by

$$R_{\alpha\beta}^{(2)} = \frac{1}{4}\left\{\bar{h}^{\mu\nu}{}_{,\beta}\bar{h}_{\mu\nu,\alpha} - 2\bar{h}^{\mu\nu}\bar{h}_{\beta\nu,\alpha\mu} - 2\bar{h}^{\mu\nu}\bar{h}_{\alpha\nu,\beta\mu} - 2\bar{h}^{\mu}{}_{\beta,\nu}\bar{h}^{\nu}{}_{\alpha,\mu}\right\}$$

$$\begin{aligned}
& + 2\bar{h}^{\mu\nu}\bar{h}_{\alpha\beta,\mu\nu} + 2\bar{h}^{\mu\nu}\bar{h}_{\mu\nu,\alpha\beta} + 2\bar{h}^\mu{}_\beta{}_{,\nu}\bar{h}_{\mu\alpha}{}^{,\nu} \Big\} \\
& + \frac{1}{4} \Big\{ \bar{h}^\mu{}_\beta{}_{,\nu}\bar{h}^\nu{}_{\nu,\alpha\mu} - \bar{h}^\mu{}_\mu\bar{h}^\nu{}_{\nu,\alpha\beta} + \bar{h}^\mu{}_\alpha\bar{h}^\nu{}_{\nu,\beta\mu} - \bar{h}^\mu{}_\mu\Box\bar{h}_{\alpha\beta} \\
& - 2\bar{h}^\nu{}_{\nu,\mu}\bar{h}_{\alpha\beta}{}^{,\mu} + \bar{h}^\mu{}_\beta{}_{,\alpha}\bar{h}^\nu{}_{\nu,\mu} + \bar{h}^\mu{}_\alpha{}_{,\beta}\bar{h}^\nu{}_{\nu,\mu} - \frac{1}{2}\bar{h}^\mu{}_\mu{}_{,\alpha}\bar{h}^\nu{}_{\nu,\beta} \Big\} \\
& + \frac{1}{8}\eta_{\alpha\beta} \Big\{ \bar{h}^\mu{}_\mu\Box\bar{h}^\nu{}_\nu - 2\bar{h}^{\mu\nu}\bar{h}^\rho{}_{\rho,\mu\nu} + \bar{h}^\nu{}_{\nu,\mu}\bar{h}^\rho{}_{\rho,\mu} \Big\}. \tag{3.35}
\end{aligned}$$

This expression for the stress-energy tensor contains terms that are not needed for the present calculations; the expression for $G_{\mu\nu}^{(2)}$ can be simplified to $R_{\mu\nu}^{(2)}$. The energy carried away is proportional to $G_{0l}^{(2)}$, but $\eta_{0l}\eta^{\alpha\beta}R_{\alpha\beta}^{(2)} = 0$ since $\eta_{\mu\nu}$ is diagonal. In the case of the angular momentum, the term $\eta^{jm}\eta^{\alpha\beta}R_{\alpha\beta}^{(2)}$ inserted in equation (3.33) results in a term of the form

$$\varepsilon_{ijk} \int_{\text{boundary}} \eta^{jm} \left\langle \eta^{\alpha\beta} R_{\alpha\beta}^{(2)} \right\rangle_t x^k dS_m = \varepsilon_{ijk} \int_{\text{boundary}} \left\langle \eta^{\alpha\beta} R_{\alpha\beta}^{(2)} \right\rangle_t x^k dS^j \tag{3.36}$$

The oriented surface element dS^j is equal to $\frac{x^j}{R}dS$, where dS is the regular surface element. Upon insertion in equation (3.36), this yields a term symmetric in j and k ($x^j x^k$), but because the Levi-Civita tensor is antisymmetric in these two indices, this term vanishes. Thus for the present calculations $R_{\mu\nu}^{(2)}$ is the only term needed in the stress-energy tensor of the radiation.

The same argument can be used to get rid of the term

$$\frac{1}{8}\eta_{\alpha\beta} \Big\{ \bar{h}^\mu{}_\mu\Box\bar{h}^\nu{}_\nu - 2\bar{h}^{\mu\nu}\bar{h}^\rho{}_{\rho,\mu\nu} + \bar{h}^\nu{}_{\nu,\mu}\bar{h}^\rho{}_{\rho,\mu} \Big\} \tag{3.37}$$

in the expression for $R_{\alpha\beta}^{(2)}$ (equation (3.35)). We include all the remaining terms in the restricted stress-energy tensor ($\tilde{G}_{\mu\nu}^{(2)}$):

$$\begin{aligned}
\tilde{G}_{\mu\nu}^{(2)} = & \frac{1}{4} \Big\{ \bar{h}^{\mu\nu}{}_{,\beta}\bar{h}_{\mu\nu,\alpha} - 2\bar{h}^{\mu\nu}\bar{h}_{\beta\nu,\alpha\mu} - 2\bar{h}^{\mu\nu}\bar{h}_{\alpha\nu,\beta\mu} - 2\bar{h}^\mu{}_\beta{}_{,\nu}\bar{h}^\nu{}_{\alpha,\mu} \\
& + 2\bar{h}^{\mu\nu}\bar{h}_{\alpha\beta,\mu\nu} + 2\bar{h}^{\mu\nu}\bar{h}_{\mu\nu,\alpha\beta} + 2\bar{h}^\mu{}_\beta{}_{,\nu}\bar{h}_{\mu\alpha}{}^{,\nu} \Big\}
\end{aligned}$$

$$\begin{aligned}
& + \frac{1}{4} \left\{ \bar{h}^\mu{}_{,\beta} \bar{h}^\nu{}_{,\alpha\mu} - \bar{h}^\mu{}_{,\mu} \bar{h}^\nu{}_{,\alpha\beta} + \bar{h}^\mu{}_{,\alpha} \bar{h}^\nu{}_{,\beta\mu} - \bar{h}^\mu{}_{,\mu} \square \bar{h}_{\alpha\beta} \right. \\
& \left. - 2 \bar{h}^\nu{}_{,\mu} \bar{h}_{\alpha\beta}{}^{,\mu} + \bar{h}^\mu{}_{,\beta,\alpha} \bar{h}^\nu{}_{,\nu,\mu} + \bar{h}^\mu{}_{,\alpha,\beta} \bar{h}^\nu{}_{,\nu,\mu} - \frac{1}{2} \bar{h}^\mu{}_{,\mu,\alpha} \bar{h}^\nu{}_{,\nu,\beta} \right\}. \quad (3.38)
\end{aligned}$$

Since we are averaging over time and integrating over the bounding surface, some identities can be used to greatly simplify the final expressions:

1. Any term in the integrand of the form $\bar{h}\square\bar{h}$ (indices were dropped for notational simplicity) is $\mathcal{O}(h^3)$ and can be discarded since they represent third order effect. To see this, recall that on the boundary at infinity the total stress energy is of the form $S_{\mu\nu} = -\frac{1}{8\pi}\tilde{G}_{\mu\nu}^{(2)} \propto \mathcal{O}(\bar{h}^2)$. The matter contribution does not appear since the source is spatially bounded. Using equation (3.12) yields

$$\bar{h}_{\mu\nu}\square\bar{h}^{\mu\nu} \propto \bar{h}^{\mu\nu}S_{\mu\nu} \propto \mathcal{O}(\bar{h}^3). \quad (3.39)$$

2. Spatial derivatives can be expressed, to leading order, in terms of time derivatives in the following way (from equation (3.15)):

$$\begin{aligned}
\bar{h}_{\mu\nu, i} &= -\frac{x_i}{R} \bar{h}_{\mu\nu, 0} - \frac{x_i}{R^2} \bar{h}_{\mu\nu} \\
&= -\frac{x_i}{R} \bar{h}_{\mu\nu, 0} + \mathcal{O}(R^{-2}), \quad (3.40)
\end{aligned}$$

where the $\mathcal{O}(R^{-2})$ term can be neglected, since upon multiplication by the field ($\bar{h}$ is $\mathcal{O}(R^{-1})$ from equation (3.15)) it becomes a $\mathcal{O}(R^{-3})$ term. All spatial derivatives are therefore proportional to $n^i = \frac{x_i}{R}$.

3. If equation (3.40) is combined with the gauge condition $(\bar{h}^{\mu\nu}{}_{,\nu})$, any derivative in the average can be integrated by parts with respect to time. For example $\langle \bar{h}_{\mu\nu, \alpha\beta} \bar{h}^{\alpha\beta} \rangle_t = -\langle \bar{h}_{\mu\nu, \alpha} \bar{h}^{\alpha\beta}{}_{,\beta} \rangle_t = \langle \bar{h}_{\mu\nu} \bar{h}^{\alpha\beta}{}_{,\alpha\beta} \rangle_t \approx 0$ is true because the field satisfies the Lorentz gauge condition. The boundary terms vanish because we are considering the average over one period of the motion of a periodic source.

By taking $\tilde{G}_{0j}^{(2)}$, inserting it back in equation (3.32) and keeping only the leading order terms, we obtain the energy loss of the matter source,

$$\frac{dE}{dt} = -\frac{1}{32\pi} \oint \left\langle \bar{h}_{\mu\nu,0} \bar{h}^{\mu\nu}_{,0} - \frac{1}{2} \bar{h}^{\mu}_{\mu,0} \bar{h}^{\nu}_{\nu,0} \right\rangle_t n^i dS_i. \quad (3.41)$$

The analogous result for the angular momentum is obtained by inserting $\tilde{G}_{jm}^{(2)}$ in equation (3.33):

$$\frac{dL_i}{dt} = -\frac{1}{32\pi} \varepsilon_{ijk} \oint dS_l x^j \left\langle \bar{h}_{\alpha\beta, .k} \bar{h}^{\alpha\beta}_{, .l} - 2\bar{h}^{\alpha\beta}_{, .k} \bar{h}_{m\alpha, .\beta} - 2\bar{h}^{\alpha\beta}_{, .m} \bar{h}^k_{\alpha, .\beta} \right\rangle_t. \quad (3.42)$$

The previous expressions (equations (3.41) and (3.42)) can be simplified quite easily if we take the surface of integration to be a large sphere enclosing the system and use the Lorentz gauge condition to express them in terms of the spatial components of $\bar{h}^{\mu\nu}$. The energy loss can be expressed in terms of the quadrupole moments of the source as (equation (3.26))[16]

$$\begin{aligned} \frac{dE}{dt} = & -\frac{1}{8\pi R^2} \oint dS \left[\frac{1}{2} (n^i n^j \partial_t^3 I_{ij})^2 - 2n_j n^k \partial_t^3 I_{ik} \partial_t^3 I^{ij} \right. \\ & \left. + \partial_t^3 I_{ij} \partial_t^3 I^{ij} + n^i n^j \partial_t^3 I_{ij} \partial_t^3 I^k_k - \frac{1}{2} \left(\partial_t^3 I^k_k \right)^2 \right], \end{aligned} \quad (3.43)$$

where dS is the surface element of a sphere of radius R , “ ∂_t^i ” is the i^{th} time derivative and the factor of $4R^{-2}$ comes from inserting equation (3.21) in equation (3.41) whenever a spatial component of the field $\bar{h}_{ij}$ appeared. The only angular dependence in the previous equation is through the vector $\mathbf{n}$:

$$\hat{\mathbf{n}} = \sin \theta (\cos \varphi \hat{\mathbf{x}} + \sin \varphi \hat{\mathbf{y}}) + \cos \theta \hat{\mathbf{z}}. \quad (3.44)$$

Taking the differential element on the sphere to be $dS = R^2 \sin \theta d\theta d\varphi$, the integration is easily accomplished, leaving only the time average to be calculated. In order to do this, information about the source is needed. This will be done in chapter 4 for coalescing compact binaries.

Expressing the result for the energy in terms of the reduced quadrupole moment $\mathfrak{I}_{ij} = I_{ij} - \frac{1}{3} \delta_{ij} I^k_k$ yields[16]:

$$\frac{dE^{GW}}{dt} = -\frac{dE}{dt} = \frac{1}{5} \langle \partial_t^3 \mathfrak{I}_{ij} \partial_t^3 \mathfrak{I}^{ij} \rangle_t. \quad (3.45)$$

To simplify the expression for the angular momentum loss requires some more thinking. Because the integrand is proportional to x^j we have to be careful when dismissing terms proportional to $\bar{h}^3$. They could be of order $\bar{h}^2$ upon multiplication by x^j . For example, $x^j \bar{h}^{\mu\nu} \bar{h}_{\mu\nu,k} = -n_k \bar{h}^{\mu\nu} [x^j \bar{h}_{\mu\nu,0} + n^j \bar{h}_{\mu\nu}]$ (from the first line of equation (3.40)) contains two terms of order $\mathcal{O}(\bar{h}^2)$ that cannot be discarded. Once all the relevant terms are included, the ones proportional to $x^j x^k$ can be dropped since their product with the Levi-Civita tensor results in null contributions. The gauge condition can be used once more to express the angular momentum loss in terms of the quadrupole moment of the system. The end result is an expression where all the angular dependence is in the vector $\hat{\mathbf{n}}$. Upon angular integration (using equation (3.44)), the angular momentum loss becomes[16]

$$\frac{dL_i^{GW}}{dt} = -\frac{dL_i}{dt} = \frac{2}{5} \varepsilon_{ijk} \langle \partial_t^2 \mathfrak{I}_{jl} \partial_t^3 \mathfrak{I}_{lk} \rangle_t . \quad (3.46)$$

Equations (3.45) and (3.46) are sufficient to describe the emission of gravitational waves and their effect on the source. They will be used in chapter 4 to calculate the emission of energy and angular momentum by compact binary systems.

Chapter 4

Inspiring Compact Binaries: The Last Minutes

4.1 Introduction

In this section, we calculate the gravitational waves emitted by a binary system composed of two bodies of mass m_1 (position vector $\mathbf{r}_1$) and m_2 (position vector $\mathbf{r}_2$). The origin of our coordinate system is located at the center of mass, so that

$$m_1 \mathbf{r}_1 + m_2 \mathbf{r}_2 = 0 \quad (4.1)$$

is satisfied[27]. In this coordinate system, the positions of the two bodies can be expressed in terms of the relative separation vector, $\mathbf{r} = \mathbf{r}_1 - \mathbf{r}_2$, as[28]

$$\begin{aligned} \mathbf{r}_1 &= \frac{m_2}{M} \mathbf{r}, \\ \mathbf{r}_2 &= -\frac{m_1}{M} \mathbf{r}, \end{aligned} \quad (4.2)$$

where $M = (m_1 + m_2)$ is the total mass of the binary.

Section 4.2 is dedicated to solving the two-body problem in classical mechanics. Subsection 4.3.1 contains a calculation of the transverse-traceless (TT) quadrupole moment

tensor ($\bar{\mathbf{I}}^{TT}$) and of its second and third time derivatives. The energy and angular momentum carried away by the radiation are most conveniently calculated in the center of mass coordinates (see equations (3.45) and (3.46)); we present a calculation of the second and third time derivatives in this coordinate system (figure (4.1)). On the other hand, the waveforms are needed in the TT gauge and are most conveniently calculated by taking a transverse-traceless projection of the reduced quadrupole moment *before* calculating the time derivatives. In section 4.4, the reaction of the system to the emission of gravitational waves is calculated using the results of sections 4.2 and 4.3. This determines the ordinary differential equations (ODE) that govern the evolution of compact binary systems. An analytical result to these equations will be presented for binaries in a circular orbit. A numerical method of integration for elliptic orbits will be described. Subsection 4.4.3 contains the waveforms as a function of time and a discussion of the differences between the radiation from circular and eccentric binary systems.

In this chapter, summation over repeated indices is assumed. Latin indices run from 1 to 3 with the identification $\{x, y, z\} \Rightarrow \{x^1, x^2, x^3\}$. Greek indices run from 0 to 3 with $t \Rightarrow x^0$. Unit vectors are denoted by bold characters with a hat (say $\hat{\mathbf{x}}$) and second rank tensor by bold characters with an over-bar (say $\bar{\mathbf{I}}$). Throughout the chapter, geometrized units are used ($G = c = 1$).

4.2 The two-body problem in Classical Mechanics

4.2.1 The equations of motion

In classical mechanics, the two-body problem is governed by the Lagrangian[28]

$$L = \frac{1}{2}m_1\dot{\mathbf{r}}_1^2 + \frac{1}{2}m_2\dot{\mathbf{r}}_2^2 + \frac{m_1m_2}{|\mathbf{r}_1 - \mathbf{r}_2|}, \quad (4.3)$$

where a " $\dot{}$ " denotes differentiation with respect to time. By taking the center of mass to be the origin of our coordinate system, the problem can be transformed into that of a body

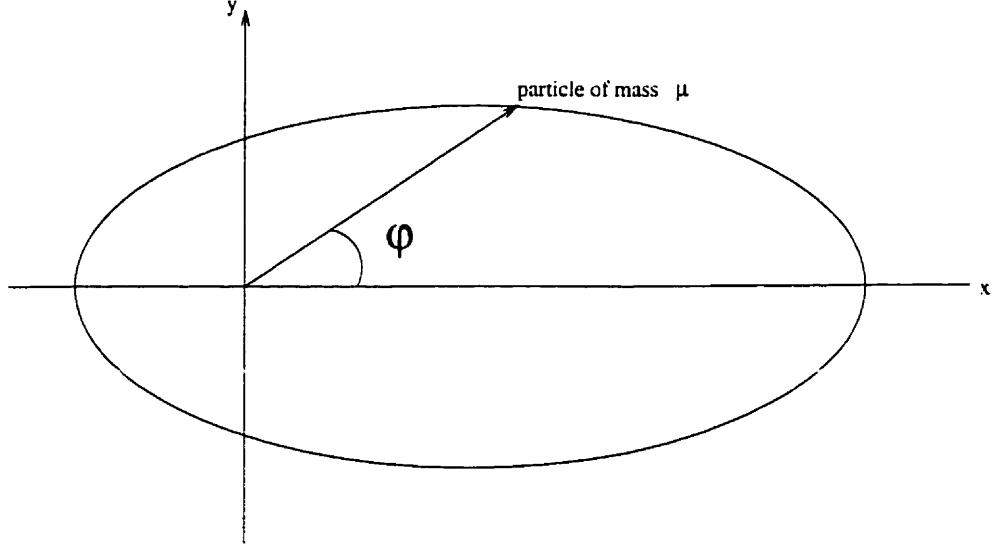


Figure 4.1: In the center of mass, the two-body problem is described by a particle of mass μ orbiting a compact object of mass M . The coordinate r is the distance between the two objects and φ is the position of the mass μ on the orbit. The z axis is pointing out of the page.

of mass $\mu = \frac{m_1 m_2}{(m_1 + m_2)}$ orbiting a compact object of mass $M = m_1 + m_2$ (see figure (4.1)).

In terms of the coordinates introduced previously (equation (4.2)) the Lagrangian is

$$L = \frac{1}{2} \mu \dot{\mathbf{r}}^2 + \frac{\mu M}{r}. \quad (4.4)$$

Since the motion takes place in a plane, θ can be set to $\frac{\pi}{2}$ without loss of generality. This choice leaves two independent variables (r and φ) and two constants of the motion provide enough first order differential equations to describe the system completely (one for r and one for φ). To obtain the two constants of the motion, we first note that the Lagrangian is of the form $L = L(\dot{\varphi}, \dot{r}, r)$, i.e. φ is a cyclic variable[27]. This provides one of the constants needed: the angular momentum. The second one is obtained by noting that we are dealing with a Hamiltonian system, which implies $\dot{E} = \dot{H} = 0 \Rightarrow E = \text{const}$ [27]. The second constant is therefore provided by energy conservation. These are the two constants required. In terms of these, the equations describing the motion are

$$\dot{\varphi} = \frac{\tilde{L}}{r^2}, \quad (4.5)$$

$$\dot{r} = \pm \sqrt{2\tilde{E} + \frac{2M}{r} - \frac{\tilde{L}^2}{r^2}}, \quad (4.6)$$

where $\tilde{L} = \frac{L}{\mu}$ and $\tilde{E} = \frac{E}{\mu}$ are the angular momentum and the energy per unit mass, respectively. These two equations can be solved immediately and exactly, since the orbit is considered to be Newtonian (no gravitational radiation emitted). To get the solutions to equations (4.5) and (4.6) in the simplest possible form, we introduce the dimensionless parameters e and p :

$$\begin{aligned} e &= \sqrt{1 + 2\tilde{E} \left(\frac{\tilde{L}}{M}\right)^2}, \\ p &= \left(\frac{\tilde{L}}{M}\right)^2. \end{aligned} \quad (4.7)$$

The ODE's are then written as

$$\dot{r} = \frac{e}{p^{1/2}} \sin \varphi. \quad (4.8)$$

$$\dot{\varphi} = \frac{(1 + e \cos \varphi)^2}{Mp^{3/2}}. \quad (4.9)$$

Note that $e = 0$ implies that $\dot{r} = 0$ and the orbit is circular. In the general case, we are dealing with a set of two coupled differential equations. They can be solved by dividing equation (4.8) by equation (4.9), which results in a differential equation for r that depends solely on φ . Solving this equation provides one with $r = r(\varphi)$:

$$r = \frac{Mp}{(1 + e \cos \varphi)}. \quad (4.10)$$

This result can then be inserted back in equation (4.9) to obtain an implicit relation for φ of the form $t = t(\varphi)$. The integration for φ is long and tedious, but because it presents no essential difficulties, only the final result is stated:

$$t = M \left(\frac{p}{1 - e^2}\right)^{3/2} \left[\arccos \left(\frac{e + \cos \varphi}{1 + e \cos \varphi}\right) - \frac{e\sqrt{1 - e^2} \sin \varphi}{1 + e \cos \varphi} \right]. \quad (4.11)$$

The period of the binary is then simply given by

$$T \equiv t(2\pi) = 2\pi M \left(\frac{p}{1 - e^2}\right)^{3/2}. \quad (4.12)$$

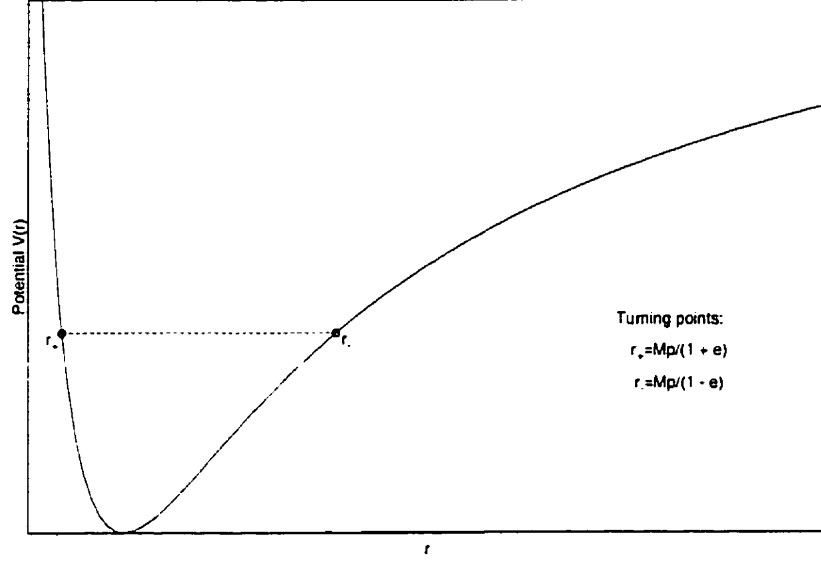


Figure 4.2: The central potential $V(r)$ of Newtonian mechanics. The turning points $r_{\pm}$ are the roots of equation (4.13). They define the eccentricity e and the size of the ellipse p .

The solution for r (equation (4.10)) holds even when the emission of radiation is included in the calculation, because even though e and p are then functions of time, they remain independent of φ . Equations (4.10) and (4.11) are the complete solution to the two-body problem in classical mechanics.

4.2.2 The turning points

A geometrical meaning can be assigned to the parameters defined in equation (4.7) by looking at the turning points of the orbit. Those are the points where the radial velocity vanishes: $\dot{r} = 0$. Inserting this in equation (4.6) yields

$$2\tilde{E} + 2\left(\frac{M}{r}\right) - \left(\frac{\tilde{L}}{M}\right)^2 \left(\frac{M}{r}\right)^2 = 0. \quad (4.13)$$

Equation (4.13) is quadratic in $\frac{M}{r}$ and the two roots are the turning points of the orbit.

Rewriting the solution in terms of e and p by using equation (4.7) reveals that

$$r_{\pm} = M \frac{p}{1 \pm e}. \quad (4.14)$$

These two equations can be seen as defining the two parameters e and p in terms of the turning points shown on figure (4.2). This equation assigns a direct physical meaning to the parameters e and p . As can be seen from figure (4.2), when $e = 0$ both roots are equal, the orbit is circular, and p is the radius of the orbit per unit mass. Thus e is a measure of the deviation from a circular orbit, i.e. of the ellipticity. In this case, p is related to the semi-major axis a of the ellipse by[28]

$$a = \frac{1}{2}(r_+ + r_-) = M \frac{p}{1 - e^2}. \quad (4.15)$$

4.3 The quadrupole approximation

4.3.1 The reduced quadrupole moment and its derivatives

We are now in a position to calculate the quadrupole moment and its time derivatives for an arbitrary binary system. They are needed to calculate the waveforms in the approximation developed in section 3.3 (see equation 3.26). The quadrupole moment is defined as[27]

$$\bar{\mathbf{I}} = \int T_{00} \mathbf{r}' \mathbf{r}' d^3x'. \quad (4.16)$$

where T_{00} is the time-time component of the stress-energy tensor (this corresponds to the energy density of the system) and $\mathbf{r}'$ is a spatial vector. Terms of the form $\mathbf{vw}$ are second rank tensors and their operation on a vector $\mathbf{u}$ is defined as $(\mathbf{vw}) \cdot \mathbf{u} = (\mathbf{w} \cdot \mathbf{u})\mathbf{v}$. For definiteness we place the binary in the xoy plane of the center of mass coordinates. This choice of origin does not restrict the generality of our calculations. If we choose another origin for our coordinate system, the difference between the quadrupole moments calculated in both coordinate systems is a constant. We are interested in time derivatives of the quadrupole moment and the constant will vanish upon derivation. This constant term does not play an essential role.

The energy density of a binary system is made of the contributions from the two masses. Since the two companions are considered to be point masses, all their energy is concentrated

at their position. In the quadrupole approximation, it is sufficient to write the time-time component of the stress-energy tensor as

$$T_{00}(\mathbf{r}) = \sum_{A=1}^2 m_A \delta(\mathbf{r} - \mathbf{r}_A), \quad (4.17)$$

where $\mathbf{r}_A$ is the position of the A^{th} mass with respect to the center of mass. This states that all the energy comes from the rest mass of the two bodies. This is obviously false for highly relativistic systems, but for consistency, the addition of higher order terms would require the waveforms to be calculated using post-Newtonian schemes[3]. Inserting equation (4.17) into equation (4.16) reduces it to

$$\bar{\mathbf{I}} = \sum_{A=1}^2 m_A \mathbf{r}_A \mathbf{r}_A. \quad (4.18)$$

In order to use the equations obtained in subsection 4.2.1, the quadrupole moment must be rewritten in terms of the coordinates defined by equation (4.2). This is easily done. Direct substitution yields

$$\bar{\mathbf{I}} = m_1 \frac{m_2^2}{M^2} \mathbf{r} \mathbf{r} + m_2 \frac{m_1^2}{M^2} \mathbf{r} \mathbf{r} \equiv \mu \mathbf{r} \mathbf{r}. \quad (4.19)$$

Just like electromagnetic waves, gravitational waves have only two independent degrees of freedom[5]. The two independent polarizations are chosen to lay in the plane transverse to the unit vector $\hat{\mathbf{n}}$ pointing from the center of mass to the observer. This vector can be described by the two angles φ_o and θ_o , where the former is the azimuthal angle between the $\hat{\mathbf{x}}$ axis and the projection of $\hat{\mathbf{n}}$ in the xoy plane and the later is the angle between $\hat{\mathbf{z}}$ and $\hat{\mathbf{n}}$ (see figure (4.3)):

$$\hat{\mathbf{n}} = \sin \theta_o (\cos \varphi_o \hat{\mathbf{x}} + \sin \varphi_o \hat{\mathbf{y}}) + \cos \theta_o \hat{\mathbf{z}}. \quad (4.20)$$

This corresponds to two successive rotations. The first one rotates the coordinates around the z axis by an angle φ_o while the second one rotates the new coordinates around the new y axis by an angle θ_o . The plane perpendicular to $\hat{\mathbf{n}}$ is spanned by the two unit vectors $\hat{\boldsymbol{\alpha}}$

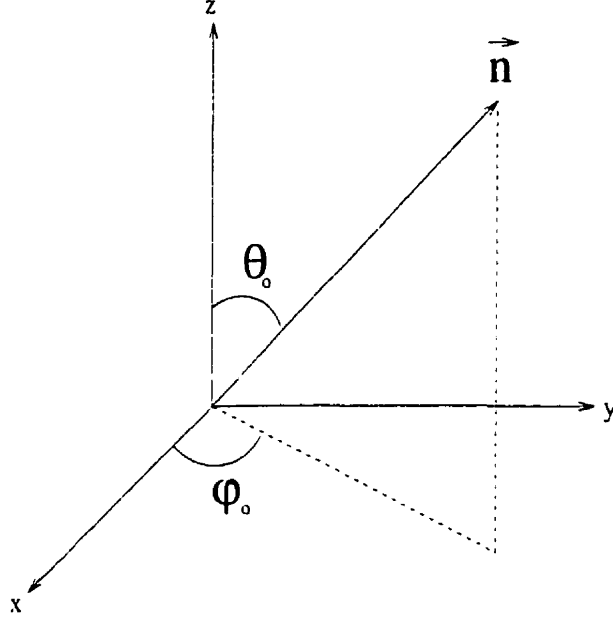


Figure 4.3: The binary is located in the xy plane. The transverse-traceless part of the gravitational waves tensor is the projection of the field obtained in subsection 4.3.1 into the plane perpendicular to $\hat{\mathbf{n}}$ (see equation (3.25)). The vector $\hat{\mathbf{n}}$ points from the binary to the observer.

and $\hat{\beta}$. The three vectors $\{\hat{\mathbf{n}}, \hat{\alpha}$ and $\hat{\beta}\}$ form an orthonormal basis. They are related to the previous basis by the transformation[29]

$$\begin{aligned}\hat{x} &= -\sin\varphi_o\hat{\alpha} - \cos\varphi_o\cos\theta_o\hat{\beta} + \cos\varphi_o\sin\theta_o\hat{\mathbf{n}}, \\ \hat{y} &= \cos\varphi_o\hat{\alpha} - \sin\varphi_o\cos\theta_o\hat{\beta} + \sin\varphi_o\sin\theta_o\hat{\mathbf{n}}, \\ \hat{z} &= \sin\theta_o\hat{\beta} + \cos\theta_o\hat{\mathbf{n}}.\end{aligned}\tag{4.21}$$

The transverse part of the quadrupole moment is obtained by projecting the tensor displayed in equation (4.19) onto the $\alpha\beta$ plane. The projection operator $\mathbf{P}$ is defined by

$$\mathbf{P} = \mathbb{I} - \hat{\mathbf{n}}\hat{\mathbf{n}} = \hat{\alpha}\hat{\alpha} + \hat{\beta}\hat{\beta},\tag{4.22}$$

where $\mathbb{I}$ is the identity matrix. It is easy to see that $\mathbf{P}$ satisfies $\hat{\mathbf{n}}\mathbf{P} = \mathbf{P}\hat{\mathbf{n}} = 0$.

The transverse projection of the quadrupole moment $\bar{\mathbf{I}}$ is given by $\mathbf{P}\bar{\mathbf{I}}\mathbf{P}$. The result is

a second rank tensor perpendicular to $\hat{\mathbf{n}}$. Its trace can be subtracted to get the transverse-traceless quadrupole moment. Because $\mathbf{P}$ is a projector, it obeys the identity $\mathbf{P}^2 = \mathbf{P}$. Accordingly, the trace can be simplified to $\text{Tr}(\mathbf{P}\bar{\mathbf{I}}\mathbf{P}) = \text{Tr}(\mathbf{P}^2\bar{\mathbf{I}}) = \text{Tr}(\mathbf{P}\bar{\mathbf{I}})$. Putting everything together, the transverse-traceless quadrupole moment tensor is given by

$$\begin{aligned}\bar{\mathbf{I}}^{TT} &= \mathbf{P}\bar{\mathbf{I}}\mathbf{P} - \frac{1}{2}\mathbf{P}\text{Tr}(\mathbf{P}\bar{\mathbf{I}}) \\ &= \mu(\mathbf{P}\mathbf{r})(\mathbf{P}\mathbf{r}) - \frac{\mu}{2}\mathbf{P}\{(\mathbf{P}\mathbf{r}) \cdot (\mathbf{P}\mathbf{r})\},\end{aligned}\quad (4.23)$$

where the superscript TT means *transverse-traceless*. The vector $\mathbf{r} = r \cos \varphi \hat{\mathbf{x}} + r \sin \varphi \hat{\mathbf{y}}$ has components along the three vectors of the rotated frame. Upon projection, only the α and β components survive:

$$\mathbf{P}\mathbf{r} = r \left[\sin(\varphi - \varphi_o) \hat{\alpha} - \cos(\varphi - \varphi_o) \cos \theta_o \hat{\beta} \right]. \quad (4.24)$$

Inserting equation (4.22) along with equation (4.24) in the definition of $\bar{\mathbf{I}}^{TT}$ (equation (4.23)), we obtain, after some simplification,

$$\begin{aligned}\bar{\mathbf{I}}^{TT} &= \frac{1}{4}\mu r^2 \left[\sin^2 \theta_o - (1 + \cos^2 \theta_o) \cos 2(\varphi - \varphi_o) \right] (\hat{\alpha} \hat{\alpha} - \hat{\beta} \hat{\beta}) \\ &\quad - \frac{1}{2}\mu r^2 \cos \theta_o \sin 2(\varphi - \varphi_o) (\hat{\alpha} \hat{\beta} + \hat{\beta} \hat{\alpha}) \\ &= \bar{\mathbf{f}}_+^{TT} (\hat{\alpha} \hat{\alpha} - \hat{\beta} \hat{\beta}) + \bar{\mathbf{f}}_\times^{TT} (\hat{\alpha} \hat{\beta} + \hat{\beta} \hat{\alpha}),\end{aligned}\quad (4.25)$$

where $\bar{\mathbf{f}}_+^{TT}$ and $\bar{\mathbf{f}}_\times^{TT}$ are the two polarization modes of the tensor. In this sense, $\bar{\mathbf{e}}_+ \equiv \hat{\alpha} \hat{\alpha} - \hat{\beta} \hat{\beta}$ and $\bar{\mathbf{e}}_\times \equiv \hat{\alpha} \hat{\beta} + \hat{\beta} \hat{\alpha}$ are the two polarization tensors[29].

The next order of business is to calculate the derivatives of the reduced quadrupole moment. At this point the system is still considered to obey Newton's law and the effect of the radiation on the orbit is not taken into account. We assume that the companions travel on the path dictated by classical mechanics, and that e and p are constants. The only dependent variables in equation (4.25) are r and φ . We use the chain rule to take derivatives; we calculate the first time derivative using $\dot{\bar{\mathbf{I}}}^{TT}$, and the second time derivative using $\ddot{\bar{\mathbf{I}}}^{TT}$ instead of $\bar{\mathbf{I}}^{TT}$. The chain rule gives the first derivative of $\bar{\mathbf{I}}^{TT}$ as

$$\frac{d\bar{\mathbf{I}}^{TT}}{dt} = \frac{dr}{dt} \frac{\partial \bar{\mathbf{I}}^{TT}}{\partial r} + \frac{d\varphi}{dt} \frac{\partial \bar{\mathbf{I}}^{TT}}{\partial \varphi}. \quad (4.26)$$

The time derivatives of r and φ are obtained from equations (4.8) and (4.9). Upon substitution, one obtains an expression for $\dot{\bar{\mathbf{I}}}^{TT}$ which again depends on time through r and φ . The scheme described above can be used again to obtain the second time derivative. The process is long and requires many verifications, but is quite straightforward. The results are:

$$\begin{aligned} \partial_t^2 \bar{\mathbf{I}}_+^{TT} = & \frac{\mu}{2p} \left\{ 2(1 + \cos^2 \theta_o) \cos 2(\varphi - \varphi_o) \right. \\ & + e \left[(1 + \cos^2 \theta_o) \left(\frac{5}{2} \cos(\varphi - 2\varphi_o) + \frac{1}{2} \cos(3\varphi - 2\varphi_o) \right) + \sin^2 \theta_o \cos \varphi \right] \\ & \left. + e^2 (1 + \cos^2 \theta_o) \cos 2\varphi_o + \sin^2 \theta_o \right\}, \end{aligned} \quad (4.27)$$

$$\begin{aligned} \partial_t^2 \bar{\mathbf{I}}_-^{TT} = & -\frac{\mu}{2p} \cos \theta_o \left\{ 4 \sin 2(\varphi - 2\varphi_o) \right. \\ & \left. + e [5 \sin(\varphi - 2\varphi_o) + \sin(3\varphi - 2\varphi_o)] - 2e^2 \sin 2\varphi_o \right\}, \end{aligned} \quad (4.28)$$

where ∂_t^i denotes the i^{th} time derivative. This result was obtained by H. Wahlquist in 1987[29]. The terms containing $\sin \theta_o$ must be multiplied by a minus sign in order for the results to agree, because of a difference of $\frac{\pi}{2}$ in the definition of the angles.

The second and the third time derivatives of the reduced quadrupole moment (see equations (3.45) and (3.46)) are needed to calculate the energy and the angular momentum carried away by the radiation. The calculations of these quantities are simplified if one adopts the center of mass coordinates. This is done by starting with equation (4.19), subtracting off its trace and taking the time derivatives in the fashion described above. A short calculation reveals that the trace is just $I^k_k = \mu \mathbf{r} \cdot \mathbf{r}$. The reduced quadrupole is then given by

$$\begin{aligned} \bar{\mathbf{I}} &= \bar{\mathbf{I}} - \frac{1}{3} \mathbb{I} I^k_k \\ &= \mu \left(\mathbf{r} \mathbf{r} - \frac{1}{3} \mathbb{I} \mathbf{r} \cdot \mathbf{r} \right). \end{aligned} \quad (4.29)$$

The procedure to calculate the time derivatives is the same as before. The final results for the second and third derivatives are:

- The second time derivatives:

$$\begin{aligned}
\partial_t^2 \mathbf{I}_{xx} &= \frac{2\mu}{p} \left[-\cos 2\varphi + e \left(\frac{1-3}{6} \cos \varphi - \cos^3 \varphi \right) + e^2 \frac{1-3}{6} \right], \\
\partial_t^2 \mathbf{I}_{xy} &= \frac{2\mu}{p} \left[-\sin 2\varphi + e (-2 \sin \varphi + \sin^3 \varphi) \right], \\
\partial_t^2 \mathbf{I}_{yy} &= \frac{2\mu}{p} \left[\cos 2\varphi + e \left(\frac{1+3}{6} \cos \varphi + \cos^3 \varphi \right) + e^2 \frac{1+3}{6} \right], \\
\partial_t^2 \mathbf{I}_{zz} &= -\frac{2\mu}{3p} e (\cos \varphi + e).
\end{aligned} \tag{4.30}$$

- The third time derivatives:

$$\begin{aligned}
\partial_t^3 \mathbf{I}_{xx} &= 2 \frac{\mu}{M} \frac{(1 + e \cos \varphi)^2}{p^{5/2}} \left[2 \sin 2\varphi + e \sin \varphi \left(-\frac{1-3}{6} + 3 \cos^2 \varphi \right) \right], \\
\partial_t^3 \mathbf{I}_{xy} &= 2 \frac{\mu}{M} \frac{(1 + e \cos \varphi)^2}{p^{5/2}} \left[-2 \cos 2\varphi + e \cos \varphi (-2 + 3 \sin^2 \varphi) \right], \\
\partial_t^3 \mathbf{I}_{yy} &= -2 \frac{\mu}{M} \frac{(1 + e \cos \varphi)^2}{p^{5/2}} \left[2 \sin 2\varphi + e \sin \varphi \left(\frac{1+3}{6} + 3 \cos^2 \varphi \right) \right], \\
\partial_t^3 \mathbf{I}_{zz} &= \frac{2}{3} \frac{\mu}{M} \frac{(1 + e \cos \varphi)^2}{p^{5/2}} e \sin \varphi.
\end{aligned} \tag{4.31}$$

All the other components are null.

4.3.2 The waveforms

The gravitational waveforms in the quadrupole approximation are derived in section 3.3.

Inserting equations (4.28) and (4.28) in equation (3.26), we get the two polarizations of the gravitational-wave tensor:

$$\begin{aligned}
h_+ &= \frac{1}{R} \frac{\mu}{p} \left\{ 2(1 + \cos^2 \theta_o) \cos 2(\varphi - \varphi_o) \right. \\
&\quad + e \left[(1 + \cos^2 \theta_o) \left(\frac{5}{2} \cos(\varphi - 2\varphi_o) + \frac{1}{2} \cos(3\varphi - 2\varphi_o) \right) + \sin^2 \theta_o \cos \varphi \right] \\
&\quad \left. + e^2 (1 + \cos^2 \theta_o) \cos 2\varphi_o + \sin^2 \theta_o \right\},
\end{aligned} \tag{4.32}$$

$$\begin{aligned}
h_\times &= -\frac{1}{R} \frac{\mu}{p} \cos \theta_o \left\{ 4 \sin 2(\varphi - 2\varphi_o) \right. \\
&\quad \left. + e [5 \sin(\varphi - 2\varphi_o) + \sin(3\varphi - 2\varphi_o)] - 2e^2 \sin 2\varphi_o \right\},
\end{aligned} \tag{4.33}$$

where R is the distance separating the binary from the observer.

Even though the time dependence of the waveforms is unknown at this point (p , e and φ are functions of time and will be calculated in section 4.4), two observations can be made. First note that in the case $e = 0$ only the term proportional to $\cos 2\varphi$ ($\sin 2\varphi$) survives in h_+ ($h_\times$). This shows that for a circular orbit, gravitational waves oscillate at *twice* the orbital frequency. For an eccentric orbit, the gravitational waves oscillate at *once*, *twice* and *three times* the orbital frequency. *The effect of the eccentricity is to introduce additional harmonics of the orbital frequency in the gravitational waves*[29].

The second observation to make is that when the observer is looking at the binary from the top ($\theta_o = 0$) both fields are equal in amplitude but out of phase by $\frac{\pi}{2}$. The polarization is circular. In the case where the observer is looking at the binary sideways ($\theta_o = \frac{\pi}{2}$), only h_+ is non-zero ($h_\times \propto \cos \theta_o = 0$) because all the motion takes place from left to right.

4.4 Radiation Reaction

4.4.1 Energy and angular momentum carried away

Our goal in this section is to derive the effect of the radiation on the evolution of the two orbital parameters e and p : we still use the quadrupole approximation. Their time evolution describes the inspiral of the two companions towards each other. In order to accomplish this, we start by evaluating the amount of energy and angular momentum carried away by the radiation. This is the same problem, since e and p are related to E and L through equation (4.7).

We start by calculating the loss of energy and angular momentum as defined by equations (3.45) and (3.46). To do this, we need to average these quantities over one period of the motion. The average over time can be replaced by an average over φ . Consider a function

$F = F(\varphi(t))$. Its average over one period is given by

$$\begin{aligned}
\langle F(\varphi(t)) \rangle_t &= \frac{1}{T} \int_0^T dt F(\varphi(t)) \\
&= \frac{1}{T} \int_0^{2\pi} d\varphi \frac{dt}{d\varphi} F(\varphi) \\
&= \frac{(1-e^2)^{3/2}}{2\pi M p^{3/2}} \int_0^{2\pi} d\varphi \frac{M p^{3/2}}{(1+e \cos \varphi)^2} F(\varphi) \\
&= (1-e^2)^{3/2} \frac{1}{2\pi} \int_0^{2\pi} d\varphi \frac{F(\varphi)}{(1+e \cos \varphi)^2} \\
&= (1-e^2)^{3/2} \langle F(\varphi) (1+e \cos \varphi)^{-2} \rangle_\varphi, \tag{4.34}
\end{aligned}$$

where the third equality follows from substituting $\dot{\varphi}$ (equation (4.5)) and T (equation (4.12)) into the second line.

In the center of mass coordinates, the energy carried away by the radiation is equal to

$$\begin{aligned}
\dot{E}^{GW} &= \frac{1}{5} \langle \partial_t^3 \mathbf{I}_{ij} \partial_t^3 \mathbf{I}_{ij} \rangle_t \\
&= \frac{1}{5} \langle (\partial_t^3 \mathbf{I}_{xx})^2 + (\partial_t^3 \mathbf{I}_{xy})^2 + (\partial_t^3 \mathbf{I}_{yy})^2 + (\partial_t^3 \mathbf{I}_{zz})^2 \rangle_t. \tag{4.35}
\end{aligned}$$

Squaring the components of the third derivatives of the reduced quadrupole moment (equation (4.31)) and dividing by $(1+e \cos \varphi)^2$ yields, for each squared element, a sum of even and odd powers of $\sin \varphi$ and $\cos \varphi$. By carrying the calculation through, one finds that the odd powers appear with odd powers of e . Since the average over one period of an odd power of a trigonometric function vanishes, the only contributing terms contain even powers of e . Grouping the different even powers of e together results in (see P.C. Peters[16])

$$\dot{E}^{GW} = \frac{32}{5} \left(\frac{\mu}{M} \right)^2 \frac{(1-e^2)^{3/2}}{p^5} \chi(\varphi),$$

where

$$\chi(\varphi) = \left\langle 1 + e^2 \left(\frac{1}{12} + \frac{71}{12} \cos^2 \varphi \right) + e^4 \left(\frac{1}{12} \cos^2 \varphi + \frac{11}{12} \cos^4 \varphi \right) \right\rangle_\varphi.$$

Our final expression for the energy carried away by the radiation is[16]

$$\dot{E}^{GW} = \frac{32}{5} \left(\frac{\mu}{M} \right)^2 \frac{(1-e^2)^{3/2}}{p^5} \left(1 + \frac{73}{24} e^2 + \frac{37}{96} e^4 \right), \tag{4.36}$$

where we used equation (4.34) and the identities $\langle \cos^2 \varphi \rangle_\varphi = \frac{1}{2}$, and $\langle \cos^4 \varphi \rangle_\varphi = \frac{3}{8}$.

The angular momentum loss is calculated in the same fashion as the energy loss. The difference is in the terms to be summed. In the center of mass coordinates, the binary is in the xy plane and the only non-vanishing component of the angular momentum is the z component. Using equation (3.46) gives this component as

$$\begin{aligned}\dot{L}_z^{GW} &= \frac{2}{5} \epsilon_{zjk} \langle \partial_t^2 \mathbf{I}_{jm} \partial_t^3 \mathbf{I}_{mk} \rangle_t \\ &= \frac{2}{5} \langle \partial_t^3 \mathbf{I}_{xy} (\partial_t^2 \mathbf{I}_{xx} - \partial_t^2 \mathbf{I}_{yy}) + \partial_t^2 \mathbf{I}_{xy} (\partial_t^3 \mathbf{I}_{yy} - \partial_t^3 \mathbf{I}_{xx}) \rangle_t.\end{aligned}\quad (4.37)$$

For the same reasons as mentioned above, the odd powers of the eccentricity are discarded. Using the same trigonometric identities and grouping the terms of the same order in e , we get[16]

$$\dot{L}_z^{GW} = \frac{32}{5} \left(\frac{\mu}{M} \right)^2 \frac{(1-e^2)^{3/2}}{p^{7/2}} \left(1 + \frac{7}{8} e^2 \right). \quad (4.38)$$

Equations (4.36) and (4.38) provides the differential equations describing the radiation reaction.

The ODE's for e and p are easily obtained from the differential equations for E and L_z . From the equations for e and p it is directly seen that $p = p(L_z)$ and $e = e(E, L_z)$. Accordingly, their time derivatives can be calculated by applying the chain rule and using equations (4.36) and (4.38). It must be remembered that the energy and angular momentum emitted as gravitational waves correspond to a decrease in the energy and the angular momentum of the binary, i.e. $\dot{E}^{bin} = -\dot{E}^{GW}$ and $\dot{L}_z^{bin} = -\dot{L}_z^{GW}$ (“*bin*” stands for “*binary*”). Executing this yields:

- one differential equation for p ,

$$\begin{aligned}\dot{p} &= \frac{dp}{dL_z} \dot{L}_z = -\frac{dp}{dL_z} \dot{L}_z^{GW} \\ &= -\frac{64}{5} \frac{\mu}{M^2} \frac{(1-e^2)^{3/2}}{p^3} \left(1 + \frac{7}{8} e^2 \right),\end{aligned}\quad (4.39)$$

- and one for e ,

$$\begin{aligned}\dot{e} &= \frac{\partial e}{\partial E} \dot{E} + \frac{\partial e}{\partial L_z} \dot{L}_z = -\frac{\partial e}{\partial E} \dot{E}^{GW} - \frac{\partial e}{\partial L_z} \dot{L}_z^{GW} \\ &= -\frac{304}{15} \frac{\mu}{M^2} \frac{(1-e^2)^{3/2}}{p^4} e \left(1 + \frac{121}{304} e^2 \right),\end{aligned}\tag{4.40}$$

where the superscript *bin* has been dropped. These results are equivalent to the results presented by P.C. Peters in 1964[16].

Equations (4.39) and (4.40) coupled with (4.9) constitute the set of first order differential equations that must be solved to obtain the evolution of the system and the waveforms (equations (4.32) and (4.33)).

4.4.2 Integration of the ODE's

In this part of the thesis, the integration of the differential equations (4.9), (4.39) and (4.40) will be accomplished. Whereas an analytic solution can be found for the circular case ($e = 0$), numerical methods of integration must be used in the general case ($e \neq 0$). The analytical solution for circular orbits is used to build templates, while the numerical solution serves to construct the signals (see chapter 2).

For the specific case where $e = 0$, we are left with a system of two differential equations to solve, since equation (4.40) for the eccentricity reduces to $\dot{e} = 0$. This implies that a circular orbit has a constant eccentricity; throughout the evolution $e = 0$. The remaining two equations are very easy to solve. The differential equation for p is

$$\dot{p} = -\frac{64}{5} \frac{\mu}{M^2} p^{-3},\tag{4.41}$$

with immediate solution

$$p = \left(\frac{256}{5} \frac{\mu}{M^2} (t_c - t) \right)^{1/4},\tag{4.42}$$

where t_c is the constant of integration defined such that $t = t_c$ at the instant of coalescence

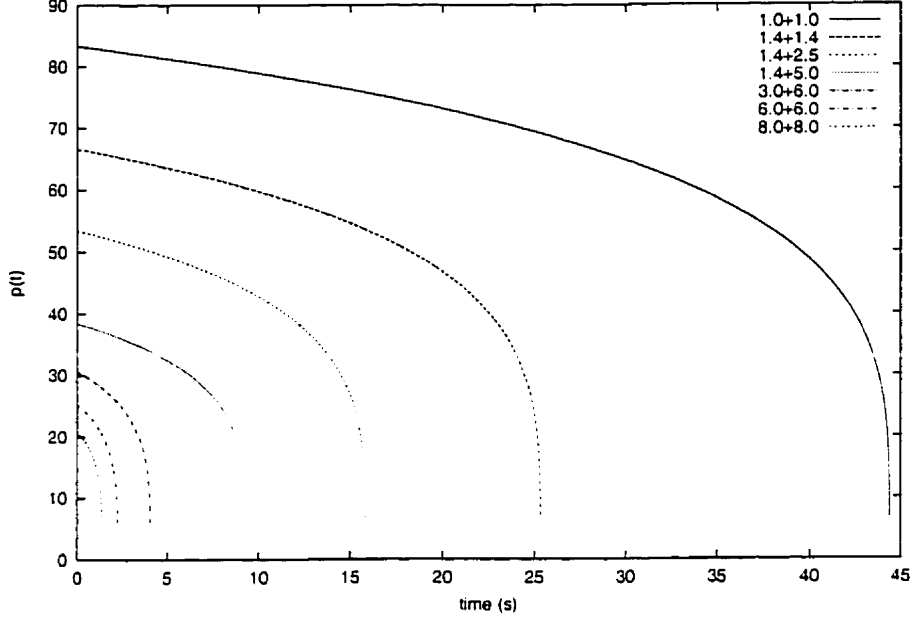


Figure 4.4: The time evolution of p is shown for various mass combinations. The various binaries are labeled by the masses, in units of solar mass, of the two companions. As the total mass of the binary increases, the two companions start their motion closer together in order to keep the initial orbital frequency constant. The increased proximity of the companions results in a faster evolution. The initial eccentricity e_{40} in all cases is 0.55.

(just prior to merger). This result can be inserted back in equation (4.9) to yield

$$\dot{\varphi} = \frac{1}{M} \left(\frac{256}{5} \frac{\mu}{M^2} (t_c - t) \right)^{-3/8}, \quad (4.43)$$

with solution

$$\varphi = \varphi_c - \left(\frac{t_c - t}{5\mathcal{M}} \right)^{5/8}, \quad (4.44)$$

where again the integration constant is chosen such that $\varphi = \varphi_c$ at the instant of coalescence. The chirp mass of the binary $\mathcal{M} = \mu^{3/5} M^{2/5}$ was introduced. The result stated in the paper by L.S. Finn and D.F. Chernoff[23] differs by a factor of two from the present result since they considered the phase of the radiation, i.e. twice the orbital phase (see subsection (4.3.2)).

The ODE's for an eccentric binary system cannot be solved analytically. The equations

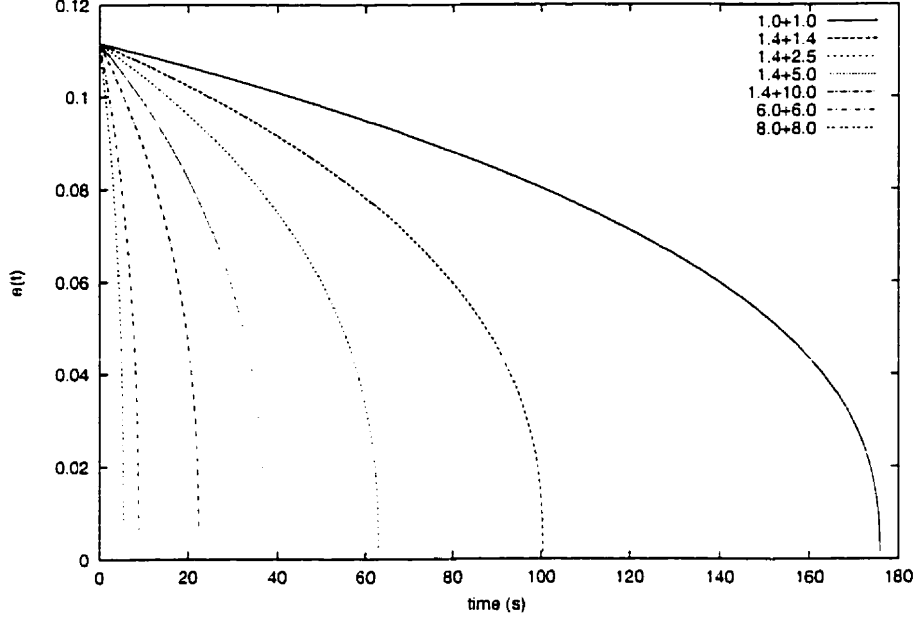


Figure 4.5: The eccentricity e_{40} when the wave's frequency is 40 Hz is chosen to be 0.1. The eccentricity at $t = 0$ is higher than 0.1 because we start the evolution when the wave's frequency is 38 Hz. Even though all the binaries start with the same eccentricity, the more massive binaries evolves faster because the two companions are placed closer together. This ensures that the initial wave's frequency is constant. The binaries are labeled by the masses, in solar mass, of the two companions.

are solved using the fourth order Runge-Kutta numerical integration routine from *Numerical Recipes in C*[30], combined with their adaptive step-size driver. The integration scheme works as follow. The driver (*odeint.c*) calls the integration routine (*rkqs.c*). The point located at $t + \Delta t$ is evaluated using the known point at t and intermediate points. The number of points chosen between t and $t + \Delta t$ determines the order of the integration. The relative error on the integration is evaluated using the sixth order terms (*rkck.c*). If the error is bigger than the relative accuracy, a smaller step-size is provided by *odeint.c*. The adaptative step-size algorithm (*odeint.c*) ensures that the numerical solution converges. The original *odeint.c* routine has been modified to end the integration at a final value of p instead of ending at a final specified time. We take $p = 6$ as the final value. For a test mass orbiting a Schwarzschild black hole, this value of p corresponds to the innermost stable

orbit.

The initial value of the parameters is set as follows. The focus of interest being the initial LIGO detector, the signals are chosen to start when the orbital frequency is equal to a third of the wave's frequency when it is first detectable by the initial LIGO interferometers ($f_{GW} = 40$ Hz):

$$T = \frac{1}{f_{orbital}} = \frac{3}{f_{GW}} = 7.5 \times 10^{-2} \text{s}. \quad (4.45)$$

This requirement ensures that we capture the entire signal. If the signal were started at a higher frequency some of the harmonics of the wave's initial frequencies (13.3 Hz, 27 Hz and 40 Hz) present in the gravitational signal (see equations (4.32) and (4.33)) would be missed. The initial value of p is then found by substituting the value of the initial eccentricity e_{40} at $f_{GW} = 40$ Hz in equation (4.12). For an initial eccentricity e_{40} and the initial period T mentioned above, the initial value of p_o is given by the relation

$$p_o = 180.42 (1 - e_{40}^2) \left(\frac{M_{\oplus}}{M} \right)^{2/3}. \quad (4.46)$$

This means that by increasing the eccentricity, or the total mass, the companions in the binary are pushed closer together. It is expected that they will experience a stronger interaction and thus decay faster. This is clearly apparent in figures (4.4) and (4.5).

The total time the binaries spend in the LIGO frequency band before coalescence is displayed in table (4.1) for the various combinations of masses considered. We see that the duration indeed decreases with increasing eccentricity. Another feature to notice is the decrease in duration as the total mass increases. This is due to the decrease in the initial separation with a mass increase, while all other factors are fixed. The proximity of the companion increases the interaction, leading to a faster coalescence. Both trends can be understood by combining equation (4.46) with equation (4.39). These two equations combined together show that initially p decays as

$$\left. \frac{dp}{dt} \right|_0 = -0.443 \text{s}^{-1} \left(\frac{\mu}{M_{\odot}} \right) \frac{(1 + \frac{7}{8}e_{40}^2)}{(1 - e_{40}^2)^{3/2}}. \quad (4.47)$$

Table 4.1: The total time (in seconds) spent in the LIGO band before coalescence. The companions are labeled by the masses of the companions, in units of solar mass.

e_{40}	1.0+1.0	1.4+1.4	1.4+2.5	1.4+5.0	1.4+10	3.0+6.0	6.0+6.0	8.0+8.0
0.00	137.97	78.75	49.24	29.04	17.60	12.65	6.96	4.31
0.05	136.71	77.94	48.81	28.78	17.42	12.53	6.90	4.26
0.10	133.02	75.90	47.48	28.01	16.95	12.19	6.70	4.15
0.15	126.99	72.49	45.32	26.73	16.19	11.64	6.41	3.96
0.20	118.91	67.89	42.44	25.04	15.16	10.90	6.00	3.71
0.25	110.15	62.31	38.95	22.98	13.92	10.00	5.50	3.40
0.30	98.03	55.95	34.99	20.63	12.50	8.99	4.94	3.06
0.35	86.06	49.10	30.71	18.12	10.97	7.89	4.34	2.68
0.40	73.65	42.05	26.29	15.50	9.38	6.75	3.71	2.29
0.45	61.28	34.97	21.87	12.89	7.81	5.62	3.08	-
0.50	49.37	28.18	17.63	10.39	6.28	4.52	2.49	-
0.55	38.31	21.86	13.67	8.06	4.88	3.51	-	-
0.60	28.41	16.21	10.15	5.97	3.61	-	-	-
0.65	19.95	11.38	7.12	-	-	-	-	-

It is clear from this equation that an increase in the mass or the initial eccentricity results in a faster initial decay rate and thus a faster evolution.

The initial value of φ is chosen to be zero. Because the two companions start their evolution closer together when the eccentricity increases, the duration of the signal is shorter. This means that the phase at coalescence will have a smaller numerical value for high eccentricities than for low ones (see figure (4.6)). For example, two compact objects of mass $1.4M_{\odot}$ each and in an orbit with $e = 0$ will go through 1959 oscillations before coalescence, while the same binary with $e = 0.75$ will go through 174 oscillations. As can be seen from figure (4.6) the evolution of the phase proceeds by steps, i.e. the orbital phase changes rapidly when the two companions approach the periastron. Kepler's law is approximately valid in the quadrupole approximation.

4.4.3 Numerical Results for the waveforms

The integrated parameters can now be inserted back in equations (4.32) and (4.33) to get the waveforms. The parameters θ_o and φ_o appearing in the waveforms are set to $\frac{\pi}{4}$ and 0,

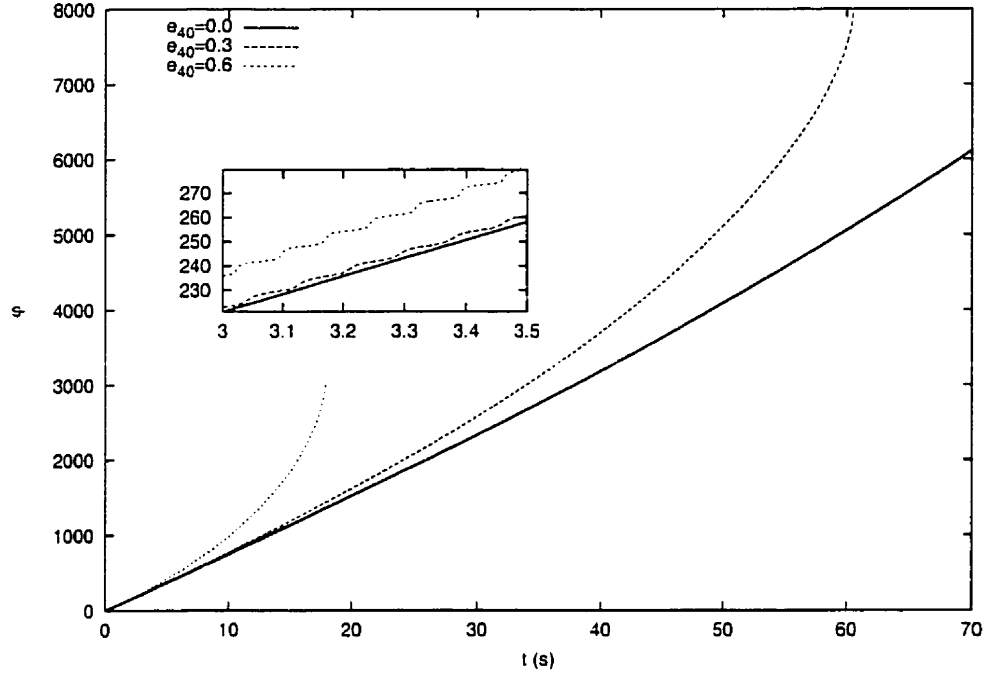


Figure 4.6: The evolution of the phase φ of a $1.4M_{\odot} + 1.4M_{\odot}$ binary as a function of time. When the eccentricity increases, the evolution of the system happens on a much shorter time scale because of the increasing proximity of the companions. Because of this faster evolution, the phase at coalescence φ_c will be smaller for an eccentric binary. Shown are the three cases $e_{40} = 0.0, 0.3, 0.6$.

respectively.

The evolution of an eccentric binary system is similar to the evolution of a circular binary, but eccentric binary systems emit radiation mostly at the periastron. This is because the two companions feel a much stronger interaction when they approach this point. Earlier in the evolution, the radiation is emitted in short bursts, resulting in sharp peaks in the waveforms, but as the system evolves the eccentricity decreases. The orbit is a little more circular, the peaks are broader and the radiation is emitted more evenly over one period of the motion. An eccentric binary never becomes completely circular, but at the end of the inspiral, most of the eccentricity has been radiated away and the gravitational waveforms resemble the waveforms from a circular binary system. This is illustrated in figure (4.7) for a $1.4M_{\odot} + 2.5M_{\odot}$ binary system.

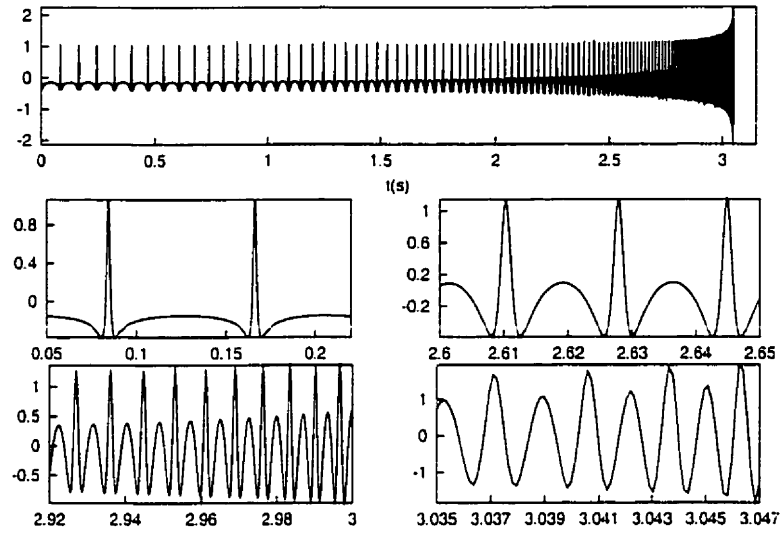


Figure 4.7: Plot of the waveform of an eccentric binary. The upper frame shows the entire signal. The lower frames show various portions of the signal. Initially, the binaries have an eccentricity of 0.75. As the gravitational waves remove energy and angular momentum from the system, the eccentricity decreases. At the end of the evolution (just before coalescence) the companions travel in an almost circular orbit. The two companions have masses $1.4M_{\odot}$ and $2.5M_{\odot}$.

The eccentric signal is similar to a circular signal in the sense that the evolution of the eccentric waveforms results in chirp signals. It sweeps upward in the frequency band until the two companions merge.

Chapter 5

Circular Waveforms as Search Templates

5.1 Introduction

In the previous chapter we obtained the explicit time-dependence of the gravitational waveforms. This, combined with the theoretical results of chapter 2, provides enough tools to evaluate the efficiency of the circular templates at capturing the radiation emitted by an eccentric binary system. In order to do this, we first need the spectral representation of both the circular and the eccentric waveforms. In subsection 5.2.1, the Fourier transforms of the circular waveforms are calculated using the stationary phase approximation. This approximation has no equivalent for the radiation emitted by eccentric binary systems, which forces their Fourier transforms to be calculated numerically (subsection 5.2.2).

Once the Fourier transforms are available, it is quite straightforward to construct the ambiguity function $\mathcal{A}(\boldsymbol{\Omega}, \boldsymbol{\vartheta})$ defined in chapter 2. The ambiguity function is evaluated numerically. A method for maximizing $\mathcal{A}(\boldsymbol{\Omega}, \boldsymbol{\vartheta})$ over the parameters $\boldsymbol{\Omega}$ characterizing the

circular templates is explained in subsection 5.3.2. Two quantities are obtained by calculating the ambiguity function: the signal-to-noise ratio and the fitting factor. While the first one provides information about the detection probability (provided a threshold is fixed on the signal-to-noise-ratio), the second gives information about the quality of the template used to model the signal. These two quantities are obtained in subsections 5.4.1 and 5.4.2, respectively.

5.2 The Fourier transforms of the waveforms

5.2.1 The stationary phase approximation

The Fourier transforms of the waveforms produced by a circular binary are obtained through the stationary phase approximation (SPA). We take these waveforms to be our templates. The SPA is based on the steepest-descent method and can be applied to any function for which the amplitude and the phase are monotonic functions of time[31]. For a function $F(t)$ written as

$$F(t) = A(t)e^{-i\psi(t)},$$

with Fourier transform

$$\tilde{F}(f) = \int_{-\infty}^{\infty} dt A(t)e^{i(2\pi ft - \psi(t))}, \quad (5.1)$$

the stationary phase approximation can be used as long as the two conditions

$$\frac{1}{\dot{\psi}} \frac{\partial \ln A(t)}{\partial t} \ll 1 \quad \text{and} \quad \frac{\ddot{\psi}}{\dot{\psi}^2} \ll 1 \quad (5.2)$$

are satisfied.

When the two conditions in equation (5.2) are realized, the maximum contribution to the Fourier transform arises when the phase of the integrand in equation (5.1) reaches an extremum. In other words, we are looking to solve the equation

$$\frac{d}{dt} (2\pi ft - \psi(t)) = 0, \quad (5.3)$$

with the trivial solution

$$\dot{\psi}(t) = 2\pi f. \quad (5.4)$$

This equation should be seen as defining a one-to-one correspondence between the frequency domain and the time domain. A single time t_o corresponds to a fixed frequency f_o . Once $\dot{\psi}(t)$ is explicitly known as a function of time and provided it has an inverse $\dot{\psi}^{-1}$, equation (5.4) can be solved to give $t(f) = \dot{\psi}^{-1}(2\pi f)$. At this point, time is defined as a function of frequency. Provided the root of equation (5.4) is unique, the Fourier transform in the stationary phase approximation is [23]

$$\tilde{F}(f) = \sqrt{\frac{2\pi i}{\ddot{\psi}(t(f))}} A(t(f)) e^{i(2\pi f t(f) - \psi(t(f)))}. \quad (5.5)$$

For signal detection with templates based on circular binary systems, we are interested in calculating the Fourier transforms of $h_+(t) \left(h_\times(t) \right)$ with $e = 0$. The only terms left are the ones proportional to $\cos(2\varphi) \left(\sin(2\varphi) \right)$ (see equations (4.32) and (4.33), respectively). We write these two functions in terms of exponentials and discard the terms containing $e^{i2\varphi}$ (these correspond to negative frequencies contributions): $\cos(2\varphi) = \frac{1}{2}e^{-i2\varphi}$ and $\sin(2\varphi) = \frac{1}{2i}e^{-i2\varphi}$. For the waveforms mentioned above, $\psi(t) = 2\varphi(t)$. Using the expression for $\dot{\varphi}(t)$ (equation (4.43)), we see that equation (5.4) becomes

$$\frac{1}{M} \left(\frac{256}{5} \frac{\mu}{M^2} (t_c - t) \right)^{-3/8} = \pi f, \quad (5.6)$$

or

$$t = t_c - \frac{5}{256} \frac{M^2}{\mu} (\pi M f)^{-8/3}. \quad (5.7)$$

The stationary phase approximation requires φ , $\ddot{\varphi}$ and p as functions of the frequency instead of functions of time. We first get $\ddot{\varphi}(f)$ by taking another time derivative of $\dot{\varphi}$ (equation (4.43)) and then substituting equation (5.7) in the resulting expression. The expressions for $\varphi(t)$ and $p(t)$ are given in equations (4.42) and (4.44). Direct substitution

of equation (5.7) yields

$$p(f) = (\pi M f)^{-2/3}, \quad (5.8)$$

$$\varphi(f) = \varphi_c - \frac{1}{32} \frac{M}{\mu} (\pi M f)^{-5/3}, \quad (5.9)$$

$$\ddot{\varphi}(f) = \frac{96}{5} \frac{\mu}{M^3} (\pi M f)^{11/3}. \quad (5.10)$$

Substituting these in the expressions for the waveforms (equations (4.32) and (4.33)) yields the Fourier transforms of the templates[23, 32]. Adopting the notation of chapter 2 for the template, we get

$$\bar{m}_{+, \times}(f) = \frac{\sqrt{30\pi}}{24} \frac{Q(\theta_o)}{2R} \mathcal{M}^2 (\pi \mathcal{M} f)^{-7/6} e^{i(2\pi f t_c + \frac{3}{128} (\pi \mathcal{M} f)^{-5/3} + \tilde{\kappa})}, \quad (5.11)$$

where

$$Q(\theta_o) = \begin{cases} 2(1 + \cos^2 \theta_o) & \text{for } h_+ \\ 4 \cos \theta_o & \text{for } h_{\times}. \end{cases} \quad (5.12)$$

$$\tilde{\kappa} = \begin{cases} 2\varphi_o - 2\varphi_c - \frac{\pi}{4} & \text{for } h_+ \\ 2\varphi_o - 2\varphi_c + \frac{\pi}{4} & \text{for } h_{\times}. \end{cases} \quad (5.13)$$

and the other symbols were defined in chapter 4. Inspection of equation (2.28) reveals that the amplitude of the template plays no role in assessing its quality, since the ambiguity function is a dimensionless ratio. The template will be of good quality if it is able to track the phase of the signal accurately. Thus, the exact form of the parameters appearing in equation (5.11) is irrelevant. The three parameters $\{t_c, \tilde{\kappa}, \mathcal{M}\}$ appearing in the phase will be varied to maximize the overlap between the eccentric signal and the circular template. Consequently, they are phenomenological parameters and cannot be expected to match the real parameters of the signal.

For detection of gravitational waves by the LIGO interferometers, it is very important to monitor the phase accurately to maximize the signal-to-noise ratio. The stationary phase approximation has been shown to be accurate enough for matched filtering[32].

5.2.2 Fast Fourier transform

We now turn to the calculation of the Fourier transforms of the eccentric waveforms, which will serve as the signals. For eccentric binary systems, the waveforms are the sum of three different terms, which oscillate at once, twice and three times the orbital frequency. If we try to calculate the Fourier transform of each term by using the stationary phase approximation, we would be trying to invert

$$\dot{\varphi}(t) = \alpha \pi f, \quad (5.14)$$

with $\dot{\varphi}$ now given by equation (4.9) as

$$\dot{\varphi} = \frac{(1 + e \cos(\varphi(t)))^2}{M p^{3/2}}. \quad (5.15)$$

The constant α equals 2, 1 or $\frac{2}{3}$ if $\psi(t) = \varphi(t)$, $2\varphi(t)$ or $3\varphi(t)$, respectively. For a given frequency, this equation has more than one root, because the angular velocity is not constant along the orbit. In other words, the eccentric waveforms do not satisfy the inequalities (5.2) because of the appearance of $\cos(\varphi(t))$ in $\dot{\varphi}$.

Instead, the Fourier transforms are obtained by using the fast Fourier transform routine *four1.c* from *Numerical Recipes in C* [30]. In order to get a constant step-size in the time domain, required by the fast Fourier transform algorithm, we use the results of chapter 4 (subsection 4.4.2) for the waveforms and interpolate between the points already available by using a cubic spline interpolation (*splint.c*).

The Fourier transform of the gravitational waveform $\tilde{h}_+(f)$ is displayed in figure (5.1). It clearly shows that a binary system spends most of its life in the low frequency regime, because this is where the amplitude is largest. Even though the binary system is chosen to have two equal mass companions ($1.4 M_\odot$ each), the Fourier transform would have the same general form for other mass combinations.

The Fourier transform of an eccentric signal is qualitatively the same, but shows more structure than the Fourier transform of a circular waveform. This is due to the presence of

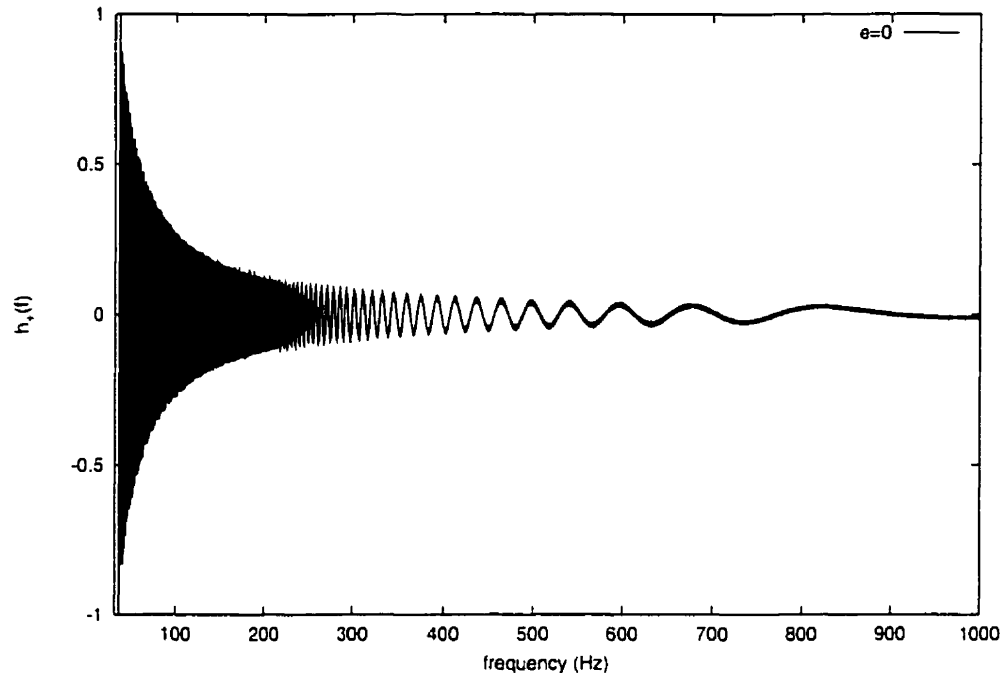


Figure 5.1: The Fourier transform of the gravitational waveform is obtained through a FFT algorithm from *Numerical Recipes in C* [30]. The shape of the Fourier transform shows that the binary system spends more time at low frequencies than at high ones. The amplitude of the waveforms decays as $f^{-7/6}$. The case displayed is for a $1.4 M_{\odot} + 1.4 M_{\odot}$ binary system with zero eccentricity

terms contributing at once, twice and three times the orbital frequency, and their subsequent harmonics. The structure of the eccentric signal is thus richer than the one of the circular template (see figure (5.2)). At high frequencies, the eccentricity of the system decreased significantly and the Fourier transforms approach the circular ones.

Figure (5.2) displays the envelope of the Fourier transform of an eccentric waveform. The figure shows the emergence of the three contributions arising at once, twice and thrice the orbital frequency. The eccentric waveforms constitute the signals we are trying to detect by matched filtering.

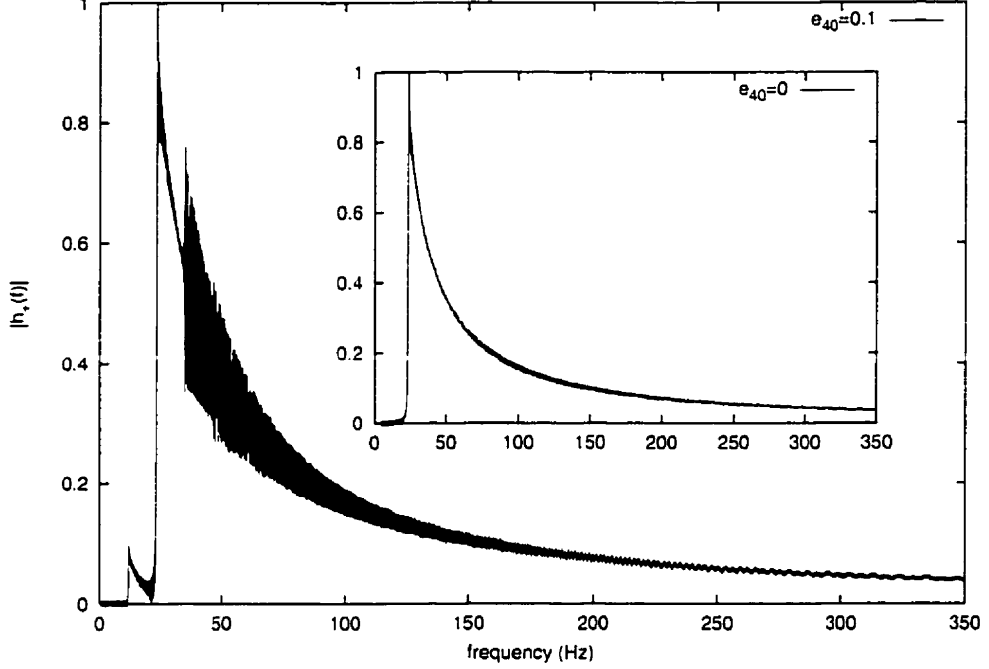


Figure 5.2: The envelope for a binary system with eccentricity $e_{40} = 0.1$ is shown. Both companions have a mass of $1.4 M_{\odot}$. The components at once, twice and thrice the orbital frequency are clearly apparent. The component at once the orbital frequency appears at 13.3 Hz, the one at twice the orbital frequency appears at 26.6 Hz and the one at thrice the orbital frequency starts at 40 Hz. The interference between the three components is obvious between 40 Hz and 150 Hz. For comparison, the inset shows the envelope of the same binary, but with $e_{40} = 0$. The only component present in this case is the one at twice the orbital frequency.

5.3 The Ambiguity Function

5.3.1 Constructing the Ambiguity function

The ambiguity function $\mathcal{A}(\Omega, \vartheta)$ defined in chapter 2, subsection 2.2.3, is built using the natural scalar product for matched filtering (equation (2.25)). We denote the Fourier transform of the eccentric signal by $\tilde{h}(f, \vartheta)$, where ϑ is the set of parameters describing the signal. For this work, we choose the detector to be located at $\theta_o = \frac{\pi}{4}$ and $\varphi_o = 0$. For each signal, the masses of the two companions have to be specified. We are looking at different mass combinations below. The template is chosen to be $\tilde{m}_+(f, \Omega)$, where $\Omega = \{t_c, \tilde{\kappa}, \mathcal{M}\}$ for a

circular template obtained using the quadrupole formula (see equation (5.11)).

From the definition of $\mathcal{A}(\boldsymbol{\Omega}, \boldsymbol{\vartheta})$ (equation (2.28)), we see that three ingredients are needed to do the maximization: The inner product of the signal $\tilde{h}(f, \boldsymbol{\vartheta})$ with itself, the inner product of $\tilde{h}(f, \boldsymbol{\vartheta})$ with $\tilde{m}_+(f, \boldsymbol{\Omega})$, and the inner product of $\tilde{m}_+(f, \boldsymbol{\Omega})$ with itself. For a given signal $\tilde{h}(f, \boldsymbol{\vartheta})$, $(h(\boldsymbol{\vartheta})|h(\boldsymbol{\vartheta}))$ is unaffected by the maximization procedure, since it only depends on the real parameters of the source. Thus we need to maximize

$$\langle \rho \rangle = \frac{(m_+(\boldsymbol{\Omega})|h(\boldsymbol{\vartheta}))}{\sqrt{(m_+(\boldsymbol{\Omega})|m_+(\boldsymbol{\Omega}))}}. \quad (5.16)$$

The numerator and the denominator will be treated separately to obtain the final expression. By using equation (2.25) with the stationary phase approximation (equation (5.11)), the denominator can be written as

$$\sqrt{(m_+(\boldsymbol{\Omega})|m_+(\boldsymbol{\Omega}))} = 2 \frac{\sqrt{30\pi}}{24} \frac{Q(\theta_o)}{2R} \mathcal{M}^2 (\pi \mathcal{M})^{-7/6} S_o^{-1/2} I^{1/2}, \quad (5.17)$$

where S_o is the amplitude of the noise in equation (2.13) and

$$\begin{aligned} I &= \int_0^\infty df \frac{1}{f^{7/3} \tilde{S}_n(f)} \\ &= 1.2223 \times 10^{-4} \text{ Hz}^{-4/3}. \end{aligned} \quad (5.18)$$

The last equality follows from the definition of the power spectral density of the noise in the initial LIGO (equation (2.13)). Because this integral is the same for all the templates, it needs to be evaluated only once.

On the other hand, the numerator of equation (5.16) has to be calculated for every eccentric signal we wish to study, since we do not have an analytical representation of the signals. With the stationary phase approximation for $\tilde{m}_+$ and leaving $\tilde{h}(f, \boldsymbol{\vartheta})$ as an unknown function, this can be written as

$$\begin{aligned} (m_+(\boldsymbol{\Omega})|h(\boldsymbol{\vartheta})) &= 2 \frac{\sqrt{30\pi}}{24} \frac{Q(\theta_o)}{2R} \mathcal{M}^2 S_o^{-1} (\pi \mathcal{M})^{-7/6} \\ &\quad [e^{-i\tilde{\kappa}} \mathcal{B}(t_c, \mathcal{M}) + e^{i\tilde{\kappa}} \mathcal{B}^*(t_c, \mathcal{M})], \end{aligned} \quad (5.19)$$

where $\mathcal{B}(t_c, \mathcal{M})$ is given by

$$\mathcal{B}(t_c, \mathcal{M}) = \int_0^\infty df e^{-i2\pi f t_c} \frac{e^{-i\frac{3}{128}(\pi\mathcal{M}f)^{-3/3}} \tilde{h}(f, \vartheta)}{f^{-7/6} \tilde{S}_n(f)}. \quad (5.20)$$

For any choice of parameters $\{t_c, \tilde{\kappa}, \mathcal{M}\}$ in the template, the signal-to-noise ratio is obtained by dividing (5.19) by equation (5.17). The result is

$$\langle \rho \rangle = I^{-1/2} [e^{-i\tilde{\kappa}} \mathcal{B}(t_c, \mathcal{M}) + e^{i\tilde{\kappa}} \mathcal{B}^*(t_c, \mathcal{M})], \quad (5.21)$$

with I defined in equation (5.18).

It is clear from equation (5.21) that the amplitude of the template does not play any role in our calculation, since it does not appear in the final expression for $\langle \rho \rangle$. Equation (5.21) also shows that if we had chosen $\tilde{m}_\times(f, \Omega)$ as our template instead of $\tilde{m}_+(f, \Omega)$, the result would have been the same. The only difference is in the definition of $\tilde{\kappa}$. Since it is a phenomenological parameter to be adjusted in the maximization procedure, this difference is irrelevant.

5.3.2 Maximizing over the parameters

The signal-to-noise ratio (equation (5.21)) has to be maximized over the three parameters $\{t_c, \tilde{\kappa}, \mathcal{M}\}$ of the circular template (equation (5.5)). First note that the maximization over $\tilde{\kappa}$ is trivially achieved by taking the magnitude of equation (5.21):

$$\langle \rho \rangle = 2I^{-1/2} |\mathcal{B}(t_c, \mathcal{M})|. \quad (5.22)$$

Because of this property, the value of $\tilde{\kappa}$ maximizing the ambiguity function $\mathcal{A}(t_c, \tilde{\kappa}, \mathcal{M})$ is not explicitly searched for in this work.

The maximization over the two parameters left is not as easily achieved because the function $\mathcal{B}(t_c, \mathcal{M})$ has to be constructed numerically. In order to maximize this discrete function we note that it is naturally evaluated by fast Fourier transform. This procedure carries the integration over the frequency to return a function of t_c . Concretely, we proceed

as follows. First we choose a chirp mass $\mathcal{M}_1$ for the template and evaluate the integral of equation (5.20) by fast Fourier transform. The function returned depends only on the template's parameter t_c . The maximization over t_c is then achieved by searching for the maximum value of $\mathcal{B}(t_c, \mathcal{M}_1)$ returned by the code. Once this is accomplished, another chirp mass $\mathcal{M}_2$ is chosen for the same eccentric signal $\tilde{h}(f, \boldsymbol{\vartheta})$. Another fast Fourier transform is taken to get $\mathcal{B}(t_c, \mathcal{M}_2)$. The same procedure of maximization over t_c is then carried out. By doing this for a range of chirp masses we construct a function $\mathcal{B}(\mathcal{M}) = \max_{t_c} \mathcal{B}(t_c, \mathcal{M})$ that depends solely on the chirp mass. At this point, the function $\mathcal{B}(\mathcal{M})$ is maximized over $\mathcal{M}$ by golden-section search (*Numerical Recipes in C* [30]).

Table 5.1: The parameter t_c obtained by maximizing the function $\mathcal{B}(t_c, \mathcal{M})$. The values extracted are in good agreement with the real time of coalescence of the binaries (see text for detail). The binaries are labeled by the masses of each companion in units of solar mass.

e_{40}	1.0+1.0	1.4+1.4	1.4+2.5	1.4+5.0	1.4+10	3.0+6.0	6.0+6.0	8.0+8.0
0.00	138.05	78.75	49.24	29.03	17.61	12.66	6.96	5.75
0.05	136.71	77.95	64.98	28.77	17.43	12.53	6.91	4.27
0.10	133.03	75.92	47.49	28.02	16.96	12.18	6.72	4.14
0.15	127.01	72.49	45.31	26.74	16.19	11.64	6.43	3.96
0.20	118.90	67.92	42.44	25.04	15.15	10.91	6.00	3.72
0.25	110.14	62.32	38.93	22.99	13.94	9.99	5.51	3.41
0.30	98.04	55.94	35.01	20.63	12.50	9.00	4.94	3.06
0.35	86.06	49.12	37.82	18.13	10.97	7.90	4.33	2.70
0.40	73.65	42.05	26.27	15.49	9.39	6.75	3.71	2.29
0.45	61.29	35.00	21.88	12.89	7.83	5.63	3.09	-
0.50	49.38	28.18	17.63	10.40	6.28	4.53	2.49	-
0.55	38.30	21.85	13.67	8.06	4.88	3.52	-	-
0.60	28.39	16.20	10.13	5.95	3.62	-	-	-
0.65	19.93	11.37	7.10	-	-	-	-	-

Table (5.1) contains the values of t_c which maximize the ambiguity function for the binary systems under study. Even though the circular templates are not a perfect match for the eccentric signals, the maximization procedure returns values of t_c that reproduce the real time of coalescence of the binaries (see table (4.1), page 59). On the other hand, the value of the chirp mass obtained by maximizing $\mathcal{B}(\mathcal{M})$ ($\mathcal{M}_{eff}$) does not reproduce

Table 5.2: The ratio of the chirp mass extracted by the maximization procedure $\mathcal{M}_{eff}$ to the real chirp mass $\mathcal{M}_r$ of the eccentric signal. The various binaries are labeled by the masses of the two companions, in units of solar mass. $\mathcal{M}_{eff}$ increases as the initial eccentricity e_{40} increases to mimic the signal (see text for details).

e_{40}	1.0+1.0	1.4+1.4	1.4+2.5	1.4+5.0	1.4+10	3.0+6.0	6.0+6.0	8.0+8.0
0.00	1.0000	0.9999	0.9999	0.9999	0.9997	0.9997	0.9994	0.9992
0.05	1.0007	1.0006	1.0007	1.0007	1.0007	1.0006	1.0004	1.0005
0.10	1.0012	1.0016	1.0017	1.0022	1.0030	1.0031	1.0033	1.0036
0.15	1.0024	1.0027	1.0031	1.0039	1.0053	1.0056	1.0076	1.0094
0.20	1.0037	1.0042	1.0048	1.0060	1.0083	1.0083	1.0113	1.0160
0.25	1.0059	1.0059	1.0071	1.0087	1.0122	1.0122	1.0165	1.0222
0.30	1.0088	1.0095	1.0106	1.0121	1.0172	1.0170	1.0228	1.0314
0.35	1.0125	1.0134	1.0149	1.0167	1.0234	1.0232	1.0312	1.0429
0.40	1.0182	1.0194	1.0210	1.0223	1.0319	1.0314	1.0418	1.0563
0.45	1.0260	1.0288	1.0302	1.0307	1.0430	1.0416	1.0562	-
0.50	1.0404	1.0412	1.0447	1.0466	1.0586	1.0574	1.0774	-
0.55	1.0598	1.0654	1.0640	1.0680	1.0842	1.0749	-	-
0.60	1.0986	1.0946	1.0986	1.0976	1.1138	-	-	-
0.65	1.1508	1.1542	1.1484	-	-	-	-	-

the real chirp mass of the signals ($\mathcal{M}_r$). Table (5.2) displays the value of the chirp mass returned by the maximization procedure compared to the actual chirp mass of the binaries. As the initial eccentricity of the signal is increased, the value of the parameter increases as well. The explanation is simple. A larger eccentricity results in a faster evolution and the circular template tries to compensate for this by increasing $\mathcal{M}_{eff}$. This corresponds to a faster evolution as can be seen from equations (4.39) and (4.40).

5.4 Numerical results

5.4.1 The signal-to-noise ratio

To construct the ambiguity function we still need to evaluate the signal-to-noise ratio one would achieve by optimal filtering (equation (2.23)). Once the Fourier transforms of the eccentric signals are known, it is trivial to construct the optimal SNR (right-hand side of

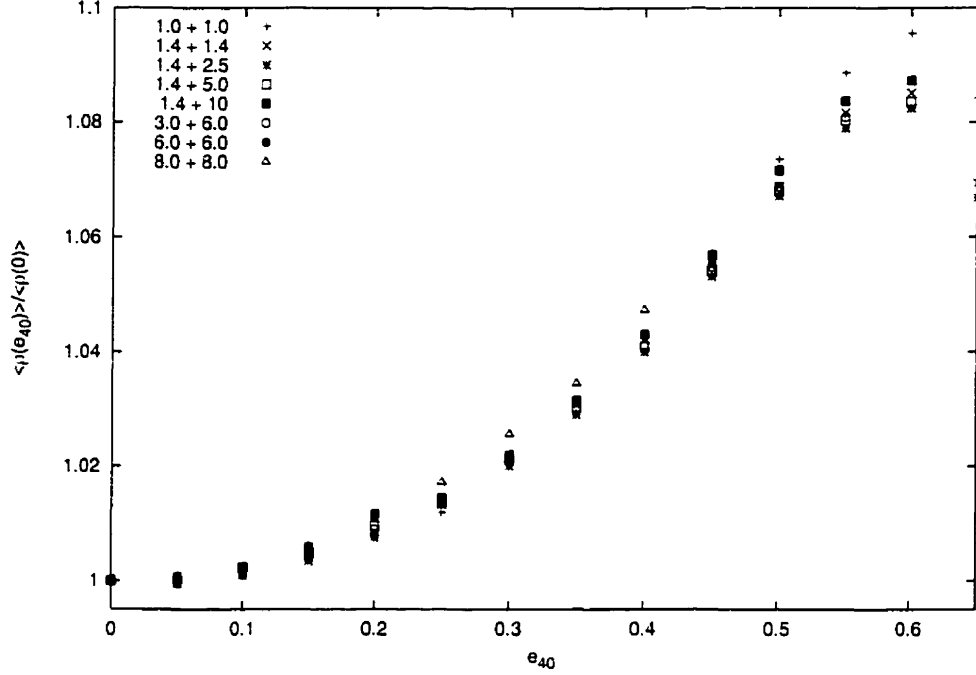


Figure 5.3: The ratio of the signal-to-noise ratio for eccentric binary systems, $\langle \rho(e_{40}) \rangle$, to the signal-to-noise ratio expected if the same binary were circular, $\langle \rho(0) \rangle$. The increase in the signal-to-noise ratio as the initial eccentricity increases is due to the presence of three terms in the eccentric signal; at once, twice and thrice the orbital frequency. The various binaries studied are labeled by the two masses of the companions, in units of solar mass.

equation (2.22)). The optimal signal-to-noise ratio is

$$\langle \rho \rangle_{max} = 4 \int_0^\infty df \frac{\bar{h}(f, \vartheta) \bar{h}^*(f, \vartheta)}{S_n(f)}. \quad (5.23)$$

The integral is evaluated numerically by using a discrete integration algorithm. The results are displayed in figure (5.3).

The increase in the SNR with initial eccentricity e_{40} is due to the presence of three terms in the eccentric signal, i.e. the Fourier transform of an eccentric signal is of the form

$$\bar{h}(f) = \bar{h}_1(f, \varphi) + \bar{h}_2(f, 2\varphi) + \bar{h}_3(f, 3\varphi), \quad (5.24)$$

where $\bar{h}_j(f, j\varphi)$ is the component of the gravitational waveform which oscillates at j times the orbital frequency. For a circular binary, $\bar{h}_2(f, 2\varphi)$ is the only term present.

To calculate the optimal SNR, we need to evaluate $\tilde{h}(f)\tilde{h}^*(f)$ (equation (5.23)). This gives terms of the form $\tilde{s}_j\tilde{s}_k^*(f)$. When $j = k$, the integral is nonzero because we are integrating the magnitude of a function, i.e. the phase of this type of term is 0. On the other hand, when $j \neq k$ the function is oscillating and vanishes upon integration between $[-\infty, \infty]$. Accordingly, the maximum SNR can be written as

$$\langle \rho(e_{40}) \rangle_{max} \approx \sqrt{\langle \rho_1 \rangle^2 + \langle \rho_2 \rangle^2 + \langle \rho_3 \rangle^2}, \quad (5.25)$$

where

$$\langle \rho_i \rangle^2 = 4 \int_0^\infty df \frac{\tilde{h}_i(f)\tilde{h}_i^*(f)}{S_n(f)}. \quad (5.26)$$

The SNR of a circular binary contains only the term $\langle \rho_2 \rangle^2$; the SNR in this case is expected to be lower than for an eccentric binary. When $e_{40} = 0$, the SNR achieved by using Wiener's optimal filter is calculated from equation (5.17) and is

$$\langle \rho(0) \rangle_{max} \approx 2.1 \left(\frac{200 \text{ Mpc}}{R} \right) \left(\frac{\mathcal{M}}{10 M_\odot} \right)^{5/6}. \quad (5.27)$$

where we used $S_o = 0.5 \times 10^{-46} \text{ Hz}^{-1/3}$ [24], and $\theta_o = 0$.

The result displayed in figure (5.3) implies that for two identical binary systems located at the same distance R , with the first binary circular and the second one eccentric, the signal from the second one is stronger. Thus, it should be easier to detect gravitational waves from eccentric binary systems, if such signals were explicitly searched for in the output of the interferometers. This would require the inclusion of eccentric templates in the bank of templates available to the LIGO interferometers. When the number of templates increases, the probability of false detection (mistaking the noise for a signal) increases as well. To compensate, the threshold on the SNR must be fixed higher[21]. It is not clear at this point if the additional computational burden and the possible loss of circular signals is an important obstacle. To answer this question requires information about the number of eccentric binaries expected to coalesce in the course of a year.

Combining the results for the optimal SNR with the non-optimal SNR (equation (5.21)) allows one to build the ambiguity function (equation (2.28), page 25). The ambiguity function is affected by two processes: the first one is the net decrease in the overlap between the eccentric signal and the circular template when the initial eccentricity increases: this affects the SNR achieved by non-optimal filtering. The second is the increase of the optimal SNR with increased initial eccentricity e_{40} . Both effects contribute to reduce the maximum value of $\mathcal{A}(t_c, \tilde{\kappa}, \mathcal{M})$. Figure (5.4) displays the ambiguity function $\mathcal{A}(\mathcal{M}) = \max_{t_c, \tilde{\kappa}} \mathcal{A}(t_c, \tilde{\kappa}, \mathcal{M})$ as a function of the effective chirp mass for various initial eccentricities e_{40} . The binary is taken to be a neutron star orbiting a 10 solar mass black hole. The net decrease induced by a higher eccentricity is clearly evident. When the signal is taken to have $e_{40} = 0$, the maximum value the ambiguity function takes is one. This was expected: the circular template models accurately the radiation coming from a circular binary.

5.4.2 The Fitting Factor

The fitting factor is the maximum value of the ambiguity function. For example, this extremum can be read off the function displayed in figure (5.4). Numerically, the maximum is found by golden-section search[30]. The results are displayed in table (5.3) and in figure (5.5). The results show two trends. First, for a given choice of input masses, the fitting factor decreases as the initial eccentricity is increased, because the signals possess additional features distinguishing them from the circular templates. The circular template is not a good model for signals coming from highly eccentric binaries. Secondly, as the total mass of the system increases, the circular templates can be used to detect binaries with higher initial eccentricities. This is mostly due to the fact that the more massive the binary gets, the shorter is its evolution. In this case, the circular template has less of an opportunity to get out of phase with the eccentric signal.

For example, 19.4% of coalescing neutron star binaries (mass of each companion: 1.4

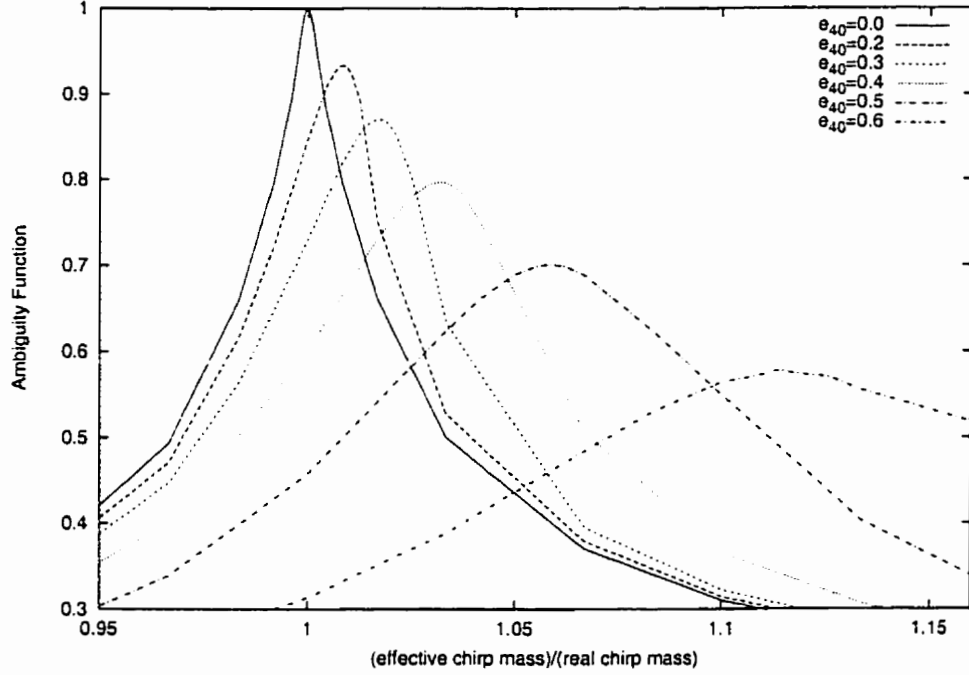


Figure 5.4: The ambiguity function maximized over $\tilde{\kappa}$ and t_c for a binary system made of a Neutron star ($1.4 M_\odot$) and a black hole ($10.0 M_\odot$). The effect of increasing the initial eccentricity is to reduce the maximum value of the ambiguity function. As expected, the circular template is not modeling the radiation from highly eccentric binaries very well.

$M_\odot$) would be missed by the detectors if the initial eccentricity were 0.1, since in this case $FF = 0.9306$ (recall that the loss in event rate is given by $1 - FF^3$). If one of the companions is replaced by a $10 M_\odot$ black hole, the loss in event rate goes down to 7.0%, and if the binary is made of two black holes of $6 M_\odot$ each, the percentage of signals missed, for the same eccentricity of 0.1, is only 2.1%. When we go to higher eccentricities, the fitting factor decreases even more dramatically. For an initial eccentricity $e_{40} = 0.5$, 93.5% of the signals from neutron star binaries will be missed, 71.8% of the signals emitted by binaries made of a neutron star and a $10 M_\odot$ black hole will be missed, and 61.4% of the signals emitted by $6.0 M_\odot$ black holes binaries will go unnoticed. This loss of events is huge and unacceptable. We adopt a threshold of 0.9 on the fitting factor such that a maximum of 27% of the signals will be missed. For the three binaries mentioned above, this correspond

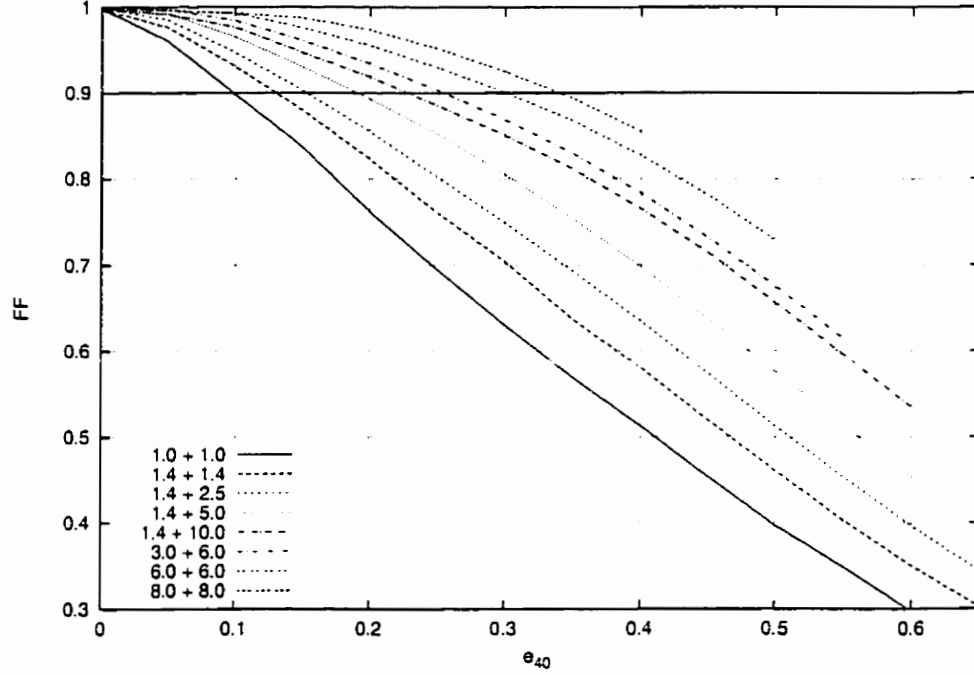


Figure 5.5: The fitting factor as a function of the initial eccentricity for various binary systems. Two trends are apparent. The first one is the net decrease in the fitting factor as the initial eccentricity is increased, while the total mass of the binary is held fix. The second one is the increase in the detection probability when the total mass of the binary system is increased.

to maximum initial eccentricities of 0.13 ($1.4 M_{\odot}$ neutron star- $1.4 M_{\odot}$ neutron star), 0.22 ($1.4 M_{\odot}$ neutron star- $10 M_{\odot}$ black hole), and 0.3 ($6.0 M_{\odot}$ black hole- $6.0 M_{\odot}$ black hole). The fitting factor for other combinations of masses are displayed in table (5.3). The trends described above are clearly apparent.

Our main conclusions are the following:

1. Eccentric binaries can be detected by the initial LIGO detector even if they are not explicitly searched for with eccentric templates;
2. Inspiring compact binaries made up of 10 solar mass black holes are easier to detect than neutron stars binaries, because the circular templates have less of a chance to go out of phase with the signal;

Table 5.3: The fitting factor for various binaries as a function of the initial eccentricity. The eccentric binaries are more easily detected when the total mass of the system is large, and when the eccentricity is low. As the total mass of the system is increased, the eccentricity up to which a eccentric coalescing binary can be observed increases.

e_{10}	1.0+1.0	1.4+1.4	1.4+2.5	1.4+5.0	1.4+10	3.0+6.0	6.0+6.0	8.0+8.0
0.00	0.9978	0.9969	0.9990	0.9975	0.9976	0.9982	0.9984	0.9996
0.05	0.9604	0.9763	0.9850	0.9923	0.9916	0.9977	0.9964	0.9959
0.10	0.8980	0.9306	0.9468	0.9648	0.9762	0.9841	0.9929	0.9929
0.15	0.8363	0.8792	0.9025	0.9296	0.9465	0.9606	0.9754	0.9873
0.20	0.7617	0.8224	0.8544	0.8926	0.9182	0.9335	0.9550	0.9731
0.25	0.6950	0.7608	0.8022	0.8518	0.8850	0.9033	0.9295	0.9517
0.30	0.6304	0.6366	0.7491	0.8051	0.8501	0.8684	0.9004	0.9246
0.35	0.5692	0.5807	0.6927	0.7534	0.8110	0.8288	0.8671	0.8933
0.40	0.5128	0.5198	0.6348	0.6983	0.7649	0.7830	0.8268	0.8545
0.45	0.4537	0.4604	0.5743	0.6373	0.7143	0.7316	0.7806	-
0.50	0.3967	0.4018	0.5126	0.5764	0.6560	0.6750	0.7281	-
0.55	0.3477	0.3496	0.4523	0.5133	0.5949	0.6135	-	-
0.60	0.2971	0.3029	0.3958	0.4524	0.5341	-	-	-
0.65	0.2568	0.2309	0.3445	-	-	-	-	-

3. If the threshold on the fitting factor is set to 0.9, then the initial eccentricity of a massive binary system ($M \leq 20 M_{\odot}$) must be less than 0.35 for a successful detection of its gravitational waves.

Chapter 6

Conclusion

6.1 Summary

The work presented in this thesis is not all original. Part of it was done before and all we did in this case was to verify the existing results. We gave a derivation for the various quantities of interest in the detection process such as the signal-to-noise ratio, the ambiguity function, and the fitting factor (see chapter 2 and the relevant sections within). The quadrupole formula and the radiation reaction for gravitationally bound systems were derived in chapter 3. We verified the results of H. Wahlquist for the eccentric waveforms[29] and of P.C. Peters[16] for the radiation reaction.

Part of the work presented here was original. We obtained the time evolution of the parameters describing eccentric binary systems, the explicit time evolution of the waveforms, and we showed that the waveforms associated with eccentric binaries possess more features than the waveforms of circular binary systems (see figure (5.2)). This made the calculations for eccentric binaries more arduous. Among other things, these features forbid the use of the stationary phase approximation to calculate the Fourier transforms of the eccentric waveforms. Instead they were obtained by using a fast Fourier transform algorithm. The fact that the gravitational waves oscillate at once, twice and thrice the orbital frequency

plays an essential role, and special attention must be paid to this effect when constructing adequate signals for the LIGO interferometers. The presence of these different terms forced us to start the orbital evolution of the eccentric binaries much earlier than is necessary for circular binaries. Once the Fourier transforms were calculated, the optimal SNR for eccentric binaries was calculated (see figure (5.3)), the ambiguity function was constructed, and the fitting factor was obtained by maximizing the ambiguity function. Our results show that massive eccentric binaries should be detected by the initial LIGO interferometers as long as the initial eccentricity is lower than 0.35. We find that the fitting factor increases as the total mass of the system increases (figure (5.5)). This is good news, since as the mass of the binary increases the radiation it emits is stronger, making the detection easier. It will be easier, for the initial LIGO, to detect massive binaries whether they are circular or not.

6.2 Future Outlook

6.2.1 Post-Newtonian corrections to the eccentric waveforms

A crucial simplification we made in this work was to assume the validity of the quadrupole approximation, thus neglecting all post-Newtonian corrections. This is an oversimplification, because the real waveforms have a slightly different behavior due to these corrections. Though it is doubtful that the results will be qualitatively affected by including these post-Newtonian corrections, to include them would be desirable if a fully quantitative analysis were necessary. The eccentric waveforms have been calculated to order $(\frac{v}{c})^4$ beyond the leading-order expression by A. Gopakumar and R. Iyer[13]. Their paper contains the basis for an improved version of the present work. By including post-Newtonian corrections into the eccentric signal and the templates, we can expect the fitting factor to increase slightly because the templates are characterized by a larger number of parameters. The new parameters might make it possible to follow the phase of the eccentric signals more accurately.

On the other hand, we cannot expect a perfect match ($FF = 1$) between eccentric signals and circular templates. There should be a quantitative difference in the fitting factor, but qualitatively, we expect the same type of curve as shown in figure (5.5).

6.2.2 Advanced LIGO

The present work was carried out for the initial LIGO detectors. These are sensitive to signals with frequencies between 40 Hz and 1000 Hz. It would be interesting to do the calculations for the advanced version of LIGO. This is the detector we are ultimately interested in, since it is far more sensitive than the initial LIGO. The frequency range for this detector will be between 10 Hz and 1000 Hz. Since coalescing compact binaries spend most of their life in the low frequency regime, the inclusion of the frequency band between 10 Hz and 40 Hz will more than double the duration of the signals. To obtain a good frequency resolution, we would have to use a huge number of points and the computing facilities presently at our disposition are not sufficient. Nevertheless we can reach some intuitive conclusions from the knowledge accumulated while working in the context of the initial LIGO. We know that the fitting factor increases when the total mass of the eccentric binary increases due to a shorter duration of the signal, i.e. the less time the eccentric signals last, the more accurately the circular templates can follow them. Thus, it is almost certain that the circular templates would not perform as well for the advanced LIGO as for the initial LIGO since the duration of the signal will be longer than the ones displayed in table (4.1). On the other hand, we can expect the same trends to appear in the fitting factor: For a given binary system, the fitting factor should decrease as the initial eccentricity increases; and for binaries with different masses and the same initial eccentricity, the fitting factor should be higher for the more massive binaries, since its signal will have to be followed for a shorter period of time.

We can also predict an increase in the optimal SNR for eccentric binaries. The increase observed for the initial LIGO is due to the presence of the three fundamental components

at once, twice and thrice the orbital frequency. The advanced LIGO will see more of the eccentric contributions at low frequencies leading to a higher signal-to-noise ratio: The three components are stronger at low frequencies than at high frequencies (see figure (5.2)) and this should be reflected in the optimal SNR.

6.2.3 Time-frequency detection of gravitational waves

We have seen that non-optimal filtering results in a net loss in signal-to-noise ratio. This is likely to occur for a variety of sources. It might be worth exploring alternative detection methods that do not rely on a prior knowledge of the sources and their evolution. Such methods exist, but very little is known about their properties and how they can efficiently be applied to the detection of gravitational waves in kilometer-size interferometers. One such method is time-frequency detection[33]. The detection process in this case is carried out in the time-frequency domain, using a filter that does not require any detailed information about the sources. This method will not result in the optimal signal-to-noise ratio, but since non-optimal filtering results in a loss of signal-to-noise ratio, it might be worth exploring whether time-frequency methods do better than matched filtering in the absence of optimal filters.

Bibliography

- [1] R.M. Wald, *General Relativity*, The University of Chicago Press, Chicago (1984).
- [2] B.F. Schutz, *A first course in general relativity*, Cambridge University Press, New York (1990).
- [3] L. Blanchet, B.R. Iyer, C.M. Will, and A.G. Wiseman, *Class. Quantum Grav.* **13**, 575 (1996).
- [4] T. Tanaka, H. Tagoshi, and M. Sasaki, *Prog. Theor. Phys.* **96**, 1087 (1996).
- [5] C.W. Misner, K.S. Thorne, and J.A. Wheeler. *Gravitation*. Freemann, San Fransico (1973).
- [6] S. Weinberg, *Gravitation and Cosmology*, John Wiley and Sons, New York (1972).
- [7] K.S. Thorne, *Three hundred years of gravitation*, Cambridge University Press, New York, 330 (1987).
- [8] B.F. Schutz, *Nuclear Physics B***35**, Proc. Suppl., 44 (1994).
- [9] B.C. Barish, <http://xxx.lanl.gov/archive/gr-qc>, 9905026 (1999).
- [10] É. É and Flanagan and S. A. Hugues, *Phys.Rev.* **D57**, 4535 (1998).
- [11] L.A. Wainstein and V.D. Zubakov, *Extraction of Signals from Noise*, Prentice-Hall, Englewood Cliffs (1962).

- [12] T.A. Apostolatos, Phys. Rev. D**54**, 2421 (1996).
- [13] A. Gopakumar and B.R. Iyer, Phys. Rev. D**56**, 7708 (1997).
- [14] C.M. Will and A. Wiseman, Phys. Rev. D**54**, 4813 (1996).
- [15] C.M. Will, *Confrontation between General Relativity and experiment: a 1998 update*,
Lecture notes from the 1998 Slac Summer Institute on Particle Physics(1998).
- [16] P.C. Peters, Phys. Rev. **136**, vol. 4B, B1224 (1964).
- [17] K. Mannheim, *AGN models:High-Energy Emission*. XXXII Rencontres de Moriond,
Les Arcs, France (1997).
- [18] S.L. Shapiro and S.A. Teukolsky, Astrophys. J. **292**, L41 (1985).
- [19] S.L. Shapiro and G.D. Quinlan, Astrophys. J. **321**, 199 (1987).
- [20] B.J. Owen, Phys. Rev. D**53**, 6749 (1996).
- [21] B.J. Owen and B.S. Sathyaprakash. Phys. Rev. D**60**, 022002 (1999).
- [22] B.S. Sathyaprakash, <http://xxx.lanl.gov/archive/gr-qc>. 9612009 (1996).
- [23] L.S. Finn and D.F. Chernoff, Phys. Rev. D**47**, 2198 (1993).
- [24] S. Droz, Phys. Rev. D**59**, 064030 (1999).
- [25] R. Balasubramanian and S.V. Dhurandar, Phys. Rev. D**50**, 6080 (1994).
- [26] J.D. Jackson, *Classical Electrodynamics*, 2nd edition, John Wiley and sons, New York
(1962).
- [27] H. Goldstein, *Classical Mechanics*, Addison-Wesley Publishing Company, Reading
(1980).
- [28] J.B. Marion and S.T. Thornton, *Classical Dynamics of particles and systems*, 3rd
edition, Harcourt Brace Javanovich College Publishers, New York (1988).

- [29] H. Wahlquist, *General Relativity and Gravitation*, vol.19, No.11, 1101 (1987).
- [30] W.H. Press, S.A. Teukolsky, and W.T. Vetterling, *Numerical Recipes in C*, Cambridge University Press, New York (1997).
- [31] C.M. Bender and S.A. Orzag, *Advanced Mathematical Methods for Scientists and Engineers*, McGraw Hill, New York (1978).
- [32] S. Droz, D. Knapp, and E. Poisson, Phys. Rev. D**59**, 124016 (1999).
- [33] W.G. Anderson and R. Balasubramanian, <http://xxx.lanl.gov/archive/gr-qc>, 9905023 (1999).

Appendix A

The Phase of the Template and the Signal in the Time Domain

In order for the circular template to mimic the eccentric signal, we have seen that the chirp mass of the template ($\mathcal{M}_{eff}$) had to be higher than the real chirp mass of the signal ($\mathcal{M}_r$). In this appendix, we want to show that the increase in the chirp mass actually allows the circular template to follow the eccentric signal more closely around 200 Hz than it would if the real chirp mass was used. We know that the massive binaries evolve faster than the less massive binaries. Accordingly, the frequency around 200 Hz for the massive binaries changes faster. This allows the template to follow the signal closely for a wider frequency range, centered around 200 Hz, than less massive binaries (one made of two neutron stars, say).

We show these effects in figure A.1 and figure A.2. We chose two binaries: one made of two $1.4M_\odot$ neutron stars ($\mathcal{M}_r = 1.219M_\odot$ and $\mathcal{M}_{eff} = 1.0946\mathcal{M}_r$), and one made of two $6.0M_\odot$ black holes ($\mathcal{M}_r = 5.223M_\odot$ and $\mathcal{M}_{eff} = 1.0418\mathcal{M}_r$). The initial eccentricities are chosen to be 0.6 for the former and 0.4 for the later. We are only looking at highly eccentric system because at low eccentricities, the signal and the templates are very similar and the

effects we wish to demonstrate are harder to see. The other problem with low eccentricity binaries is that $\mathcal{M}_{eff}$ is very close to $\mathcal{M}_r$, i.e. the template built using $\mathcal{M}_{eff}$ is very close to the template built using $\mathcal{M}_r$. This does not make for an easy comparison of the two templates.

The top portion of figure A.1 shows the eccentric signal and the template obtained using $\mathcal{M}_r$, while the bottom portion shows the same signal with the template obtained using $\mathcal{M}_{eff}$. The $1.4M_\odot + 1.4M_\odot$ binary is a low mass system and its evolution happens on a large time scale. This means that the frequency does not change rapidly around 200 Hz ($f=200$ Hz at $t = 15.88s$). The template with $\mathcal{M}_{eff}$ is in phase with the signal for 7-8 cycles, while the one with $\mathcal{M}_r$ is in phase for only 3-4 cycles. Thus the increased chirp mass allows the template to follow the eccentric signal more closely than when $\mathcal{M}_r$ is used.

For the $6.0M_\odot + 6.0M_\odot$ binary, the same phenomena is seen, but here the effect is sharper. The frequency of the signal reaches 200 Hz at $t = 3.66s$. The template with $\mathcal{M}_r$ is in phase with the signal for 2-3 cycles, while the template with $\mathcal{M}_{eff}$ remains in phase with the signal for 6-7 cycle. During the short time interval where the template and the signal are in phase, the frequency sweeps from 185 Hz to 217 Hz. The signal and the template are in phase for a wider frequency interval when the mass is large (two black holes) than when it is low (two neutron stars).

This shows that an increase in the chirp mass in the system tends to compensate for the eccentricity of the signal. It also shows that for massive binaries, the template follows the signal for a wider frequency interval.

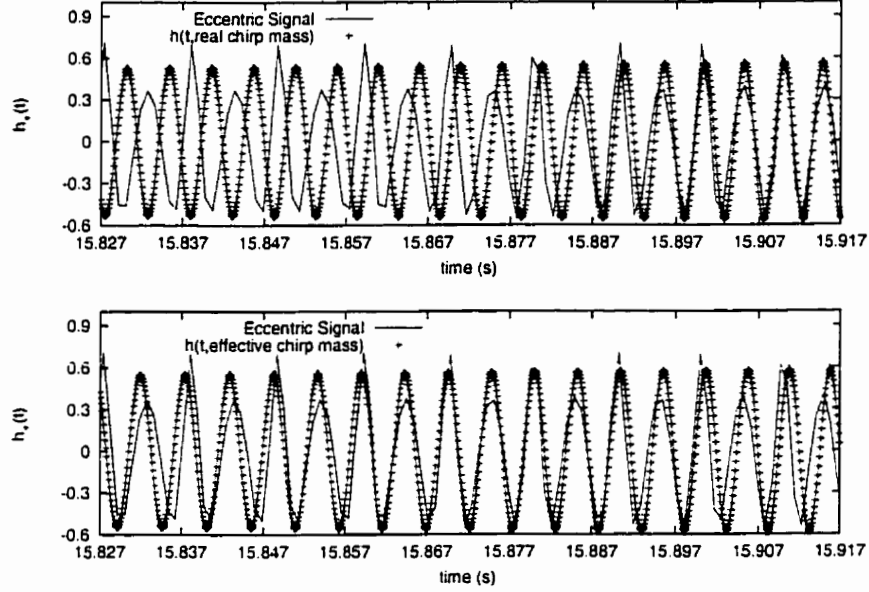


Figure A.1: Comparison of the phase in the time domain of the signal from a $1.4M_\odot + 1.4M_\odot$ binary with $e_{40} = 0.6$ with the circular template. The top portion shows the template built using $\mathcal{M}_r$ and the signal, while the bottom portion shows the template built using $\mathcal{M}_{eff}$ and the same signal. The template with $\mathcal{M}_{eff}$ follows the eccentric signal during more cycles than the template with $\mathcal{M}_r$.

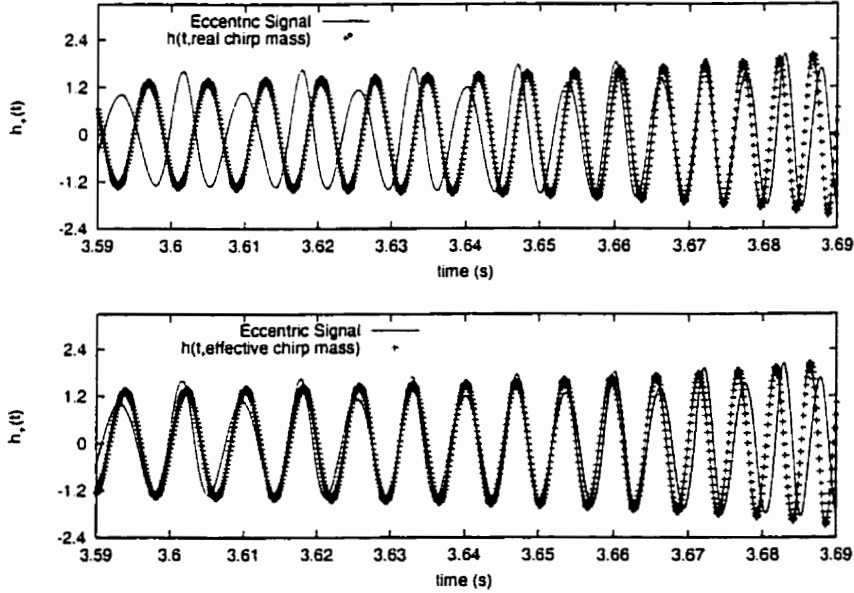


Figure A.2: Comparison of the phase in the time domain of the signal from a $6.0M_\odot + 6.0M_\odot$ binary with $e_{40} = 0.4$ with the circular template. The top portion shows the template built using $\mathcal{M}_r$ and the signal, while the bottom portion shows the template built using $\mathcal{M}_{eff}$ and the same signal. The template with $\mathcal{M}_{eff}$ follows the eccentric signal during more cycles than the template with $\mathcal{M}_r$.

Appendix B

Orbital Parameters: Effect of the Interpolation

In order to obtain accurate results for the fitting factor, it was important to monitor accurately the evolution of the orbital parameters. The numerical integration scheme for p , e , and φ is robust, but because we used a cubic spline interpolation algorithm to get a constant step-size in the time domain, oscillations were introduced around the real time dependence of the parameters. The purpose of this appendix is to show that these oscillations are never large enough to significantly affect the results presented previously.

The effect of the interpolation can be evaluated in two different ways. The first test we make is to compare the numerical results with the analytical solution for circular binary systems. This allows us to evaluate how important the oscillations in the parameters p and φ are, when $e_{40} = 0$. This method can not be applied when $e_{40} \neq 0$, because in this case, no analytical solution is available. We test our algorithm by using an equation that constrains the value e can take for a given value of p . This constraint equation will be derived below. We do not explicitly evaluate the quality of the phase when $e_{40} \neq 0$, but if the interpolation is done properly for p and φ in the circular case, and if the interpolation is done properly

for e and p in the general case, it is reasonable to assume that the phase is accurate.

For the circular case, we present the difference between the numerical results and the analytical solutions for a $1.0M_{\odot} + 1.0M_{\odot}$ binary and a $1.4M_{\odot} + 10.0M_{\odot}$ binary. For the first binary, the maximum difference is $\Delta\varphi \equiv |\varphi_{\text{analytical}} - \varphi_{\text{numerical}}| = 0.1$ radians (see figure (B.1) and $\Delta p \equiv |p_{\text{analytical}} - p_{\text{numerical}}| = 0.00047$ (see figure B.2). These values are reached at the end of the orbital evolution where the signal is heavily weighted by the power spectral density of the noise. For this reason, this part of the signal does not contribute largely to our results. For the phase, a difference of 0.1 corresponds to a relative error of $4.6 \times 10^{-4}\%$, since the phase at this point is 21570.62 radians. The final value of p at the end of the evolution for this system is 7.6, leading to an error of 0.006%. The same remarks can be made about the $1.4M_{\odot} + 10.0M_{\odot}$ binary, because in this case, $\Delta\varphi = 0.0025$ radians (see figure B.3) and $\Delta p = 0.00014$ (see figure B.4). These lead to a maximum relative error of $9.1 \times 10^{-5}\%$ for φ , and $2.4 \times 10^{-3}\%$ for p .

We now examine the error introduced in the evolution of the parameters when $e_{10} \neq 0$. The constraint equation mentioned previously is obtained by taking the ratio of $\dot{e}$ to $\dot{p}$ (equations (4.39) and (4.40)):

$$\frac{d \ln e}{d \ln p} = \frac{19}{12} \left(1 + \frac{121}{304} e^2 \right) \left(1 + \frac{7}{8} e^2 \right)^{-1}. \quad (\text{B.1})$$

Once the differential equation is solved, we are left with a constraint equation that must be satisfied during the whole orbital evolution. The parameters p and e are constrained by

$$p = K \left(1 + \frac{121}{304} e^2 \right)^{\frac{870}{2299}} e^{\frac{12}{19}}, \quad (\text{B.2})$$

where the constant of integration K is fixed once initial values for p and e are provided. We test the interpolation scheme by verifying that K keeps the same value throughout the numerical evolution. In other words, we evaluate K initially (using the initial values of p and e we chose) and we compare it with the value of K obtained from the numerical values of p and e , i.e. we evaluate $\Delta K = |K_{\text{initial}} - K_{\text{numerical}}|$. If the difference between the two values is small at all times, we can be confident that the algorithm is working properly.

We consider ΔK for two values of e_{40} for a $8.0M_{\odot}+8.0M_{\odot}$ binary. The two eccentricities considered are $e_{40} = 0.2$ and $e_{40} = 0.5$. For $e_{40} = 0.2$ we have $K_{initial} = 76.33$, whereas for $e_{40} = 0.5$, $K_{initial} = 32.45$. In the first case, the maximum difference $|K_{initial} - K_{numerical}|$ is 0.0012 (see figure B.5), corresponding to a relative error of $1.6 \times 10^{-3}\%$. In the second case, the maximum difference is 0.00018, corresponding to a maximum relative error of $5.5 \times 10^{-4}\%$. Thus, we are confident that the algorithm is working properly when $e_{40} \neq 0$.

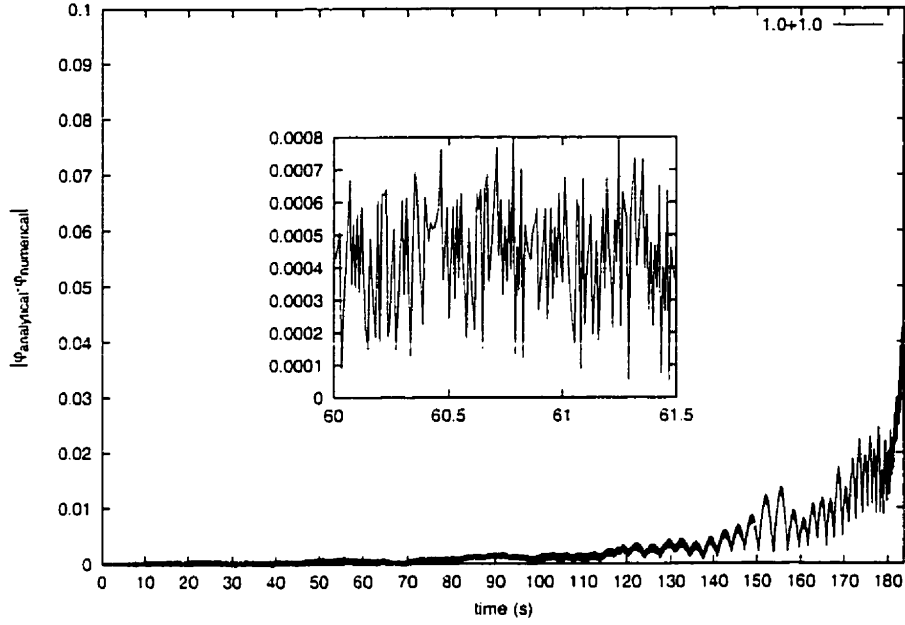


Figure B.1: The difference between the analytical solution and the numerical results for φ for a $1.0M_{\odot} + 1.0M_{\odot}$ binary. The maximum value the difference takes is 0.1. This is a small error compare to the large value of the phase at this stage in the evolution: the relative error is $4.6 \times 10^{-4}\%$.

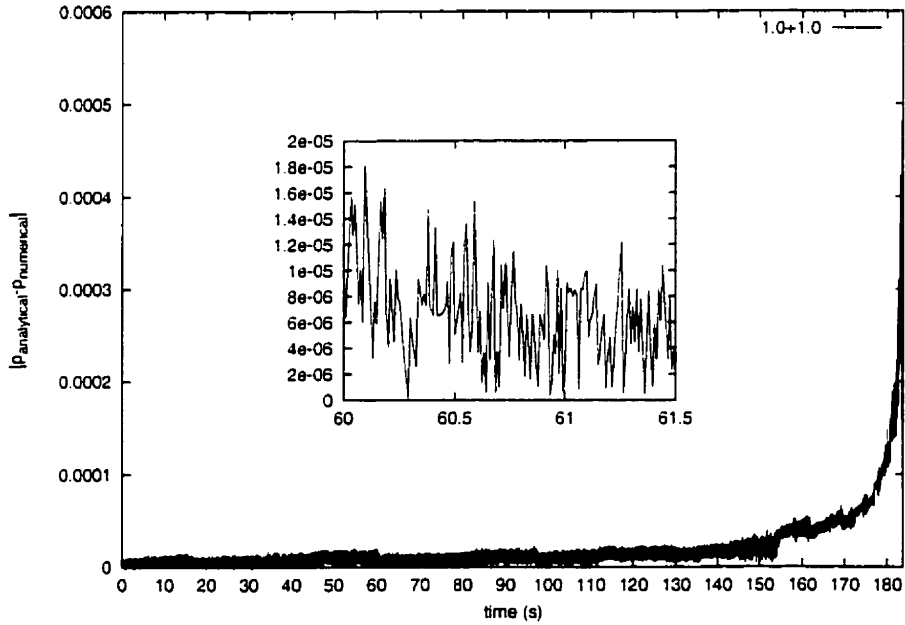


Figure B.2: The difference between the analytical solution and the numerical results for p for a $1.0M_{\odot} + 1.0M_{\odot}$ binary. The maximum value the difference takes is 0.00047. This leads to an error of 0.006% in the value of p .

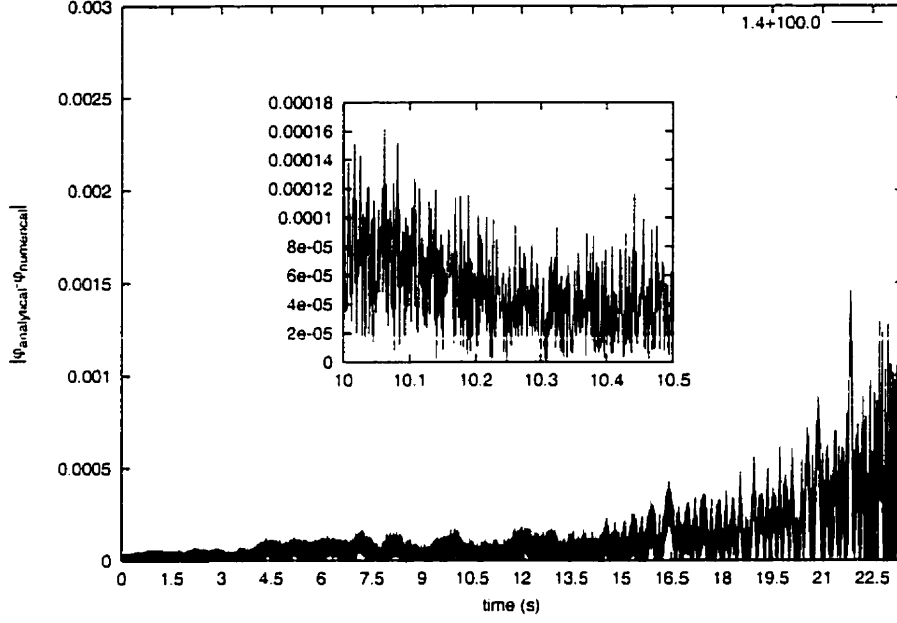


Figure B.3: The difference between the analytical solution and the numerical results for φ for a $1.4M_{\odot} + 10.0M_{\odot}$ binary. The maximum value the difference takes is $\Delta\varphi = 0.0025$. This corresponds to a relative error of $9.1 \times 10^{-5}\%$.

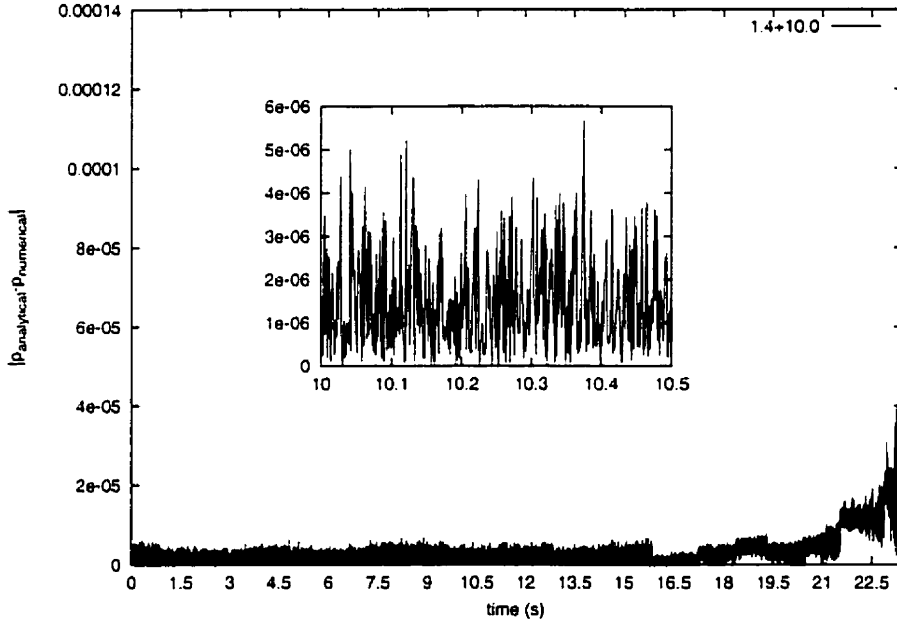


Figure B.4: The difference between the analytical solution and the numerical results for p for a $1.4M_{\odot} + 10.0M_{\odot}$ binary. The maximum value the difference reaches is $\Delta p = 0.00014$. This leads to a maximum relative error of $2.4 \times 10^{-3}\%$ at the end of the inspiral.

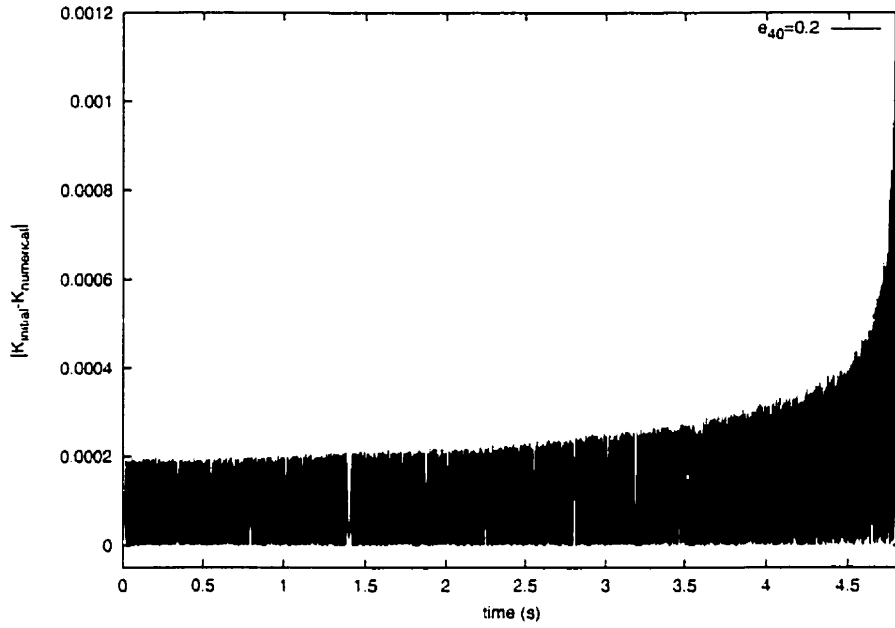


Figure B.5: The difference $|K_{initial} - K_{numerical}|$ for a $8.0M_{\odot} + 8.0M_{\odot}$ binary with $e_{40} = 0.2$. The maximum value the difference takes is 0.0012. The initial value of K is 76.33.

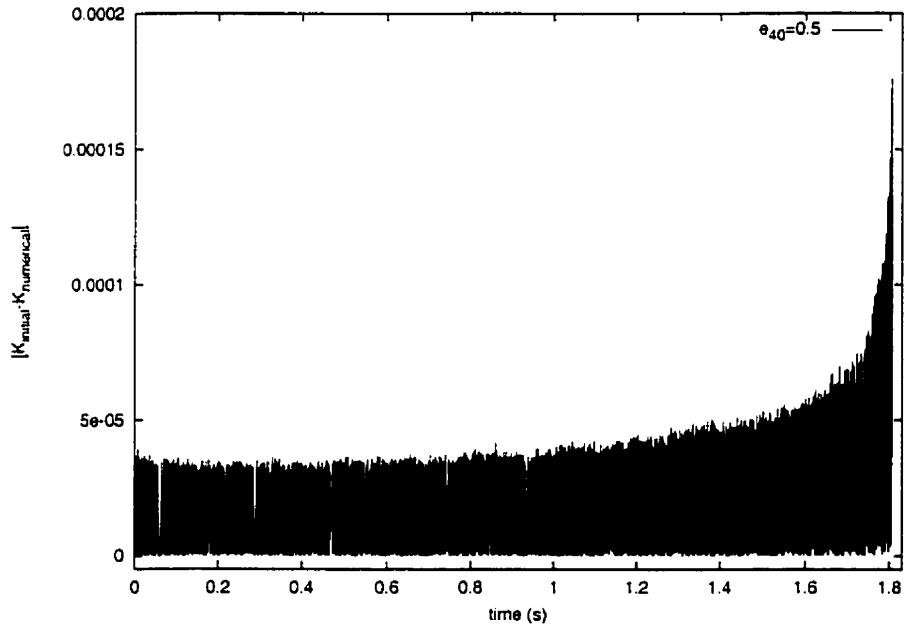


Figure B.6: The difference $|K_{initial} - K_{numerical}|$ for a $8.0M_{\odot} + 8.0M_{\odot}$ binary with $e_{40} = 0.5$. The maximum value the difference takes is 0.00018. The initial value of K is 32.45.